United Nations Development Programme

Project Document template for projects financed by the various GEF Trust Funds

<table>
<tr><td colspan="3">Project title: Niger national child project under the Africa Minigrids Program</td></tr>
<tr><td>Country:
Niger</td><td>Implementing Partner (GEF Executing Entity): Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)</td><td>Execution Modality:
Full NIM</td></tr>
<tr><td colspan="3">Contributing Outcome (UNDAF/CPD, RPD, GPD):
UNDAF Outcome: By end of 2022, the target rural populations will have access to innovative supply chains that generate decent jobs, participate in crisis and disaster management mechanisms related to food and nutrition insecurity and sustainable management of natural and energy resources adapted to climate change
By 2022, targeted rural populations have access to innovative value chains for decent job creation, participate in preventive mechanisms for the management of food and nutrition insecurity crises and disasters and the sustainable management of natural resources and energy adapted to climate change.
● Output 1.2.: Vulnerable communities in targeted rural areas, particularly women and youth, are empowered to sustainably manage natural resources by improving access to renewable energy and alternative technologies Vulnerable communities in targeted rural areas, particularly women and youth, have required capacities to sustainably manage natural resources with increased access to renewable energy and alternative technologies.</td></tr>
<tr><td colspan="2">UNDP Social and Environmental Screening Category:
Substantial</td><td>UNDP Gender Marker:
2</td></tr>
<tr><td colspan="2">Atlas Award ID: 00143728</td><td>Atlas Project/Output ID: 00131725</td></tr>
<tr><td colspan="2">UNDP-GEF PIMS ID number: 6659</td><td>GEF Project ID number: 10832</td></tr>
<tr><td colspan="3">LPAC meeting date: 20 October 2022</td></tr>
<tr><td colspan="3">Last possible date to submit to GEF: 31 August 2022</td></tr>
<tr><td colspan="3">Latest possible CEO endorsement date: 19 December 2022</td></tr>
<tr><td colspan="3">Project duration in months: 48 months</td></tr>
<tr><td colspan="2">Planned start date: 1 November 2023</td><td>Planned end date: 31 October 2027</td></tr>
<tr><td colspan="2">Expected date of Mid-Term Review: 1 November 2025</td><td>Expected date of Terminal evaluation: 31 April 2027</td></tr>
<tr><td colspan="3">Brief project description:
As part of the UNDP-supported, GEF-financed Africa Minigrids Program (AMP), this project seeks to increase access to accessible, reliable, affordable, clean energy in Niger rural areas by increasing the financial viability, and promoting scaled-up commercial investment, in renewable energy (RE) minigrids, with a focus on cost-reduction levers and innovative business models.</td></tr>
</table>

This will be achieved through 4 project outcomes: (i) Stakeholder ownership in a national minigrid delivery model is advanced, and appropriate policies and regulations are adopted to facilitate investment in RE minigrid development; (ii) Innovative business models based on cost reduction are operationalized, with strengthened private sector participation in RE minigrid development; (iii) Financial sector actors are ready to invest in a pipeline of low-carbon minigrids and concessional financial mechanisms are in place to incentivize scaled-up investment; and (iv) Digitalization and data are mainstreamed into local minigrid market development, and knowledge, awareness, and networking opportunities in the minigrid space are improved and facilitated. The project is expected to bring about direct commissioning of at least 0.334 MW in solar photovoltaic (PV) generation capacity and 0.814 MWh of battery storage. The lifetime greenhouse gas (GHG) emissions reduction from project activities, particularly investment in minigrid pilots, is estimated at 17,015 metric tons of carbon dioxide equivalent (tCO2eq) (direct) and 2,052,000 tCO2eq (indirect). The number of direct beneficiaries is estimated at 13,534 people, of which at least 51 percent are women, as a result of 4 new and/or improved minigrid connections.

FINANCING PLAN	
GEF Trust Fund grant	USD 1,601,376
Confirmed cash co-financing to be administered by UNDP	n.a.
(1) Total Budget administered by UNDP	**USD 1,601,376**
(2) Total confirmed co-financing to this project not administered by UNDP	**136,757,017**
(3) Grand-Total Project Financing (1)+(2)	**USD 138,358,393**

SIGNATURES:		
Signature: print name below	**Agreed by Government Development Coordination Authority**	**Date/Month/Year:** *within 6 months of GEF CEO endorsement*
Signature: print name below	**Agreed by Implementing Partner**	**Date/Month/Year:** *within 6 months of GEF CEO endorsement*
Signature: print name below	**Agreed by UNDP**	**Date/Month/Year:** *within 6 months of GEF CEO endorsement*

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LIST OF ACRONYMS

AFD	French Development Agency (*Agence Française de Développement)*
AfDB	African Development Bank Group
AMDA	Africa Minigrid Developers Association
AMP	Africa Minigrids Program
ANERSOL	National Agency for Solar Energy (*Agence Nationale d'Energie Solaire*)
ANPER	National Rural Electrification Agency (*Agence Nationale Pour l'Électrification Rurale*)
APE Solaire	National Association of Solar Energy Professionals *(Association Nationale des Professionnels de l'Energie Solaire)*
ARSE	Regulatory Authority for the Energy Sector *(Autorité de Régulation du Secteur de l'Énergie)*
BOAD	*West African Development Bank*
BPPS NCE	Bureau for Policy and Programme Support, Nature, Climate and Energy
CEMG	Clean Energy MiniGrid
CSR	Corporate Social Responsibility
DFI	Development Finance Institution
DREI	Derisking Renewable Energy Investment
E&S	Environmental & Social
ECOWAS	Economic Community of West African States
ECREEE	ECOWAS Center for Renewable Energy and Energy Efficiency
ESIA	Environmental and Social Impact Assessment
ESMAP	Energy Sector Management Assistance Program
ESMP	Environmental and Social Management Plan
EU	European Union
FISAN	Investment Fund for Food and Nutrition Security (*Fonds d'investissement pour la Sécurité Alimentaire et Nutritionnelle)*
GEF	Global Environment Facility
GEFSEC	Global Environment Facility Secretariat
GIS	Geographic Information System
GMG	Green minigrid
GoN	Government of Niger
IP	Implementing Partner
IPP	Independent Power Producer
IsDB	Islamic Development Bank
IRENA	International Renewable ENergy Agency
LCOE	Levelized Cost of Electricity
LDC	Least Developed Country
MERE	Ministry of Energy and Renewable Energies
MF	Ministry of Finance
MFF	MiniGrid Funding Facility
MFI	Microfinance Institution
M&E	Monitoring & Evaluation
MG	Minigrid
MSP	Medium Sized Project

NESAP	Niger Solar Electricity Access Project
NIGELEC	Electricity Utility in Niger
O&M	Operations & Maintenance
PASDEL	Project for Access to Energy Services and Promotion of Local Development (*Projet de l'Accès aux Services Energétiques et de promotion du Développement Local)*
PAYGO	Pay-As-You-GO
PDAE	Master Plan for Access to Electricity (*Plan d'Accès à l'Électricité*)
PERAN	Off-Grid Autonomous Rural Electrification Projects
PFD	Program Framework Document
PIF	Project Identification Form
PIR	GEF Project Implementation Report
POPP	Programme and Operations Policies and Procedures
PPG	Project Preparation Grant
PTFM	Multifunctional Platform
RE	Renewable Energy
RMI	Rocky Mountain Institute
SEFA	Sustainable Energy Fund for Africa
SESA	Strategic Environmental and Social Assessment (SESA) approach
SHS	Solar Home System
SNAE	National Strategy for Electricity Access (*Stratégie Nationale d'Accès à l'Électricité*)
STAP	GEF Scientific Technical Advisory Panel
TOC	Theory of Change
ToT	Training of Trainers
TSE	Special Electricity Tax (*Taxe Spécifique d'Électricité*)
WAEMU	West African Economic and Monetary Union
WB	World Bank

I. DEVELOPMENT CHALLENGE

General Context: Niger is West Africa's largest country (1,267 million km2) with one of the fastest population growth rates on the planet at 3.8%[1] reaching 24.2 million inhabitants in 2020[2]. However, this large landlocked country has one of the lowest population densities on the continent. About 80% of its area is barely inhabited desert and most of the population lives in rural areas in the southern parts of the country. Niger, classified as Least Developed Country (LDC), is one of the poorest countries in the world with 42% of its population living in extreme poverty in 2021 (under $1.90 a day)[3] out of which 75% are girls and women[4]. The population is comprised by 61% of working age women, however 84% men are part of the active population[5], therefore there is room to integrate women as part of the workforce. It has a weak economic diversification and competitiveness where agriculture accounts for almost 40% of its Gross Domestic Product (GDP) - mainly subsistence farming and herding. About 90% of its population works in the agricultural sector. About 90% of Niger's export revenues are related to mining including gold, uranium and thorium ores and concentrates[6].

Niger has seen its first democratic transfer in history with the election of Mohamed Bazoum in February 2021 and made some substantial change in terms of political governance, security issues and the number of refugees and displaced persons keep rising especially at its borders with neighboring conflicted countries such Burkina Faso, Mali and Nigeria. In March 2021, according to UNOCHA, there were about 250,000 refugees and over 276,000 displaced persons in Niger (over half are estimated to be women and almost 66% are under 18 according to UNHCR).

The country, one of the hottest on the planet, is also increasingly vulnerable to natural disasters and climate change effects while barely contributing to Greenhouse Gas (GHG) emissions globally (0.009% in 2019[7]). Climate shocks lead to a high rainfall variability and negatively impact the socio-economic situation. Water constraints, desertification, reduced agricultural yields, land deterioration, food insecurity, rising food prices, rapid population growth and increased pressure on arable lands become commonplace.

COVID-19 has compounded the structural fragility of the country, in addition to security and climate issues, and leading to a lower annual economic growth rate (shrinking to 3.6% in 2020 compared to 5.9% in 2019)[8]. While investment efforts around oil are boosting the economic growth rate in 2021 and will continue in the coming years with the exploitation of new oil fields, the livelihoods of the population especially in rural areas will remain at risk.

Energy situation in Niger: The power mix in Niger is highly dependent - close to 80% - on cheaper hydroelectricity imports from neighboring Nigeria (on average 90% cheaper than the production by the national power utility NIGELEC according to a report in 2016). Another 18% comes from costly diesel thermal generation by NIGELEC. The utility gets its fuel tax-free to nourish its diesel generator.

Niger has considerable untapped renewable energy (RE) potential, and the Government wants to increase its share of RE in the power mix. Currently, solar energy accounts for 0.9% of power mix, the country is well endowed with solar irradiation throughout the year (6 Kwh/m2/day with a daily insolation of 9 hours) and therefore, the production is expected to increase to 30% by 2030 according to the Sustainable Energy for All (SE4All) Investment Prospectus (2019).[9]

Moreover, the Kandadji 130 MW hydropower plant is planned to be operationalized by 2027.

[1] https://data.worldbank.org/indicator/SP.POP.GROW?locations=NE, visited on 4 May 2022

[2] https://data.worldbank.org/indicator/SP.POP.TOTL?locations=NE, visited on 4 May 2022

[3] https://www.worldbank.org/en/country/niger/overview#1 visited on 4 May 2022

[4] https://www.concernusa.org/story/poverty-in-niger/ visited on 4 May 2022

[5] https://data.worldbank.org/indicator/SL.TLF.CACT.FE.ZS?locations=ML-NE-BF-TD-MR visited on 4 May 2022

[6] https://www.worldstopexports.com/nigers-top-10-exports/#:~:text=The%203%20biggest%20exported%20products,of%20Niger's%20overall%20export%20revenues visited on 4 May 2022

[7] https://ourworldindata.org/co2/country/niger, visited on 4 may 2022

[8] https://www.coface.com/Economic-Studies-and-Country-Risks/Niger, visited on 4 may 2022

[9] IRENA, Niger Renewables Readiness Assessment 2013, available at http://www.irena.org/DocumentDownloads/Publications/RRA_Niger.pdf

National electricity capacity today represents only 272 MW (excluding mining companies Independent Power Producers IPP) with an import capacity of 194 MW. The existing national grid has a limited capacity in addition to being fragmented with 4 grids across the country that are not connected to each other.

Niger's electricity demand is expected to grow by 10% per annum[10] in the coming years - one of the highest growth rates in the world. Growing population and low electricity access rates are the main levers of this increase. The electricity access rate is one of the lowest in Sub-Saharan Africa at around 19.3% with 19.5 million people lacking access to electricity[11]. Based on a Multi-Tier Framework (MTF) survey done in 2018, 15.8% are connected to NIGELEC's national grid. In addition, discrepancies are relatively high between urban and rural areas; 48% in urban areas and only about 13% in rural areas (2022)[12].

Niger has taken some critical steps to improve the electricity access rate, especially in rural areas and energy markets by creating a regulatory body, *Autorité de Régulation du Secteur de l'Energie* (ARSE), to increase transparency and fair competition among numerous energy actors. The Government also created *Agence Nigérienne de Promotion de l'Electrification en milieu Rural* (ANPER), which is mandated to design, implement, and monitor rural electrification programs throughout the country.

Government has lately developed the Green MiniGrids (GMG) market through a variety of efforts to:

- Improve the legal and regulatory framework and create a conducive environment to promote access to electricity and the promotion of green minigrids (GMG), including via programmes/projects such as the World Bank's Electricity Access Expansion Project (NELACEP) and the African Development Bank's (AfDB) Green minigrid support Programme (funded by SEFA))[13], among others. However, a clear minigrid delivery model including the tariffication aspects is not yet available in this relatively nascent GMG market in Niger. The *Plan Directeur d'Accès à l'Électricité* (PDAE) for 2021-2025 identified clear criteria for rural localities which should benefit from MGs for their electrification (about 400 MGs). There is no active multi-stakeholder concertation framework yet in place around minigrids (MG) leading to limited experience sharing and some double work.
- Promote of green and hybrid minigrids development and management in rural areas in Niger. A variety of MG feasibility studies supported by development partners in collaboration with ANPER such as Power Africa, Liptako-Gourma Development Authority (ALG), World Bank, AfDB, etc. as well as the construction of MGs itself.
- Identify and test different delivery and business models for MGs. So far, these MGs adopt various business models financially sustainable and but are not supported by any relevant regulatory framework yet.
- Grant import duties exemptions on some solar MG equipment but are not considered sufficient by the local developers requesting for VAT tax exemption.
- Setup of some financial mechanisms to support rural electrification (e.g. via the Niger Solar Electricity Access Project NESAP project) including PAYGO for Solar Home Systems (SHS) and productive use, a dedicated credit line for local solar entrepreneurs available at partnering financial institutions but so far with a relatively high interest rate. For instance, *Plan International* partnered with a local microfinance institution, MFI Capital Finance, to provide guarantee lines for end-users to access *inter alia* solar ancillary equipment for productive use. The recently approved World Bank's Niger Accelerating Electricity Access Project (HASKÉ)[14] project will offer some subsidy for solar ancillary products to rural end-users (households, productive use) (World Bank, n.d.).

[10] https://www.se4all-africa.org/fileadmin/uploads/se4all/Documents/Country_IPs/PI_SeforALL_Niger_FINAL__1_.pdf viewed on 15 August 2022

[11] IEA, IRENA, UNSD, World Bank, WHO. 2022. Tracking SDG 7: The Energy Progress Report. World Bank, Washington DC)

[12] https://iea.blob.core.windows.net/assets/e2e30d7e-3051-4a25-9720-d608c7d6b2f4/TrackingSDG7TheEnergyProgressReport2022.pdf

viewed on 15 August 2022

[13] https://www.afdb.org/en/news-and-events/sefa-grants-us-1-million-to-promote-green-mini-grids-in-niger-15990

[14] https://projects.worldbank.org/en/projects-operations/project-detail/P174034

- Digitalize the energy space. Smart meters have been added for a few MGs including one operated by the private sector developer 'Africa Green Tech'. The HASKÉ project is considering an e-platform for tendering processes around MGs

Despite all the efforts made by the Government and the development partners through these projects, local private developers still lack sufficient capacities to promote the MG industry due to various risks: limited technical competencies, lack of financial capacities, missing suitable financial mechanisms provided by financial institutions in the country, limited payment capacities of very poor rural households in Niger, a conducive regulatory framework not yet available, MG profitability not yet clear, etc. An additional major hurdle lies in the low purchasing power of rural households across Niger unable to either access finance or to pay back their loans (including. PAYGO mechanisms).

Table 1 - Policy context for renewable energy minigrids in Niger

Policy / planning document		**Relevance**
Sector Policies	Strategic Vision / Development Plan: ***Stratégie de Développement Durable et de Croissance Inclusive 2035 (SDDCI)***, *2017 and* ***Plan de Développement Economique et Social (PDES)***	Electricity considered as crucial for human capital development as well as rural economy (access to water, irrigation, productive use/ income generating activities, women empowerment, youth employment) Acknowledgement that women are more vulnerable especially towards access to economic activities.
	National electricity policy: ***Document de Politique Nationale de l'Electricité (DPNE)***, 2018	Access to electricity for all; development of national energy resources; mobilization of the private sector. By 2021, the country aimed to electrify 1,400 localities, thereby achieving a 22% household access rate to electricity at the national level, including 10% in rural areas.
	Decree n°2019-027/PRN/MESU/DD of January 11, 2019 on the application modalities of the law n°2018-28 of May 14, 2018 determining the fundamental principles of environmental assessment in Niger	Determines the administrative environmental assessment procedure.
Strategies and Plans	National Electrification Strategy: ***Stratégie Nationale d'Accès à l'Électricité (SNAE)***, *2018*	Access to affordable, reliable and modern electricity services for all Nigeriens, based on the principle of social justice and equity. The target is at least 80% of the population by 2035 with the support of the private sector. Universal electricity coverage should be achieved as follows: 85% by grid extension from NIGELEC, 5% by MG and 10% by individual kits. Electricity services can be delegated, in the form of concession, leasing/affermage, interested management company or any other form of delegation, according to the conditions fixed by the application decree of the present law particularly in the field of green minigrids. This possibility is open not only to the State but also to local authorities. Pre-financing of meters to connect households, especially the most vulnerable ones, including for MG.
	Energy and Electricity Sector Master Plan : ***Plan Directeur d'Accès à l'Electricité (2019-2035 / PDAE)***, 2019	Increase the electricity access through 3 main options (i) densification (ii) new electrification via grid extension and MG development (iii) individual solutions for remote localities. Location ranking and prioritization to be electrified, according to the 3 options above. Budget planning.

Policy / planning document		Relevance
		About 400 projects covering around 1,000 localities have been identified for MG development.
	Nationally Determined Contribution (NDC) - ***CDN révisée***, *2021*	Key priorities for mitigation measure mainly around agriculture, forestry, water and energy. Regarding energy measures planned are around: improve access to electricity to 60% by 2030, reach 402 MXc for renewable energy production by 2030, reach 100MW for the off grid capacity by 2030, energy efficiency in the residential industry, household, transport and tertiary sectors; reduction of transmission and distribution losses; These measures in the energy sector should lead to a reduction of GHG emission of 5 324 ktCO2 by 2030.
	National strategy and actions plan of renewable energy, adopted in 2004	Aims to have 10% of the national power mix RE by 2020
	National Action Plan for RE - ***Plan d'Actions Nationale des Energies Renouvelables***, 2015	Increase contribution of renewables in the power mix up to 30%; Increase off-grid technologies capacity; Create a Renewable Energy national policy and a fiscal regime for renewables; Develop a strategic framework for PPPs
Laws	Electricity Sub-sector Law(s) and its application decrees ***Code de l'Électricité***, 2016	Ends the monopoly of NIGELEC, opening up the sector to private sector participation, particularly in generation and rural electrification Various application texts designed but not adopted at the same time or not yet adopted leading to misunderstandings
	Environmental legislation (pertinent to the energy sector): **Environmental Management Law**, 1998	Article 3.1. Activities, projects and development programs which, because of their size or impact on the natural and human environment, may affect the latter, are subject to prior authorization by the minister responsible for the environment. This authorization shall be granted on the basis of an assessment of the consequences of the activities, project or program updated by an environmental impact assessment prepared by the promoter and approved by the Ministry in charge of the environment.
	National Dialogue - ***Arrêté sur la création, organisation et composition du cadre national de concertation sur les énergies domestiques et alternatives***, 2002	National Dialogue focusing on domestic and alternative energy supply.
	Specific electricity tax, 2015	A tax applicable to all grids including off-grid that are under a delegation of services agreement. This tax of 2f/kWh is used at 60% to finance rural electrification and 40% to finance development and maintenance of public lighting, traffic lights, grid extension, payment for power costs of municipalities
	Decree around homogenized tariffs for NIGELEC, 2017	A homogenized tariffication grid for all – urban and rural areas – has been developed and applied by NIGELEC across the country
Dedicated regulations	Tariff-setting regulation (tariff level, tariff structure, including connection fees) & subsidy design	Not applicable for minigrids *per se*. Only in the ***projets d'Electrification Rurale Autonome hors réseaux au Niger*** PERAN (see below), minigrid developers are allowed to offer

Policy / planning document		Relevance
(minigrids)		cost-reflective tariffs (CAPEX & OPEX).
	Décret portant sur les Modalités de réalisation des projets d'Électrification Rurale Autonome hors réseaux au Niger (PERAN), 2019	Partially specifies the different delivery models around minigrids and individual kits with (1) public funding, (2) private funding and (3) private funding with a spontaneous application (without tender). In all cases the national PPP unit is involved. Minigrids should be based in priority on RE or hybrid solutions. They should be built complying with the state of the art and norms and regulations in Niger (which are non-existent yet for minigrid components) and should be designed to be potentially interconnected with the grid. When there is public financing involved, the MG developer can produce, transport, and distribute electricity in the given locality through a delegation of services for 5-10 years (renewable). The selection of the MG operator is based on a transparent tendering process. Minigrid developers are allowed to offer cost-reflective tariffs (CAPEX & OPEX).
	Market exit regulation at grid arrival	Not yet existent.
	Concessional Regime for minigrids - ***Conditions et modalités de conclusion des conventions de délégation et d'attribution des licences***, 2020	States all conditions and modalities for agreements around the delegation of electricity public services and the attribution of licenses in the energy sector including for minigrids
	Environmental review process for minigrid projects	The need for an environmental and social impact study for CEMG (Clean Energy MiniGrids) projects could be lifted according to the pilot size and location according to the relevant office at the Ministry in charge of Environment. A request for lifting (based on objective arguments) should be provided for each pilot in advance and approved by this office.
Standards	Generation technology requirements for minigrids	Not clearly stated but added to a certain extent on the specifications in tenders
	Quality standards for solar minigrid components	
	Standard waste management and recycling procedures	
	Specific grid codes for technical operation of minigrids	
	Interconnection standards	
	Quality of service standards	
Cross-cutting policy and regulation	Fiscal policy for minigrid components : ***Arrêté conjoint N°0029/ME/MF du 13 septembre 2017 portant liste des équipements et matériels à énergies renouvelables à exonérer des droits et taxes perçus en douane***, 2017	List of RE MG components (not exhaustive) that are exempted from any custom taxes and duties. Each year the list is updated. As quality standards are not clearly set, some components of lower qualities are imported and compete with prices that can go up to 7 times less than others.
	Public-Private Partnerships Law or Framework: PPP Contracts Law, 2018	Minigrids with public financing go through a PPP contract and the dedicated unit to support and monitor the PPP. It is also stipulated that the sectoral law (Electricity Code) for the energy industry prevails over the PPP Contracts Law. The original financial eligibility threshold of 5bn Francs has been taken out after a revision of the law. While a maximum contribution of the State of 5% to any PPP is set, for MGs, a partnership with a subsidy of 50-80% has been designed whereby the Ministry of Energy signs the convention and transfers the subsidy then to ANPER.

Policy / planning document		Relevance
	National Gender Policy - *Politique Nationale de Genre*, *2017* *Plan d'Actions National pour l'intégration du genre dans l'accès à l'énergie, 2020.*	In its analysis chapter, the document acknowledges the crucial role of access to energy and its negative impact on income generating activities, additional burden for domestic activities and health of women. It is critical to have a gender-based approach towards access to modern energy. There are no sex disaggregated data around energy and electricity in Niger.
	National Climate Change Policy - *Politique Nationale en matière de Changements Climatiques*, 2012	The policy relies on 6 key pillars all relevant to AMP Niger. (i) improvement of knowledge, promotion of R&D, production and dissemination of information on climate change; (ii) strengthening and developing the adaptive capacities of populations (iii) strengthening and developing actions to mitigate GHG emissions and promote (iv) the integration of climate change issues into national, regional and local planning tools; (v) capacity building of actors on climate change and (vi) the strategy for mobilizing financing for the implementation of the policy.

Risks & Barriers to renewable energy minigrid development

During the PPG, the preliminary conclusion of the currently ongoing Derisking Renewable Energy Investment (DREI) was made available and is presented in the Table 2 below. The DREI exercise is estimated to end by September 2022. It is expected to have a proper risk evaluation of minigrid expansion in the country at the start of the project.

Table 2 - Preliminary Conclusions of the DREI

N°	Risk Category	Description of underlying barrier	Risk Level	Preliminary recommendations from DREI analysis
1	**Energy market risk:** Risk arising from limitations and uncertainty in the energy market (off- and on-grid) regarding market outlook, access, price and competition	*Market access, competition and grid expansion:* Limitations and inability, including due to government regulations, of minigrid developers to access the electrification market; uncertainty regarding potential future competition in electrification; unclear, or lack of, grid planning and expansion policies. *Tariffs:* Uncertainty or inflexibility in electricity tariff regulations for minigrids. *Technical standards:* Lack of clarity, uncertainty and/or inconsistent government technical requirements for minigrids regarding (i) quality of service and (ii) grid integration, should it occur.	High	**Policy measures** ● Publication of the Rural Electrification Master Plan and regular updating with information on concession areas ; ● Establish a regulatory approach of 2 cohabiting regimes (light touch and comprehensive) for the granting of licenses, with a clear methodology for setting up required tariffs in the comprehensive regime and clear and balanced technical standards (revision of the PERAN decree). **Financial measures :**

N°	Risk Category	Description of underlying barrier	Risk Level	Preliminary recommendations from DREI analysis
				● Compensation system through a subsidy per kWh in the event of connection to the national grid → AMP Niger could consider supporting this measure after an assessment as stated in project activities Output 1.4 below).
2	**Social acceptance risk:** Risks arising from lack of awareness and resistance to renewable energy and minigrids in communities	Resistance by general public and local communities due to unfamiliarity with electricity and renewable energy sources; mis-information/perceptions and lack of awareness for minigrid offerings; resistance from incumbent businesses (e.g., diesel based generation) and users (e.g., SHS), disrupted by minigrids	Medium	**Policy measures :** ● Community and public awareness impact campaigns → AMP Niger will cover this under Component 4

N°	Risk Category	Description of underlying barrier	Risk Level	Preliminary recommendations from DREI analysis
3	**Hardware risk:** Risk arising from limitations in the quality and availability of mini-grid hardware, as well as the customs treatment of hardware	*Quality of hardware:* Lack of access to information on quality, reliability (performance) and cost of hardware; lack of clarity or uncertainty regarding government technical standards to ensure safety of minigrid hardware; lack of availability of warranties for components *Availability of hardware*: Lack of a competitive market for buying hardware (from both international and domestic suppliers); where appropriate, lack of locally tailored hardware *Customs:* Cumbersome customs/clearing process for importing hardware, leading to delays in delivery; punitively high customs tariffs on minigrid hardware, particularly in comparison to other sectors.	High	**Policy measures :** ● Development of standards and certifications for equipment; ● Streamlined customs procedures - Clarification of equipment exempt from import taxes and VAT. → AMP Niger could consider supporting this measure after an assessment as stated in project activities Output 1.4 below.
4	**Digital risk:** Risks arising from use of cellular networks for remote monitoring and payments; the use of software; and abuse of consumer data	*Cellular networks and mobile money:* lack of cellular coverage in rural areas, where electrification needed; over-dependence on a single operator for reliable cell service and payment processing; lack of mobile money, or limitations relating to fees on mobile money transactions *Software:* Limited standardization of software and interfaces on minigrid developers' back-end data and operations, and mobile money payment platforms *Abuse of consumer data:* possible abuse of consumer data privacy on payments and usage; lack of understanding/clarity on uses of consumer information	High	**Policy measures :** ● Telecommunications regulations for universal coverage and access to mobile money ; ● Support capacity building for industrial associations for the sharing of best practices and standards. → AMP Niger will contribute to this under Components 2 (building the capacity of the industry association) and 4 (communities of practice in Niger and sharing knowledge with other communities of practice embedded activity with the AMP Regional project)
5	**Labour risk**: Risks arising from the lack of skilled and qualified potential employees	Lack of a competitive labor market of educated, skilled and qualified potential employees, leading to higher costs, hiring non-local staff and suboptimal performance	Medium	**Policy measures** : ● Set up programs to develop a competitive and qualified labor market in renewable energies→ AMP Niger will support the analysis of existing curricula and programs for higher education and vocational training across Niger to identify gaps and support training activities at targeted institutions under Component 1.

N°	Risk Category	Description of underlying barrier	Risk Level	Preliminary recommendations from DREI analysis
6	**Developer risk:** Risks arising from limitations in the mini-grid operator's management capability, and its creditworthiness and cash-flow.	*Management capability:* lack of C-suite talent and experience to ensure effective execution (business planning, financial structuring, plant design (resource and demand assessment), installation, operations, and maintenance), and to manage challenges (limited information, unforeseen events) *Developer credit worthiness and cash flow strength:* Inability of developer to secure low-cost financing from investors due to lack of credit worthiness, or insufficient cash flows to meet investors' return requirements	High	**Policy measures :** ● Improvement of network effects and information flow / technical assistance to promoters → AMP Niger will support targeted capacity building for local private sector operators based on a need assessment under Component 2. **Financial measures :** ● Public loans, guarantees, public equity → AMP Niger will contribute with technical assistance to advance financing mechanisms to scale up CEMGs.
7	**End-user credit risk:** Risk arising from customers' willingness, ability, and methods of payment for electricity	*Lack of information on end-user credit worthiness:* Lack of end-user credit data with which to assess the ability of end-users to pay for the initial connection fees, ongoing electricity bills and ancillary equipment (e.g., lights and appliances) *Poor credit worthiness and non-payment:* Risk of delayed, reduced or non-payment by customers due to poor credit worthiness, lack of funds available, electricity theft and social dynamics	High	**Policy measures :** ● Promotion of the productive use of electricity (PUE) → AMP Niger will strongly support PUE as a main focus for any pilot site. In addition, a national study will be conducted on opportunities to boost economic activities through access to electricity and productive use (Output 2.2.) ● Facilitation of access to consumer credit / micro-credit → AMP Niger will facilitate consumer credit/micro-credit as part of Component 3 via technical assistance on relevant financing mechanisms and capacity building of financial institutions representatives in Niger ● Government mandates to insure solvent key customers (ex: telecommunication tower) **Financial measures :** ● Public loans, guarantees, public equit y→ AMP Niger will contribute with technical assistance to advance financing mechanisms to scale up GMG.

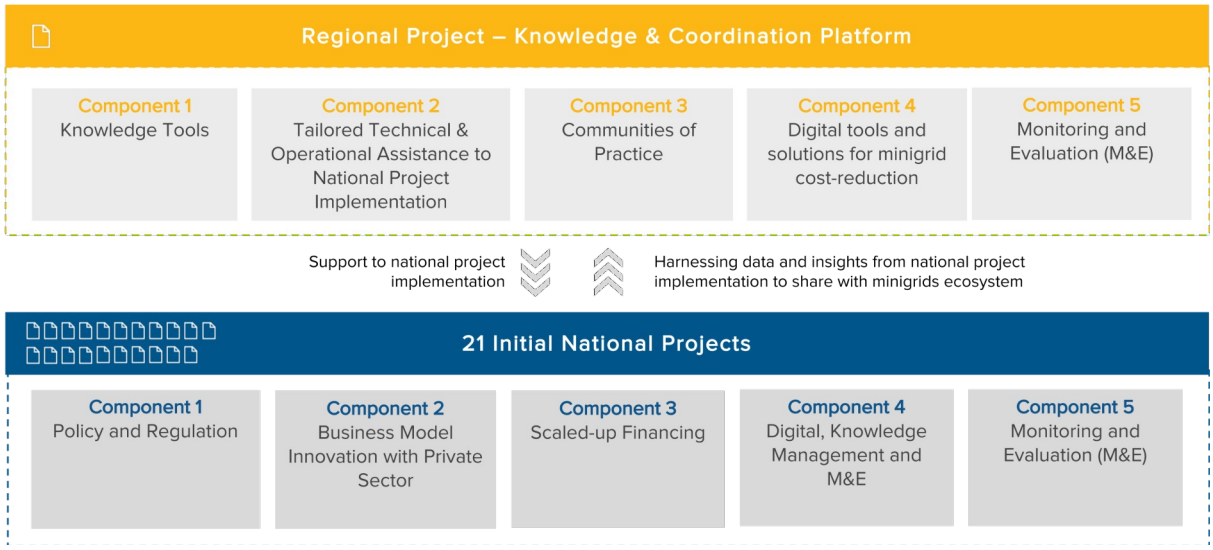
N°	Risk Category	Description of underlying barrier	Risk Level	Preliminary recommendations from DREI analysis
8	**Financing risk:** Risks arising from scarcity of domestic investor capital (debt and equity) for minigrids, and domestic investors' lack of familiarity with minigrids and appropriate financing structures	Capital scarcity - liquidity constraints in domestic banking: Limited availability of long-term domestic loans due to high banking reserve requirements Capital scarcity - under-developed domestic financial sector: Low number of well-capitalized actors (debt, equity, insurance, pensions); lack of regulatory clarity on new types of financial products Capital scarcity - competing incentives/ mandates: existing policies incentivize or mandate domestic financial sector (banks, pension funds) to invest in alternative, competing sectors to minigrids Limited domestic investor experience with minigrids: Lack of information, assessment skills and track-record for minigrid projects amongst domestic investor community; lack of network effects (investors, investment opportunities) found in established markets; lack of familiarity and skills with appropriate finance structures.	High	**Policy measures :** ● Liberalize the national financial sector (reforms for new types of financing, incentives) ● Technical assistance to financial institutions and investors → AMP Niger will provide technical assistance to these actors under Component 3 **Financial measures :** ● Public loans, guarantees, public equity → AMP Niger will contribute with technical assistance to advance financing mechanisms to scale up CEMGs.
9	**Currency risk:** Risks arising from currency mismatch between domestic currency revenues and hard currency financing	Uncertainty due to volatile local currency; unfavorable currency exchange rate movements resulting in domestic currency revenues not being sufficient to cover hard currency debt/equity servicing; inability to economically hedge FX exposure due to illiquid FX derivative markets.	Low	**Policy measures :** ● Support for the development of a foreign exchange market derivative products.
10	**Sovereign risk:** Risk arising from a mix of cross-cutting political, economic, institutional and social characteristics in the particular country which are not specific to mini-grids	Limitations and uncertainty related to conflict, political instability, economic performance, weather events/natural disaster, legal governance, ease of doing business, crime and law enforcement, land tenure and infrastructure in the country.	High	**Financial measure :** • Political risk insurance

III. STRATEGY

Programmatic approach. This project is part of the broader **Africa Minigrids Program (AMP)**, a regional technical assistance and investment program with the objective of supporting access to clean energy by increasing the financial viability and promoting scaled-up commercial investment in renewable minigrids, with a focus on cost-reduction levers and innovative business models. The programmatic approach aims to achieve greater impact by creating new minigrid markets across the African continent, which, in aggregate, will create scale and momentum, attracting private sector interest and investment. It will also allow for a broader sharing of knowledge and good practice, and create economies of scale in providing program services.

Program design. As shown in **Figure 1** below, AMP is comprised of two main elements: (i) a **Regional Project**, acting as the knowledge, advocacy and coordinating platform of the Program; and (ii) a cohort of an initial 21 **AMP National Projects** that share a common approach, seeking to reduce minigrid costs via five country-level components: (i) policy and regulations, (ii) business model innovation with private sector, (iii) scaled-up financing and (iv) digital, knowledge management and (v) monitoring & evaluation (M&E).

Figure 1 - Africa Minigrids Program's Architecture

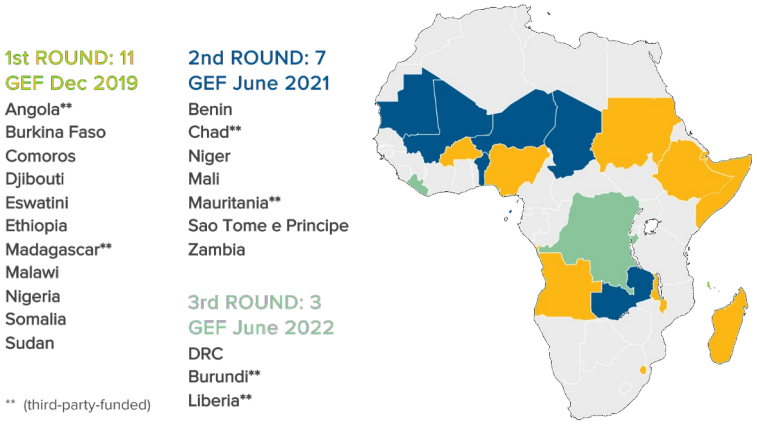


The program is initially supporting three rounds of national projects, totaling 21[15] in number, which together host an estimated total of **396 million people without electricity**,[16] or more than two thirds out of the 587 million total people without access to electricity in Africa. The initial AMP participating countries are shown in **Figure 2** below.

[15] A first round of 11 national projects approved at the concept stage in the GEF December 2019 work programme (Angola, Burkina Faso, Comoros, Djibouti, Ethiopia, Eswatini, Madagascar, Malawi, Nigeria, Somalia and Sudan). A second round of 7 national projects have been approved at the concept stage in the GEF June 2021 work programme (Benin, Chad, Niger, Mali, Mauritania, Sao Tome & Principe, and Zambia). A third round of 3 national projects (Burundi, DRC, Liberia) have been approved at the concept stage in the GEF June 2022 work programme.

[16] IEA (2022), Tracking SDG7: The Energy Progress Report, 2022, IEA, Paris https://www.iea.org/reports/tracking-sdg7-the-energy-progress-report-2022

Figure 2 - Africa Minigrids Program's participating countries list and map[17]



Country Strategy. Niger has been selected as one of the AMP National Projects as presented in Figure 2 above. The country is a poverty-stricken country with a large rural population (80% of total). A combination of climate, political instability and security shocks and crises has hampered the growth of Niger's economy. Niger must address a few the negative constraints if it hopes to reduce its structural fragility and achieve the sustained pace of economic growth needed to reduce development gaps. The country's economy is vulnerable to climate change and its population, more than 70% of whom work in subsistence agriculture, is affected by volatile weather conditions. The GoN prioritizes the power sector for both socio-economic development and climate adaptation. Unfortunately, current installed RE capacity is less than 1% of the power mix and the electricity access rate is less than 20% (and under 4% in rural areas). Yet, Niger has high and reliable solar irradiation intensity, and a significant opportunity for green mini-grids to play a major role in increasing electricity access. GoN has articulated in its National Electricity Policy (2018) an objective to have electricity access for all by promoting renewable energy. The National Electrification Strategy also mentions access to affordable, reliable and modern electricity services for all Nigeriens, based on the principle of social justice and equity with the support of the private sector. Moreover the '*Plan Directeur d'Accès à l'Électricité 2019-2035*' states that the electricity access increase should be done through 3 main options (i) densification (ii) grid extension and MG development (iii) individual solutions for remote localities.

So far, the minigrid market is at an early stage with 14 minigrids installed but not all are operational. Out of which 10 are owned by the Government (financed by Exim Bank of India), 1 by *Plan International* and 3 by private sector developers (1 offering a containerized MG solution). Other MGs are in the pipeline via the Niger Solar Electricity Access Project (NESAP – WB project)[18] , 7 MGs with 2 Mauritanian companies, 2 MG with ANPER, 9 MG with Plan International (connected to women-led multi-functional platforms) as well as IsDB, BOAD, etc. The market development is still very nascent, with a lot of actors, there is room for testing delivery and business model improvement.

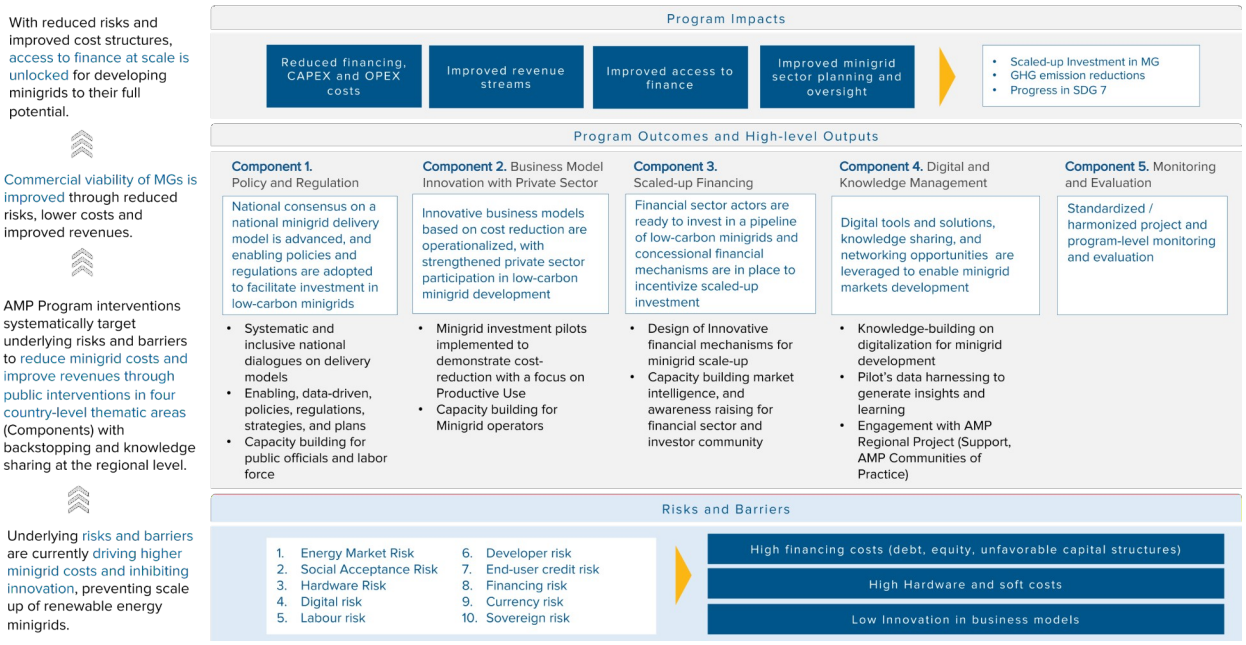
Program's Theory of Change (TOC). AMP Niger will follow the Regional AMP Theory of Change (TOC), developed in the Program Framework Document (PFD) and set out in **Figure 3** below. This TOC is premised on a baseline context where, while good progress is being made, several risks and barriers are driving high financing costs (equity and debt) and reducing the competitiveness of minigrids with respect to fossil-fuel based alternatives. All else being equal, the need for higher returns that reflect these risks translates into higher energy prices that, in turn, adversely affect affordability for the end-user, or lead to a larger subsidy requirements for rural electrification programs. As a result, renewable energy minigrids do not get financed and built at scale. By focusing on cost-reduction levers and innovative business models, the project can improve the financial viability of renewable energy minigrids which in turn can accelerate and scale up their adoption as part of the effort towards achieving universal energy access. When renewable energy minigrids are more

[17] The designations employed and the presentation of material on this map do not imply the expression of any opinion whatsoever on the part of the Secretariat of the United Nations or UNDP concerning the legal status of any country, territory, city or area or its authorities, or concerning the delimitation of its frontiers or boundaries.

[18] https://projects.worldbank.org/en/projects-operations/project-detail/P160170

competitive, private capital and commercial financing will then flow, resulting in various program benefits: investment at scale, GHG emission reductions, and electrification and lower tariffs for end-users.

Figure 3 - Africa Minigrids Program's Theory of Change



Alignment with GEF focal areas. The proposed strategy is aligned with the GEF Strategic Focal Areas CCM-1-1 "Promote innovation and technology transfer for sustainable energy breakthroughs for decentralized renewable power with energy storage", and CCM-1-3 "Promote innovation and technology transfer for sustainable energy breakthroughs for accelerating energy efficiency adoption".

UNDP's Derisking Renewable Energy Investment (DREI). The Program's TOC draws on UNDP's Derisking Renewable Energy Investment (DREI) Framework by focusing on cost reduction levers across the themes of policy and regulation, business model innovation and private sector as well as innovative finance that can be employed to reduce risk (e.g., policy derisking), compensate for risk (e.g., financial incentives) or transfer risk (e.g. financial derisking).

DREI is an innovative, quantitative framework to support policymakers to cost-efficiently promote private investment in renewable energy. A central focus of the DREI framework is on private sector financing costs. Developing countries often exhibit high financing costs for renewable energy due to investment risks that can exist in early-stage markets. The primary risks and barriers for renewable minigrid development and scaling up, as identified in the DREI Framework, are summarized in the Table 2 above. From an investor's perspective, these risks result in higher financing costs (equity and debt) and reduce the competitiveness of minigrids relative to alternative sources of energy (e.g., diesel generators). All else being equal, the need for higher returns that reflect these risks translates into higher energy prices that, in turn, or require larger subsidy requirements for rural electrification programs.

An opportunity is for policymakers to systematically address these investment risks, lowering financing costs and leading to competitive investment. Although there are both public and private strategies to address investment risks, the DREI framework is concerned mainly with public strategies, and identifies three central ways – often used in combination – that the public sector can improve the risk-return profile of private sector investment opportunities: (i) Reducing risk, targeting underlying barriers that create investment risk. These instruments are typically policies, such a legislation, or technical programmes ("policy derisking"); (ii) Transferring risk, shifting risk from the private to public sector. These include instruments such as guarantees, or credit lines to commercial banks for on-lending ("financial derisking"); and (iii) Compensating for risk, increasing the return of investments. These are typically targeted subsidies for renewable

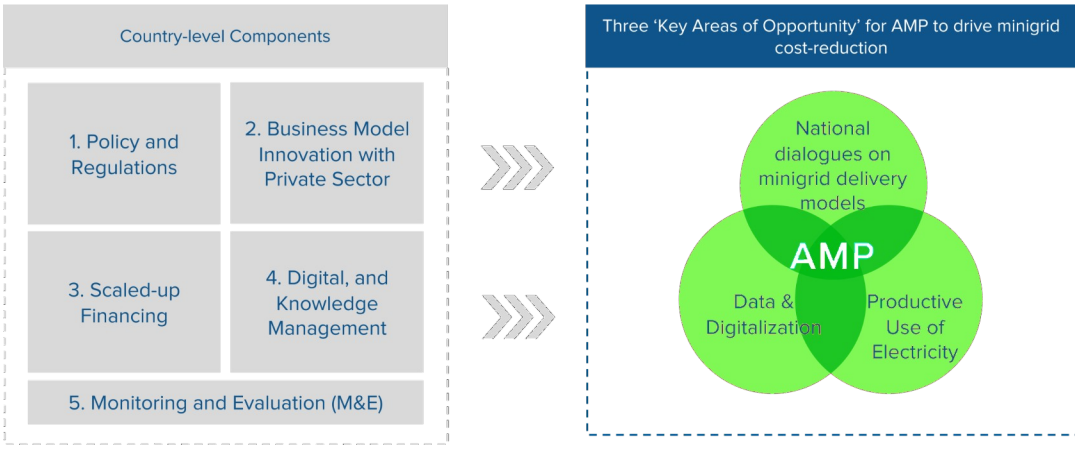
energy ("direct financial incentives"). As RE minigrid markets mature, an opportunity also exists for diversifying risk through aggregation of multiple mini-grid assets ("portfolio derisking").

As regards AMP, UNDP's DREI framework will be applied either qualitatively and quantitatively at various points in the project cycle, both at the national level in each country, and then aggregated into regional knowledge products by the AMP Regional Project and disseminated widely. The DREI framework, both at the national and regional level (in aggregate), will act as the program's mechanism to harvest and disseminate data on changes in the financing costs, hard and soft costs, and resulting costs for minigrids.

Program's 'Key Areas of Opportunity'. The AMP has adopted a common architecture of five key components - a combination of enabling policy and regulations, business model innovation with private sector involvement, innovative financing and digital innovation - as the levers to lower investment risks, thereby reducing financing, hardware and soft costs while increasing revenues and improving system efficiencies. The last component is related to monitoring and evaluation (M&E).

Within this architecture, AMP will emphasize - and seek to develop comparative advantages - in three 'key areas of opportunity': (i) an emphasis on advancing national dialogues on minigrid delivery models, (ii) promoting productive uses of electricity, and (iii) leveraging data and digital solutions for minigrid cost-reduction. Collectively these three areas can guide AMP's overall direction, creating a niche identity for the program. This approach, illustrated below in Figure 4, is structured to advance the program objectives of cost-reduction and innovation for minigrids and give effect to the TOC. The way in which this project will address these areas of opportunity is described in detail further below.

Figure 4 - AMP's Key Areas of Opportunity



National dialogue on minigrid delivery models. A delivery model (refer concept in Box 1 below) that is suitable to country expectations and context for minigrids has proven critical to establish an enabling and attractive investment environment for minigrids. Equally, a delivery model that has not been defined or is not consistent with the national context, will be an impediment to scaled-up investment. An important focus of the AMP is therefore to encourage a national dialogue between key stakeholders in support of a suitable delivery model being defined.

Today, there is no clear delivery model(s) for MGs in Niger. Some key aspects are not yet clearly defined and are being tested. The *Decree on Modalités de réalisation des projets d'Electrification Rurale Autonome hors réseaux au Niger* (PERAN) (see Table 1 in the Development Challenge chapter above) gives a broad and flexible framework for MGs. In practice, existing and future minigrids are based either on concessions, others are fully private, others, lately, are based on affermage-lease and some are under ANPER's supervision and without any (private) operator yet.

In addition, based on the social equity principle promoted in a variety of guiding laws and strategies, the government has expressed to offer an average national uniform tariff across the country, for on- or off-grid. On the other hand, the PERAN Decree for CEMG (Clean Energy Minigrids) operators stipulates a cost-reflective tariffs. The Government informed that

the discrepancy between these two tariffs would need to be compensated by ANPER. Therefore, a compensation mechanism is essential (see component 3).

Given the critical importance of the minigrid delivery model with its associated ownership, tariff structures and subsidy mechanisms in attracting private sector investment, Niger will have the opportunity to draw on the resources of the AMP Regional Project and the GoN's recent experience to identify the delivery model(s) most relevant to the country context, most feasible and with the best prospects of success. The possible options for each aspect need to be thoroughly understood by stakeholders and substantiated with real examples. The decisions for/against certain options must be openly discussed and weighed up in terms of the interplay between the aspects and the resulting consequences for the sector.

To this end, one of the first activities envisaged in the project is to get all relevant stakeholders on board and initiate a process of national dialogue to weigh up all aspects of minigrid delivery models (Output 1.1.).

Box 1 - The Concept of a Minigrid Delivery Model

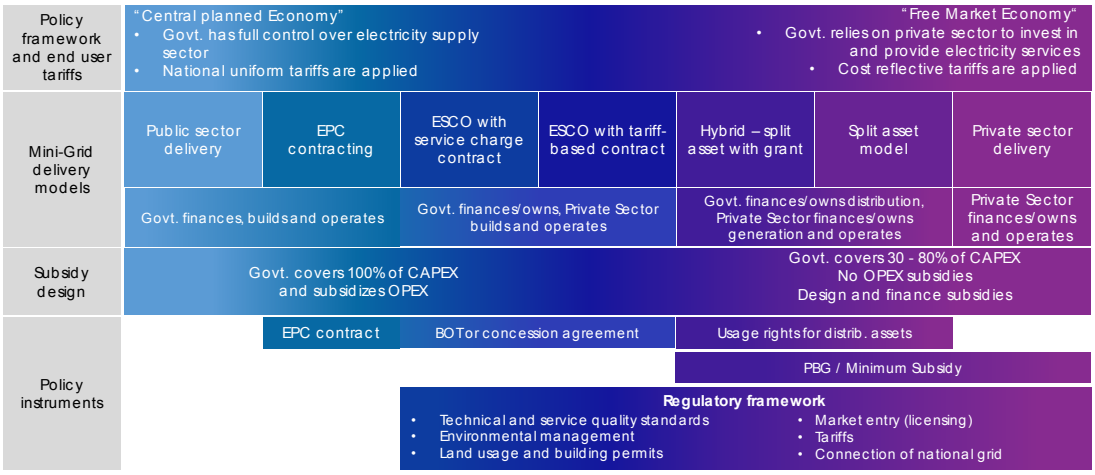
The concept of a minigrid 'delivery model' is a key concept for the AMP. This text box seeks to set out a common understanding of the concept and its importance to the national projects and overall program.

Definition: A minigrid delivery model, determined by the national government, is the cornerstone of a country's over-arching minigrid regulatory framework. It defines who finances, builds, owns and who operates and maintains the minigrids. Where applicable, it seeks to engage the private sector. A minigrid delivery model is closely associated to other key components of a minigrid framework, including tariff structures/mechanisms and subsidy levels/mechanisms.

In each country, identifying one (or more) delivery models will provide a framework for all sector stakeholders to plan for the longer term, particularly with regard to mobilizing private investment as one of the main objectives of the project. Figure 1 below describes the spectrum of design options for delivery models, across a number of different elements (ownership, policies, finance etc.)

This decision-making process around identifying a delivery model is complex and countries will be encouraged to establish a national dialogue for this purpose, involving all relevant stakeholders to varying degrees (different ministries such as energy, finance, health and environment, local authorities, the public, the media, the beneficiary communities, utilities, the private sector, and other key stakeholders) in order to build a national consensus on the basis of which large-scale deployment of mini-grids can be accelerated and have a sustainable impact.

Figure 5. Conceptual outline of minigrid delivery models



Source: JAKOB SCHMIDT-REINDAHL, Mini-grids Policy Expert, INENSUS

Digitalization (digital solutions and tools) and harnessing the data opportunity. The emergence of minigrids as a viable solution to electrify remote and isolated communities relies strongly on digital tools and solutions[19]. Digital technologies and solutions are fundamental to enabling off-grid electrification and offer significant potential to lower minigrid costs, reduce risks, and address barriers to scale. Many of the opportunities around digitalization are related to leveraging the large amount of data generated by minigrid projects to surface insights, learning and optimization. Data is a tremendously valuable asset in the minigrid sector that remains underutilized. The programmatic approach allows the AMP to make an impactful contribution to growing a data asset and harmonized digitalization in the sector. Employing digital integration as a catalyst for the minigrid sector reflects the UNDP digital transformation strategy that initiated a comprehensive process of connecting knowledge within the organization and across networks, creating opportunities, improving operational efficiencies, and building and maintaining partnerships and alliances. It also echoes the broader UN data-driven strategy and commitment to advance global "data action" with insight, impact and integrity.

In Niger, a digital platform centralizing all relevant data does not exist yet. However, the World Bank's HASKÉ project aims to cover some of this aspect with ANPER. A collaboration with other development partners will be sought, including UNDP, as discussed with the World Bank during PPG. This should lead to a comprehensive and integrative digital tendering and monitoring platform for CEMGs.

[19] Solutions are commonly grouped into four categories: (i) digital planning, (ii) digital operations, (iii) digital aggregation platforms, and (iv) digital payments

Box 2 - Digitalization and Minigrids

Digital opportunity for minigrids The figure below shows different categories of digital solutions in the minigrid sector: (i) digital planning, (ii) digital operations, (iii) digital aggregation platforms, and (iv) digital payments. In common to all these is the potential of digital technologies – whether used by policy makers, financiers or minigrid developers - to lower minigrid costs, reduce risks, and address barriers to scale.

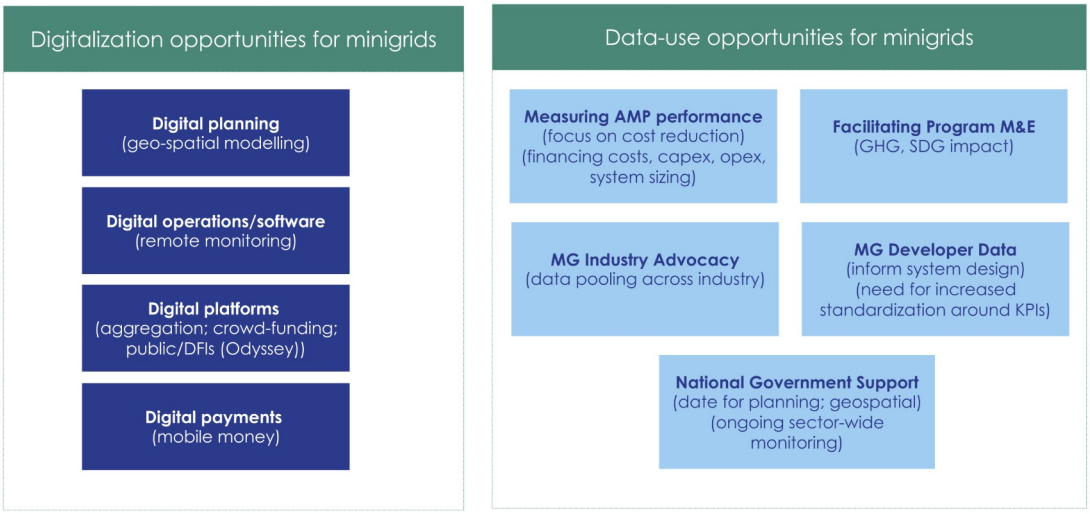


Figure 6. Digital and data opportunities for minigrids in the AMP

Data use opportunity for minigrids. Many opportunities around digitalization are related to leveraging the large amount of data generated by minigrid projects to surface actionable insights, learning and optimization to consolidate business models and technical solutions for scaling-up minigrids. For instance, the use of operational performance information from existing systems to forecast demand and design future minigrids can help avoid a very common pitfall of many minigrid systems which are significantly oversized and hence not financially viable.

Opportunities across the Program, and with the AMP regional project. The AMP provides a unique opportunity to develop a single set of metrics and guidelines for data collection, and use them to collect data from minigrid investment pilots across different national projects which the AMP Regional Project can then aggregate, derive insights from, and systematically disseminate knowledge with participating AMP countries and with the broader minigrids sector in Africa. At the same time, the link between the regional project and the total of eighteen (18) national child projects provides a unique 'distribution channel' opportunity across Africa for AMP to mainstream the use of digital tools and solutions for minigrids cost-reduction and scale-up.

Productive uses of energy (PUE). While PUEs are widely recognized as a key element to improve the viability and sustainability of minigrids, the AMP focus is uniquely tailored, taking a deliberate, integrated approach across a broad ambit of the Sustainable Development Goals (SDGs)[20]. With the benefit of global experience and best practices, the AMP pursues solutions where productive uses are embedded in agricultural value chains or around which economic activity can be anchored. The AMP's emphasis on energizing agricultural production is based on the demonstrated impacts and amplified benefits resulting from (i) improved product quality and increased yields, (ii) contributions to value addition, (iii) increased value retention within the rural communities, and (iv) contributions to socio-economic developmental objectives for rural areas, which in turn has a positive effect on the minigrid revenue model. Further recognizing that these multiple benefits cannot be assumed with energy access, but depend on wider development programmes, the AMP approach combines the delivery of electricity infrastructure with innovative business models and various

[20] As described by SEforAll, access to energy is not the end in itself, but a means to many ends. Reliable and affordable energy is needed to improve living standards, increase rural incomes, support delivery of health and educational services, and improve gender and social inequality. It also enables access to clean cooking technologies and fuels – essential to reducing indoor air pollution and associated health risks. https://www.seforall.org/sites/default/files/Beyond-Connections-Introducing-Multi-Tier-Framework-for-Tracking-Energy-Access.pdf

interventions aimed at encouraging economic activity, support business development and stimulate rural economic transformation with an emphasis on improved wellness, empowering women and youth as well as ensuring sound social and environmental stewardship.

The role of productive use is increasingly acknowledged by different stakeholders as critical in the delivery model in Niger. Power Africa's feasibility studies, for instance, highlight this. This includes anchor clients such as telecom towers as well as the role of cooperatives or technologies such as solar pumping irrigation, food processing equipment, lighting, etc.

Minigrid investment pilots' contribution to the Program's TOC. AMP Niger includes funds, under Component 2 (Business model innovation and private sector), for supporting minigrid investment pilots seeking to demonstrate innovative business models and cost-reduction opportunities. Minigrid pilots have a key role within AMP by contributing to demonstrate cost-reduction which can be leveraged to improve the financial viability of renewable energy minigrids. Minigrid pilots are aligned with one or more of the three key areas of opportunity mentioned above by demonstrating: (i) a particular delivery model or elements of a delivery model around which the government wishes to build capacity and engage with minigrid developers; (ii) productive uses of electricity and their potential to reduce costs and enable minigrid development at scale; and/or (iii) opportunities around digitalization and the use of data for minigrid cost reduction. Feedback loops to other national project activities (e.g., national dialogue, capacity building) and with the AMP Regional Project (e.g., Community of Practice) are intended to actively disseminate the learnings from the pilots to inform both the policy and regulatory environment as well as technical capacity building.

At the project preparation stage, considerable initial consultations, analysis and planning have been performed to advance the design of the minigrid pilots. This is described in Section IV below. At the beginning of project implementation, as an initial preparatory step, the PMU, in consultation with key stakeholders, and with support from the AMP regional project, will update and finalize the proposed approach to the design of the minigrid pilots, compiling a 'Minigrid Pilot Plan'. The project's 'Minigrid Pilot Plan' will include specifications on the pilot's approach that is aligned with key design principles set out in **Box 3** below. The project's Minigrid Pilot Plan will then (i) need to be reviewed and cleared by UNDP Niger CO and BPPS NCE and (ii) be shared with the Project Board.

Box 3 - Key principles for minigrid pilot implementation

Principle 1. Digital platforms. The use of digital platforms for tendering the pilots is a central element of the overall AMP and digital strategy for the project. With digital platforms emerging as critical enabler for procurement and operation of cost-effective and viable minigrids, using a digital platform for pilot projects provides an opportunity to build capacity of key stakeholders in using this facility which can then set the foundation for later using digital platforms for sector-wide, large-scale tenders or results-based financing programs. Digital platform software can manage the selection, Monitoring and Evaluation (M&E) and payments of pilots including capacity building of the Implementation Partner, Project Management Unit and minigrid developers. The digital tendering platform in Niger will be covered by World Bank's new HASKÉ project.

Principle 2. Productive use: third party ownership model. For pilots that will financially support the purchase of productive use equipment using an allocation under the GEF INV, it is required that the project will only provide its support via a third-party ownership model, as opposed to a self-ownership model. Third party ownership models involve the minigrid asset owner purchasing the productive use equipment, and then effectively leasing it back to the end-user, as part of an "energy as a service" offer.

Principle 3. Clear methodological basis for additionality for calculating the level of (GEF INV and/or UNDP TRAC)/financial support for capital expenditures (CAPEX). It is critical that the appropriate use of grant funding to the pilots be ensured, requiring a methodological basis for which the level of CAPEX subsidy will be determined during implementation using the principle of minimal concessionality. Suitable methodologies for minimal concessionality can include calculating the level of GEF INV support on the basis of achieving (i) Levelized Cost of Electricity (LCOE) parity with a diesel minigrid, (ii) LCOE parity with pre-existing residential tariffs, or (ii) LCOEs based on the willingness to pay of the end-users (via surveys etc.). During implementation, the AMP regional project may also provide updated guidance on suitable minimal concessionality methodologies.

Principle 4. Minigrid pilots data sharing. Pilot beneficiaries (e.g., minigrid operators) receiving support from the project will be required to share minigrid performance data with the national and regional project. Specific terms and conditions for data-sharing and how best to operationalize the commitment and its adoption by the beneficiaries will be defined and agreed upon with minigrid operators during project implementation, with support from the AMP Regional Project.

Principle 5. Compliance with UNDP Social and Environmental Safeguards and Gender requirements. Pilot projects receiving project funding are required to comply with all the relevant national standards of the country as well as UNDP standards as it pertains to social and environmental safeguards and gender equity. In support of this principle, an Environmental Safeguards Management Framework (ESMF), developed for the program, and a gender action plan accompany this ProDoc (Annexes 10 and 11). The ESMF will have to be incorporated and considered in developing the environmental and social management plans for pilot projects. A critical consideration under this framework is the need to ensure environmentally sound management of replaced equipment, including batteries, inverters and solar panels, after their usage. The responsible handling of waste with recycling of batteries and other recyclable equipment, should be clearly documented, budgeted and monitored in compliance with national and UNDP safeguards requirements.

Linkages to the AMP Regional Project: The project will align with the AMP Regional Project to foster knowledge sharing, learning, and synthesis of experiences in a multi-directional manner– i.e. flowing from the AMP Regional Project to the Niger project, and vice versa, and between the Niger project and other national projects within the Program. The main role of the AMP Regional Project is to make best practices in regulations and policies, innovative and inclusive business models, digitalization and technical support available to all AMP beneficiary countries, while providing technical and operational support for national projects' on-the-ground implementation. **Box 4** below provides a summary of the technical and operational support that will be available to the project. A full detailed elaboration of these offerings and the protocols attached to each service will be communicated to the project at the Niger project's inception workshop. The areas of support, listing of available firms/individual consultants under contract by the regional project and protocol for how the project can request and/or access such expertise (if needed/requested) will be elaborated in the first year of regional project implementation and disseminated to this project and the staff of all other participating AMP national projects.

Box 4 - AMP Regional Project Indicative Service Offering

Digital, Knowledge management and monitoring and evaluation (M&E)

Knowledge building/sharing. The regional project will curate, develop and share knowledge with the project on program's thematic areas (Policies and regulation, innovative business models, financing, digitalization).

Insight Briefs development and dissemination. National projects will gather data and audio-visual content (video footage, photos, etc.) highlighting national project activities which will be the subject of an 'insight brief' to be developed and widely disseminated by the AMP Regional Project.

Communities of Practice. One of the primary ways national project staff will interface with the regional project is via the 'Communities of Practice' (CoPs) and associated activities/platforms. While it is expected that many of the activities will be undertaken virtually (via internet-based platforms, webinars or digital platforms) it is also expected that the CoPs will include actual in-person workshops, meetings or training events that project staff will participate on.

Common M&E Framework/QAMF. The AMP Regional Project will develop, with inputs from national projects, a Quality Assurance and Monitoring Framework (QAMF) for measuring, reporting and verification of the sustainable development impacts of all minigrid pilots supported by national projects, including GHG emission reductions.

Data aggregation platform. The AMP Regional Project will deploy and use a web-based data management platform to aggregate data from all national project pilots based on the QAMF to track Results Framework indicators as well as program objectives, SDG impacts and GHG emission reductions.

Systematic data analytics and insights. The regional project will harness data shared by the national projects to extract insights and learnings which will be disseminated across all national projects and within the broader minigrids ecosystem.

Technical and operational support for national projects' implementation

Access to specialized expert international consultants in selected areas hired, retained, contracted and paid for by the AMP regional project and made available to all participating national project staff and selected beneficiaries on as needed basis. This support may range from virtual assistance to in-country missions.

Database of qualified international consultants and firms provided for information purposes to the project in an effort to assist in identifying high-quality experts and firms who may be available for contracting by national governments under their own procurement rules and modalities.

Generic terms of reference (ToR) for various standard activities will be provided to projects for information purposes.

Specialized advisory support for implementing UNDP's minigrid DREI analyses. During project implementation, the UNDP DREI Core team, working with the regional project, will make available to national teams and consultants the resources and tools to conduct full quantitative DREI applications, and will provide ongoing support and quality assurance.

Operational support for national projects. The AMP Regional Project will provide support to the project, on an ad-hoc and as-needed basis, through its PMU staff or by hiring or recommending subject matter experts, for the project to execute activities. Further details on specific support around M&E activities provided in Section VI (Monitoring and Evaluation Plan).

IV. RESULTS AND PARTNERSHIPS

Expected Results:
Five components and relevant outputs and activities have been developed to increase energy access in rural areas and stimulate the uptake of low-carbon minigrids (mainly solar-based) in Niger by improving the financial viability and promoting scaled-up commercial investment in such systems. The project focuses on the cost-reduction (hardware, soft and financing costs) and innovative business models for minigrids. UNDP's derisking approach will be adopted to catalyze private sector investments in the off-grid rural energy market. In doing so the activities proposed under the five project outcomes will seek to:

1. Advance national consensus on a national minigrid delivery model and adopt enabling policies and regulations to facilitate investment in low-carbon minigrids
2. Operationalize innovative business models based on cost-reduction, with strengthened private sector participation in low-carbon minigrid development
3. Facilitate financial sector actors' readiness in investing in a pipeline of low-carbon minigrids and ensure that concessional financial mechanisms are in place to incentivize scaled-up investment
4. Leverage digital tools and solutions, knowledge sharing, and networking opportunities to enable minigrids market development
5. Conduct the relevant project monitoring and evaluation

Greenhouse Gas (GHG) Emissions Mitigated. This project will result in GHG emissions reductions which will be measured via the GEF7 Core indicator 6: Greenhouse Gas Emissions Mitigated. This indicator captures the amount of GHG emissions expected to be avoided through the project's investment in renewable energy minigrid pilots and will be measured above a baseline value considering that in the absence of the project, the end users would have been supplied by fossil-fuel-based minigrid(s). Mitigation benefits include both (i) direct emissions reductions attributable to the minigrid pilot investments made during the project's implementation period, totaled over the lifetime of the investments (20 years); and (ii) Indirect emissions reductions resulting from the increased uptake of minigrids for off-grid electrification of rural areas due to replication, scaling-up and market change to which the project has contributed by creating a general enabling investment environment for minigrid market development, and facilitating subsequent investment flows. Annex 12 describes the methodology used to define targets for direct and indirect GHG emissions mitigated.

10% of the estimated indirect GHG mitigated of this project have been removed from the project and allocated to the AMP regional project, in line with the apportioning of the overall program budget and reflected in the PFD allocation of GHG emissions reductions across the different AMP national projects. This reflects the benefits of AMP national projects accessing the regional project's support which is expected to contribute and enhance the enabling conditions required for minigrids development across AMP countries.

Project components, outcomes, outputs and activities:

Some changes have been made regarding project components, outcomes and activities between the PIF (Concept Note) and the PPG (Project formulation). They are indicated in the table below:

Table 3 – Changes to the project based on baseline assessments

Outputs with GEF budget at Concept Note		Outputs with GEF budget at CEO ER		Change	Justification
1.1 An inclusive national dialogue identifying minigrid delivery models is facilitate and supports an integrated approach to off-grid electrification	*$ 195,206*	*1.1. An inclusive national dialogue to identify minigrid delivery models is facilitated, clarifying priority interventions for an integrated approach to off-grid electrification*	*$ 250,428*	Slight wording change	
1.2 Minigrid DREI techno-economic analyses updated to propose most cost-effective basket of policy and financial derisking instruments and contribute to AMP Flagship Report on Cost Reduction		*1.4. Targeted policy and financial derisking instruments of the Minigrid DREI techno-economic analyses implemented and contributing to AMP Flagship Report on Cost Reduction*		Scope change as DREI analyses conducted in 2021-2022 on the "Support for rural electrification by renewable energy systems in the Liptako-Gourma region - Pilot phase" project executed by UNDP and financed by SIDA	The DREI analyses were conducted from late 2021 to June 2022 as part of the ALG project (Integrated Authority for the Development of the Liptako-Gourma, a region shared by 3 countries: Niger, Mali and Burkina Faso) executed by UNDP under SIDA financing. The derisking instruments were identified. Some of them are considered and will be implemented as part of AMP Niger. A workshop sharing the targeted de-rising instruments of DREI under AMP will be held in Year 1. An update of the DREI will be conducted in Year 4.
n.a.		1.2. A review of the political and regulatory frameworks on the possible minigrid delivery models and suitable incentives is proposed in close collaboration with the National Dialogue Platform members and other development partners		Added an output on designing a suitable minigrid framework and what it should encompass	Focus will be put on clearly translating the suitable minigrid delivery model(s) into the political and regulatory frameworks. Relevant aspects of the framework will need to be identified at project inception as there are current progress made with ongoing projects from development partners. Therefore, an adaptive management approach will be required as changes will happen between PPG and implementation. This could include supporting cost-reduction efforts of equipment through existing but insufficient customs duties and tax exemptions on the one hand, and VAT reduction on the other hand as well as grid interconnection risks for private sector CEMG operators.
1.3 Design of a tendering program and all required bidding documentation, support for tender management (including		*1.3. Analysis of existing (pre)-feasibility studies conducted for selected minigrid sites to enhance sector planning and decision-making on a delivery*		Tendering program is already in place. The focus is put on leveraging the many existing (pre-)feasibility studies for pilot MG sites across the country. This	As a nascent market especially, it is critical to elaborate suitable tender documents and contracts – homogenized. During PPG phase and through consultations especially with ANPER and World Bank, the "tendering toolbox" design is ongoing. BIdding documentation templates have already been developed and. a

Outputs with GEF budget at Concept Note		Outputs with GEF budget at CEO ER		Change	Justification
use of e-tendering systems, proposal evaluation, and transaction advisory)		*model for minigrid development*		will help the entire ecosystem, including the private sector, to select and operate MG in a sustainable and viable manner	digital tendering platform should be provided by the new World Bank HASKÉ project)
1.4. Capacity building provided to public official (regulator, ministries) specifically to design procurement/tender processes that incorporate cost-reduction levers and innovative business models		*1.5. Capacity building provided to public officials (regulator, ministries) to support cost-reduction levers and innovative business models*		Focus of the capacity building for public officials expanded	During PPG, the need for institutional capacity building around minigrids (in every aspect) has been raised at inception and validation workshops. Capacity building efforts for public officials will encompass a larger spectrum of topics around CEMG and related rural electrification. A rapid assessment of the current capacity, the needs and gaps will need to be done. This will inform the relevant training required and develop a modular based approach on each participants' knowledge level and needs.
n.a.		*1.6. Domestication of quality standards and norms for solar minigrid components, and institutional capacity of ANERSOL strengthened*		Additional output on standards as barely considered in this nascent CEMG market	Only a few standards have been yet designed around some solar equipment with ANERSOL and World Bank. To ensure good quality standards for both CEMG equipment and ancillary devices, it is critical to define relevant standards and enforcement measures. Customs duties and taxes could be remained for lower quality products (to be discussed during project implementation)
n.a.		*1.7. Public programmes (apprenticeships, certificates, university programs) to develop competitive, skilled labor market in minigrids facilitated*		Additional output on public programmes around minigrids	The limited competencies at national level on minigrids represent a hurdle for Niger's minigrid market development. National programmes, with a modular approach and at different levels to cover specific needs, will be facilitated, based on the existing local offer, the AMP regional e-training platform, South-South collaboration including with other AMP projects, etc.
2.1. Pilots developed, including on productive use/innovative appliances and modular hardware/system design, leading to cost reduction in minigrids	*$545,591*	2.1. Pilots developed, including on productive use/innovative appliances and modular hardware/system design, leading to cost reduction in minigrids	*$773,433*	n.a.	More funding allocated to pilot projects as key for the development of a nascent minigrid market in NIger, as well as to prove its value proposition and enabling the potential of scaling up to be unleashed. The initial (PIF) $300k allocated for the Long Term Low Emissions Development Strategy (LT-LEDS) has been reallocated to the CAPEX costs for CEMG implementation as other funds are sought to cover this activity (probably Climate Promise).
n.a.		*2.2. National report on*			Given the fact that Niger's rural population is facing extreme

Outputs with GEF budget at Concept Note		Outputs with GEF budget at CEO ER		Change	Justification
		opportunities to boost economic activities through electricity access and productive use		Additional output	poverty, the entry point to contribute to the MGs sustainability is the productive use of energy. This study will be useful for the pilot sites of AMP as well as other projects to identify value chains in rural areas where PUE could be significant.
2.2. Capacity of winning tender bidders strengthened to develop and implement cost reduction levers and innovative business models		*2.3. Capacities of private minigrid developers and communities are strengthened*		Changes in terms of scope of the capacity buildings efforts based on stakeholder consultations changes	Capacities of local private minigrid developers (winning tender bidders are not) as well as communities will be built and/or reinforced. A modular approach based on knowledge level, needs and capacities required will be put in place for better results, adoption and application.
n.a.		*2.4. Support provided to establish and grow a national industry association for private sector developers*		Additional output	AMP is focusing on the private sector's role in supporting the development of the MG market in Niger. The need to have a local industry association for advocacy, business collaboration and knowledge sharing purposes would make a difference for local private sector players, as well as the National Dialogue platform and the development of the CEMG market in Niger (and the project's success).
3.1. Design support, including development of operational guidance, provided for Minigrid Funding Facility (MFF, or equivalent financial mechanism) under rural electrification agencies/funds	*$175,000*	*3.1. Support financing mechanisms to scale-up RE minigrids investment*	*$147,844*	The variety of existing and potential mechanisms need to be assessed to determine which mechanisms(s) the project could support through technical assistance	Based on stakeholder consultations, including with IP ANPER, a sectoral fund would be welcomed including to ensure that the identified delivery model (Component 1) is applied in practice Other existing financing schemes for minigrids and RE, as well as other sectors in Niger on the one hand, and a benchmark of successful financing mechanisms abroad could be considered. A comprehensive study is envisaged to assess the different options suitable for Niger. AMP Niger could then select where technical assistance could be provided to scale up financing for RE minigrids across the country.
3.2. Innovative financing solutions for minigrid development are identified and implemented through the MFF (or equivalent) with supporting human and		n.a.		Innovative financing included under Output 3.1.	Innovative financing mechanisms are considered as potential financing mechanisms and will be evaluated under Output 3.1. Some innovative financing are already used in Niger incl. crowdfunding for a 100% private-sector led 100% RE minigrid (Africa Green Tech).

Outputs with GEF budget at Concept Note		Outputs with GEF budget at CEO ER		Change	Justification
institutional strengthening					
3.3. Financial sector capacity-building on business and financing models for renewable minigrids		*3.3. Domestic financial sector capacity-building on business and financing models for minigrids*		Slight rewording	
n.a.		3.4. Replication plan (including investment plan) for scaling up rural energy access developed		Added	A replication plan based on best practices and lessons learned, including from AMP Niger pilot projects as well as other MG projects in the country, will be designed to scale-up MG market development while leveraging well-functioning innovative business models and more. The replication plan will also include a financing plan linked to the financing mechanisms supported by the project under component 3
4.1. A project digital strategy is developed and implemented, including linkages to (and following guidance from) the AMP Regional Project	*$ 240,000*	*4.1. A project digital strategy is developed and implemented, including linkages to and following guidance from the AMP Regional Project*	*$263,960*	Slight rewording	
		4.2. A 'Minigrids Digital and Data Management Platform' is implemented to run tenders and manage data from pilots, and to support minigrids scale-up and cost-reduction		Additional output as the digital platform is a critical aspect of the digital strategy to be implemented Output 4.2. moved to Output 4.3.	In line with Output 4.1. as well as with the entire project's components, a minigrids data management platform suitable to Niger' needs will be implemented. This should be covered by World Bank's HASKÉ project. It shall facilitate the tendering process from A to Z, monitoring of pilot sites and other indicators, cost optimization. AMP will ensure that this platform is adapted to the requirements. The project shall complement this platform with a repository of all relevant data, studies, reports and more around rural electrification in general and minigrid in particular, at national level. Indeed, during project formulation, the lack of a centralized repository was a challenge to gather all relevant, reliable and most recent data. Stakeholder consultations confirmed this need with all actors' types (government, technical and financial partners, private sector)

Outputs with GEF budget at Concept Note		Outputs with GEF budget at CEO ER		Change	Justification
4.2. Quality Assurance and Monitoring Framework for measuring, reporting and verification of the sustainable development impacts of all minigrids pilots, including GHG emission reductions is adopted and operationalized based on standardized guidance from the regional project		*4.3. A Quality Assurance and Monitoring Framework for measuring, reporting and verification of the sustainable development impacts of all minigrids pilots supported, including GHG emission reductions, is adopted and operationalized based on standardized guidance from the regional project*			
4.3. Engage with regional project, including, but not limited to, via (i) Communities of Practice and (ii) capturing sharing lessons learned		4.4. Engage with regional project, including, but not limited to, via (i) participating in Communities of Practice and (ii) capturing and sharing lessons learned			
4.4. Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, (iii) Mid-Term Evaluation and (iv) Terminal Evaluation				Output 4.4. moved to Component 5 as per new guidelines from UNDP/GEF to have a separate M&E component	.
n.a.		*4.5. Awareness raising campaigns, including lessons learned, are developed and disseminated at all levels nationally (including intervention zones) and with the regional project*		Added	To support visibility, adoption and minigrid market development and scale-up, targeted awareness raising campaigns at national (including political sphere and general public), and AMP regional levels will be designed and rolled out. This will include climate change risks and mitigation efforts.
	n.a.	*5. Monitoring & Evaluation*	$27,000		A dedicated component on monitoring & evaluation has been

Outputs with GEF budget at Concept Note		Outputs with GEF budget at CEO ER		Change	Justification
					added to comply with GEF and UNDP processes and facilitate potential corrective measures to achieve the expected project's results. The budget related to M&E is split between GEF funds ($7,000) and UNDP TRAC funds ($105,256) for at total of $112,256

Component 1. Policy and Regulation

Outcome 1: Stakeholder ownership in a national minigrid delivery model is advanced, and appropriate policies and regulations are adopted to facilitate investment in low-carbon minigrids.

As stated in the baseline, Niger is a relatively early-stage market for low-carbon minigrids. While 14 minigrids have been installed across the country under the supervision of ANPER, a clear national minigrid delivery model has not yet been clearly determined and adopted. The PERAN decree has broadly described different financing options but missing some concrete information on delivery and business models. Currently, a variety of models are on the ground with respect to ownership, tariffs, financing, etc. The need for a national delivery model with stakeholder ownership is crucial to support the market development with private sector involvement and investment.

The existing policy and regulatory framework for electricity in general and minigrids in particular is in place but still needs to be adapted, fine-tuned and adopted on the ground. The National Strategy for Electricity Access (SNAE) and the Master Plan for Electricity Access (PDAE) were prepared with GIS and techno-economic modelling electrification analysis across the country (including rural areas) leading to identifying the most suitable option for every locality across Niger: from grid expansion (NIGELEC), minigrids to individual kits. The PDAE 2021-2025 aims at enabling about 6,000 localities to have access to modern energy services out of which 300 minigrids. A DREI analysis is currently ongoing and to be completed by September 2022, it will determine which de-risking instruments to be supported by the GoN to facilitate the uptake of the low-carbon minigrids in Niger (see preliminary findings under Table 2 - Preliminary Conclusions of the DREI).

Output 1.1. An inclusive national dialogue to identify minigrid delivery models is facilitated, clarifying priority interventions for an integrated approach to off-grid electrification

The project will facilitate the establishment of a multi-stakeholder working group or platform to initiate systematic discussions that will inform the national government's identification of the optimal minigrid delivery model to meet the country's rural electrification objectives. The national dialogue will be centered around key issues regarding who finances, builds, owns and who operates and maintains the minigrids, and the related key components of a minigrid framework, including tariff structures/mechanisms and subsidy levels/mechanisms. It should lead to the official adoption of a national delivery model and a buy-in from all relevant stakeholders. Such a national dialogue platform has been highly welcomed by all stakeholders during the various consultations to facilitate coordination and avoid 'reinventing the wheel' and 'double work'.

This output includes the following activities:

Activity 1.1.1. **Establish multi-stakeholder national dialogue group/platform**

Support the establishment of a working group or a similar platform that includes a diverse and representative mix of stakeholders from Government, technical and financial partners, local authorities (collectivités), civil society, private sector, rural populations, local media and others, and initiate a national dialogue to identify the optimal minigrid delivery model based on the GoN's vision, the feedback from first MGs implemented in Niger as well as lessons learned from other AMP countries and beyond. The National Dialogue will be based on the Concertation Framework of energy sector players (*Arrêté portant création d'un cadre de concertation des acteurs intervenants dans le secteur de l'énergie*) that would be amended to include questions related to minigrids. After the review of this *Arrêté* a collaboration with the national dialogue platform around energy in Niger will also be envisaged. Efforts can be made to support and amplify women's and youth voices, for example, by ensuring at least 40% of participants are female and youth, and that at least one dialogue co-chair is female, that events are held at convenient times/places with childcare provided, and that communication barriers are addressed (i.e., linguistic barriers, digital/technology barriers, norms governing speaking in front of elders and members of the opposite sex).

Activity 1.1.2. **Provide inputs for national dialogue decision making**

Provide input to the discussion in the form of gap analysis, best practice reports, and suggestions for delivery models and make sure that the probable consequences of any decision taken for the overarching framework are evaluated and well understood. Support for ongoing gender mainstreaming of the dialogue may also be required.

Activity 1.1.3. **Establish feedback loop between national dialogue and the project**

Align the ongoing dialogue with activities implemented in parallel under the other outputs and loop respective (pre-)results back into the discussion. This should include, but not be limited to, activities which can shed light on trends and progress regarding minigrid cost reduction (e.g. DREI analyses and tracking of minigrid costs, resulting subsidy levels and electricity tariffs that will apply for minigrid pilots).

Output 1.2. A minigrid regulatory framework, including tariff model, tax regime, and grid expansion risk, is developed in close coordination with the authorities concerned and other development partners

Quite some efforts have already been made by various partners to support the GoN in creating a conducive policy and regulatory framework for minigrids and especially green minigrids, including AfDB and World Bank. At the time of writing, several other initiatives and activities are planned by different stakeholders, including GoN, ANPER, World Bank, AfDB and EU, around the general electricity regulatory framework including the minigrids regulatory framework.

Limits of the existing framework have been identified by many stakeholders and included in a recent diagnostic study on the policy and regulatory framework of the Electricity sector in Niger, including minigrids, financed by the EU. A revision of relevant policies and regulations starting from the Electricity Code and including minigrids and private investments is planned with the support of the World Bank's HASKÉ project. This revision is a prerequisite for the EU's involvement in the energy sector in Niger by 2023. A first draft of the revision should be available by December 2022 and a final version latest by April 2023. As such, UNDP Niger will participate in the dialogue on the revision prior to AMP Niger's project management unit (PMU) team being in place by early 2023.

The World Bank HASKÉ project will develop an online platform for the management of license applications located at ANPER. The digital platform will be used from designing to monitoring minigrid projects. This platform will be available to all relevant partners, including AMP.

This output includes the following activities:

Activity 1.2.1. Conduct a brief assessment of the actions taken in terms of minigrid regulatory framework since project formulation

A rapid assessment of actions taken and planned by other stakeholders will be conducted. The results and gaps of the assessment will be shared and discussed during the National Dialogue Committee. AMP's activities will then be adapted, where necessary, using an adaptive management approach and in line with the minigrid delivery model identified and selected by the national government and the National Dialogue Committee

Based on the stakeholder consultations as well as desk reviews, this *could* include - *if not covered yet at project launch* by t he Electricity Code revision or other initiatives– the following:

- Develop the tariff model and calculation tool for isolated minigrids in collaboration with ARSE and taking the following into account:
    - ECREEE RE tariffication tool (financed by EUEI PDF 2017)
    - World Bank study conducted under the NESAP project
    - UNDP study recommending off-grid tariffication integrating payment capacities of rural populations and CAPEX realities for a private sector operator in Niger in 2016 (*cadre et schéma de gouvernance de l'électrification rurale*)
    - Results from the DREI study Analysis conducted during the project preparation phase
    - Lessons learned from the first implemented minigrids on national grid tariffs, business models, system sizing, CAPEX related aspects and recovery, payment capacities of end-users, types of end-users (residential, commercial, social), living/working arrangements of end-users (e.g., polygynous households, cottage businesses, child-headed households, displaced or resettled households) etc.
- Assess the grid expansion risks: the PDAE clearly defines which localities would be covered by a national grid expansion/densification by 2035 with specifics on off-grid solutions such as minigrids and individual kits. The

risk of grid expansion and interconnection issues remain as the compensation schemes for private sector developers and operators represent an additional hurdle for the private sector to invest in minigrids.

- Set up standardizing and categorizing interventions of minigrid operators (technology type, etc.) - see output 1.6

Activity 1.2.2. Formulate articles to be added to relevant minigrid regulatory texts and support endorsement

The results and gaps assessment (Activity 1.2.1) will help identify legislation, policies, regulations, sector plans and strategies that require further analysis and development to properly address minigrid development challenges and risks. This activity will support further analysis and development of the policy derisking instruments as needed. The project, including through the National Dialogue platform, will support the endorsement of the revision of the relevant legal texts.

One such area of intervention could be the revision of a 2017 decree (updated in 2022) establishing tax exemptions applicable to solar PV technologies which has not yet addressed specific challenges faced by the minigrids sector.

Importing equipment (including RE related ones) are taxable in three ways: customs taxes and duties, VAT and other tiny taxes (including ECOWAS, WAEMU and statistics taxes - all together under 10%). In Niger, a 2017 joined act (*Arrêté conjoint portant liste des équipement et matériels à énergies renouvelables à exonérer des droits et taxes perçus en douane*) lists some solar components (mainly related to individual kits) that are exempted from custom taxes and duties. While this list should be updated once every year, the first review was done early 2022 encompassing some solar ancillary equipment (eg. solar pumps and solar refrigerators). The VAT (19% in 2022) was not included in that decree while it is hampering private sector energy services players.

Other areas of intervention could include:

- structuring arrangements after the Electricity Code revision
- adding VAT reduction or exemption to the relevant 2017 joint act (as stated above)
- revising standards and business models of minigrid operators (including technology type, etc.) - see output 1.6.
- adopting a Strategic Environmental and Social Assessment (SESA) approach where the role of local authorities and communities engage localities and highlighted in the regulatory framework. The "*Code des Collectivités*" and its relevant clauses will be considered where necessary.

Output 1.3. Analysis of existing (pre)-feasibility studies conducted for selected minigrid sites to enhance sector planning and decision-making on a delivery model for minigrid development

Feasibility studies have been undertaken by various donors and partners in coordination with ANPER in approximately 350 out of 500 localities identified as potentially relevant for electrification via minigrids. Component 2 will address activities to analyze and summarize the main findings and impacts on a suitable minigrid delivery and business model to be tested. Moreover, some additional (pre)-feasibility studies could be undertaken in the remaining villages (i.e.,150 villages) and which would nourish the decision-making on a delivery model and relevant financially sustainable business models for minigrids (see Component 2).

In addition to the demand assessment and trends, ability and willingness to pay, productive use (commercial and social), and other aspects, the economic development potential will be further assessed along with a holistic approach of all stakeholders and their needs (not only in terms of electricity).

This output includes the following activities:

Activity 1.3.1. Cluster existing (pre-) feasibility studies by type of village and needs and identify missing pre-feasibility studies per type of village/potential business model

Existing pre-feasibility studies undertaken by various partners (including private sector players) and available at ANPER will be collected, analyzed and grouped based on common specificities leading to specific delivery and business models. Objective criteria will be identified to categorize each locality. Consultations with selected stakeholders including on their feedback on existing minigrids and more will be conducted. Relevant updates and hypotheses (e.g., demand evolution, etc.) will be done where deemed necessary for the analysis as some studies have been conducted a few years ago already. Where possible, gender aspects can be added to the clustered feasibility studies (e.g., cultural groupings, women's land

tenure, presence of micro-finance institutions and women's agricultural and artisanal co-ops, etc.) as well as environmental and social considerations.

Activity 1.3.2. Draw recommendations for sector planning and decision-making on a delivery model

Based on Activity 1.3.1. and benchmarks within AMP countries and beyond, different options will be elaborated with their respective strengths and weaknesses. These options will nurture the discussions and conclusions for the national delivery model within the National Dialogue platform (Output 1.1.).

Output 1.4. Targeted policy and financial de risking instruments of the Minigrid DREI techno-economic analyses implemented and contributing to AMP Flagship Report on Cost Reduction

As regards to AMP, UNDP's DREI framework will be applied either qualitatively and quantitatively at various points in the project cycle, both at the national level in each country, and then aggregated into regional knowledge products by the AMP Regional Project and disseminated widely. The DREI framework, both at the national and regional level (in aggregate), will act as the program's mechanism to harvest and disseminate data on changes in the financing costs, hard and soft costs, and resulting costs for minigrids.

DREI, by interviewing the private sector, is a well-suited tool to evaluate the risk environment and suggest instruments at various stages of the project to ultimately lower the project development costs. The DREI financial modeling stage captures hardware and soft costs to determine the levelized cost of electricity (LCOE) of the technology being assessed.

The preliminary conclusions of the DREI can be found under Table 2 above. The report will be shared and made available on the Regional AMP Knowledge Platform and will inform Niger's National Dialogue (Output 1.1) and the PMU to advance the scaling up of the market through derisking measures.

This output includes the following activities:

Activity 1.4.1. Identify and disseminate specific DREI derisking measures

Based on the national DREI analyses and the derisking measures, the PMU will select relevant DREI derisking measures specific for the AMP Niger project. This might lead to adapting some project outputs/activities accordingly. In Year 1, the project will build on the DREI workshops undertaken in 2022 and organize 1 round-table workshop with government, private sector and other key stakeholders to disseminate the identified DREI derisking measures to be covered by the project.

Activity 1.4.2. Coordinate with regional project on national DREI analysis update (Year 4)

In the final year, or year 4, of the national project's implementation period, whichever happens first, the original national-level DREI analyses will be refreshed to track evolutions in financing costs as well as in hardware and soft costs. For administrative efficiency, the regional project will fund and execute this update (a 'light quantitative DREI analysis'), on behalf of the national project. The deliverable will be a brief note of 2-5 pages on the DREI national update. The data from the national refreshed DREI analysis will be fed into an update note to the year 2 AMP flagship regional DREI knowledge product, which will provide an end-of-program overview of the evolution in minigrid costs across AMP countries. The national project's contribution to this activity will be: (i) facilitating the DREI national update (to be funded and executed by the regional project); and (ii) disseminating the findings of the national DREI update note, and the update to the regional flagship DREI product.

Output 1.5. Capacity building provided to public officials (regulator, ministries) to support cost-reduction levers and innovative business models

While public officials benefited and will benefit from various capacity building around low-carbon minigrids with the support of various partners, fact is that:

1. At various ministries and agencies, new nominations and appointments were made leading to a loss of institutional memory. Ensuring that knowledge is acquired and transferred to new staff is critical.
2. Knowledge disparities exist within a same unit/department as well as between entities (ministries/agencies/utilities)

3. Some aspects in terms of capacity building were not yet covered and AMP Niger aims at bridging these gaps (be it at regulatory, managerial or technical levels)
4. Training should target not only the ministry and the agencies related to energy but also other members of the National Dialogue Platform as access to electricity is a cross-cutting topic and is critical to support economic, social and environmental development across Niger.

This output includes the following activities:

Activity 1.5.1. **Analyze provided capacity building activities and identify gaps**

The following sub-activities would be conducted:

- List all capacity building activities around low-carbon minigrids in the past years with the support of public officials and technical & financial partners, and which ministry or agency benefited from it.
- Gather all training materials to avoid reinventing the wheel and upload them on a specific platform (as part of the Community of Practice under Component 4) so all public officials and beyond can access it.
- Assess the needs of targeted public institutions
- Identify gaps based on the training materials collected along with the feedback of public officials.
- Draw recommendations on capacity building efforts to be provided for public officials
- Coordinate with other development partners and beyond on their planned trainings around minigrids

A special focus will be put on incorporating gender aspects in the entire minigrid project life cycle including the tendering process (including gender scoring of tenders) as well as SES aspects.

Activity 1.5.2. Design comprehensive training materials and conduct workshops with a gender-diverse selection of public officials

Based on Activity 1.5.1., specific comprehensive training materials supporting the understanding and ownership of the identified national minigrids delivery model(s) by public officials will be developed and rolled out. These capacity building efforts lie on multiple approaches: (i) modular whereby participants will attend trainings based on their needs and knowledge level (ii) holistic aspect taking a village with all its activities around minigrids, i.e. including representative members of households, businesses, and social institutions (schools, health centers, etc.) and their respective needs, (iii) participatory where trainings are not purely academic but with a large space left to apply the knowledge and practical exercises.

The multi-country initiative around Solar Renewable Energy Training and Certification Program implemented by Burkina Faso's 2iE institute targeting engineers and staff from Ministries and national utilities (including 20% women) should be leveraged.

Activity 1.5.3. Conduct Training of Trainers (ToT)

A training for both, male and female trainers will be provided to targeted training institutions, ANPER and the Ministry of Energy and Renewable Energies, as well as the trainers of the Ministry of Technical Education and Vocational Training to contribute to ownership and sustainability of the various training modules. Institutional memory will hence be supported along with sharing these training materials with the national and regional knowledge practices around AMP.

Output 1.6. Domestication of quality standards and norms for solar minigrid components, and institutional capacity of ANERSOL strengthened

Today, there are no real quality standards for solar minigrid components. Article 5 of the PERAN decree stipulates that minigrids are build complying with state-of-the art standards and that the technologies selected should be interconnectable to the national grid. ANERSOL, the newly created agency (2018) supporting solar energy promotion, had benefited from technical assistance of WB, AfDB and AFD including building their quality control capacities.

This output includes the following activities:

Activity 1.6.1. Review existing standards and identify gaps

Together with ANERSOL, in collaboration with ECREEE, a review of existing national, regional and international standards around CEMGs will be conducted. Gaps will be identified and clear recommendations will be drawn. The results of this analysis will be presented to ANERSOL.

Activity 1.6.2. **Develop adapted standards**

Based on the review under Activity 1.6.1., ANERSOL and the PMU, with ECREEE's and Verasol's support, will design relevant standards and a plan to enforce these standards. Thresholds could be used around quality standards based on the installed capacity, e.g., under 350kW the standards would be simplified to avoid jeopardizing local efforts to electrify. Such a double standards system has been implemented for instance in Sierra Leone since 2020.

These standards should as much as possible take E&S aspects into consideration through SESA.

ANERSOL's mandate includes the definition of standards as well as the exclusivity for quality control of such standards. However, the capacities of the newly created ANERSOL related to standards and their enforcement are limited.

Activity 1.6.3. Reinforce ANERSOL's capacities

A workshop to present the review (Activity 1.6.1.) and standards (Activity 1.6.2) along with a training on how to reinforce these norms will be provided to ANERSOL's staff. Relevant training materials and documentation will be provided by the project.

Output 1.7. Public programmes (apprenticeships, certificates, university programs) to develop competitive, skilled labour market in minigrids facilitated

Climate and environmental risks and mitigation efforts as well as gender considerations will be included in the activities below.

This output includes the following activities:

Activity 1.7.1. Identify minigrid market needs in terms of competencies and jobs

A comprehensive analysis of the needs of the minigrids market, including efficient end-user appliances and equipment adapted to minigrid environments, in terms of competencies and jobs to contribute to its scaling up should be conducted. This preliminary study is critical to pave the way for the development of suitable training programmes and modules by academics to the market's needs. Such training should integrate a large part of practice and hands-on work, especially to include women who on average have less formal education, to facilitate ownership and direct application of learnings.

Activity 1.7.2. Analyze existing university and higher learning institutions programs and gap analysis

An assessment of the studies portfolio (including modules) around renewable energy and minigrids in particular at the University of Niamey, École des Mines, de l'Industrie et de la Géologie (EMIG) and other high learning institutions across Niger is carried out. Potential gaps compared to the market's needs (see 1.8.1) are identified, and recommendations are made. Bridging these gaps include strategic partnerships with Burkina Faso's 2iE, the soon to be relaunched Côte d'Ivoire's École Supérieure Interafricaine de l'Électricité and targeted higher learning institutions across the ECOWAS region and beyond, as well as ECREEE, IRENA and AMP's partner RMI. The analysis should encompass the inclusion of gender aspects in the curricula.

The analysis' findings will be introduced and shared to universities and higher learning institutions as an opportunity, as well as to the authorities in charge of providing budgets to these institutes.

Activity 1.7.3. Analyze existing vocational training programs and gap analysis

An assessment of the training portfolio (including modules) around solar PV installation and maintenance, battery maintenance, wastage &recycling, becoming a rural electrician and other relevant trainings will be conducted. This will include:

- vocational training courses designed by Plan International's CEMGs project financed by the EU and the Schneider Foundation and validated by the Ministry in charge of Vocational Trainings in Niger. This activity included a training of Trainers (ToT) aspect where 2 inspectors at the Ministry and 19 trainers in 2 vocational training centers (Dosso/Tilabéry and Marady/Zinder).
- training provided by energy service providers for their staff.

A particular focus will be put on certification aspects and their enforcement as this is key to raise awareness of, and ensure ownership by end-users (residential, commercial and social) of reliable quality cleantech solutions as a suitable alternative, especially minigrids. Lessons from Barefoot College-inspired experiences in West Africa will be used to

determine the extent to which paper-based qualifications, even literacy, are prerequisites for electricity sector vocational training, hopefully lowering barriers to female enrollment. Gaps compared to the market's needs (see 1.8.1) will be identified, and recommendations drawn. The latter shall include discussions with ECREEE, IRENA, RMI (one of AMP's Regional Project implementers) as well as with AMP regional project and national child projects on how they go about for instance with the certification of solar PV and minigrids installers. Successful capacity building initiatives in other countries including 2iE institute in Burkina Faso etc. will also be leveraged, this includes also sharing best practices and lessons learned in promoting CEMG in the given country and relevant capacity building efforts.

The analysis' findings will be introduced and shared to vocational training centers (including the 2 training centers already capacitated by Plan International) as an opportunity, as well as to the authorities in charge of providing budgets to these institutes.

Activity 1.7.4. Train the trainers of relevant institutions in Niger

Based on the findings of activities 1.8.3. and 1.8.4., and the support of the targeted training institutions, the project will support some ToT activities of these institutions for both male and female trainers in collaboration with identified partners (see above).

Component 2. Business Model Innovation with Private Sector

Outcome 2: Innovative business models based on cost reduction are operationalized, with strengthened private sector participation in low-carbon/renewable energy minigrid development

Given Niger's electricity, and specifically minigrids' situation, policy and regulatory framework, and being an early-stage minigrid market, the project aims at enabling the proof of concept of minigrids with private sector engagement in rural areas. Thanks to innovative business models of demonstration pilots, rural communities will gain access to reliable, affordable and clean electricity. Lessons learned in Niger (with the support of a study conducted by Power Africa with ANPER) and in other countries have highlighted, especially in LDCs, that a minigrid can only become profitable and sustainable when based on:

- productive use
- and cost-reduction.

Such players, be it commercial (for-profit) or social (health centers, schools), are energy intensive during the day (or up to 24/7 such) where the sun is largely available and represent a relatively stable and significant electricity demand source. The project will support the identification of relevant energy intensive value chains in rural areas across the country (Output 2.2).

In terms of cost-reduction efforts, 3 levers are available:

- Sector levers - related to legal requirements (e.g., legal registration, importation license, tariff approvals, environmental impact assessment, land usage rights, village level MOU) and mainly covered in Component 1
- Supply levers - related to the site preparation costs (e.g., site visits, community engagement, transports and logistics), CAPEX (e.g. civil works, electricity generation and storage equipment, distribution infrastructure, metering and monitoring equipment, VAT and duties) and OPEX (recurring infrastructure expenses, salaries and other HR related costs, O&M costs)
- Demand levers - related to customer uptake and demand stimulation (including flexible tariff regimes)

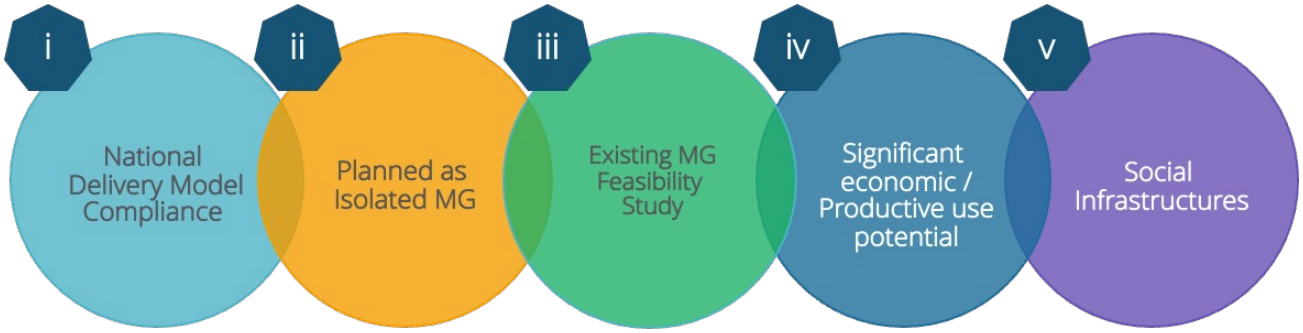
Output 2.1. Pilots developed, including on productive use/innovative appliances and modular hardware/system design, leading to cost-reduction in minigrids

The specific type and locations of pilot projects will be determined during implementation phase as:

- Additional information to carefully select pilot sites and identify the type of pilots would need to be gathered including a significant stakeholder engagement to ensure relevance and additionality of the pilot project
- For pilot projects led by the private sector, a competitive procurement process needs to be followed to select the private sector developers who will receive support for minigrid investments

- For pilot projects in combination with other initiatives such as multifunctional platforms supported by UNDP or other players, the relevance will need to be reassessed.
- Realities at PPG phase for a given site may drastically evolve (e.g., CEMG developed by other players, insecurity issues, etc.)
- Taking into consideration the exclusion criteria for minigrid locations stated in the ESMF (see Annex 8)

Nevertheless, some criteria have been identified with the Implementation Partners (IP), ANPER, and the responsible party for Component 1, the Ministry of Energy and Renewable Energies, in addition to the pilot project principles stated in the previous chapter:



(i) National Delivery Model Compliance

Niger's minigrid national delivery model is relatively mixed as in most nascent minigrid market. Today the pilot assets can be owned either the Government/the public utility NIGELEC, or the private sector, or communities. The country started with a concession model and now shifting towards an affermage model in a trial & test kind of approach. Given the very low electrification rate in rural areas, greenfield minigrids will be favored to test and demonstrate different business models (including also different technologies) in an early-stage market. It is possible that other delivery models be considered during implementation as the legal and regulatory framework around electricity and minigrids in particular will be undertaken by the World Bank (HASKÉ) by April 2023 in collaboration with relevant partners (including UNDP), and hybridization of existing diesel minigrids in rural areas would be welcome to - as this is the case for instance of health centers covered by World Bank's NESAP project. A combination of different RE sources minigrids (biogas and solar) will be considered. Each pilot project will also consider environmental, social and climate risks and mitigation measures, as well as gender considerations.

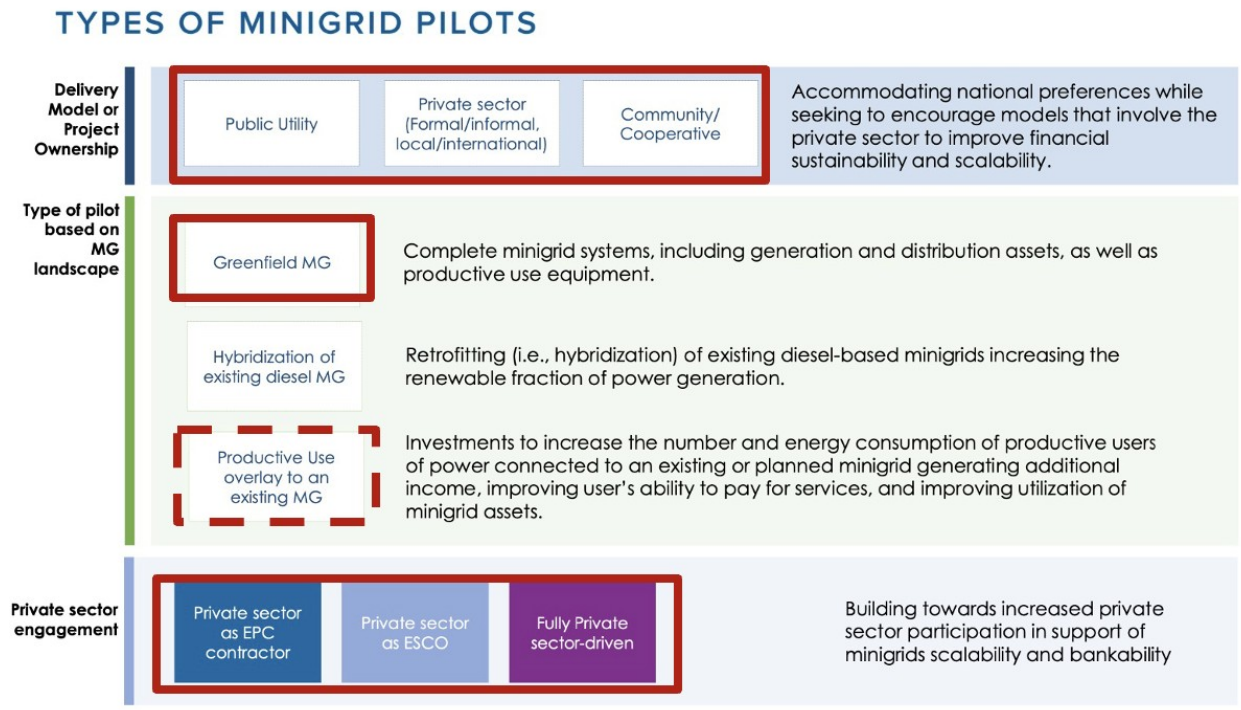


Figure 1- Proposed Delivery Model for Pilots

(ii) Planned as isolated MG

According to the PDAE, as indicated in the baseline and in Component 1, electrification efforts were categorized (grid extension/densification by NIGELEC, isolated minigrids, and individual kits based on specific criteria). Pilot projects under AMP Niger shall be part of the list of villages identified for an electrification via isolated minigrids and that should not be interconnected to the grid before 2035. It could also include, as discussed with ANPER, existing CEMG pilot sites where there is no operator yet and owned by ANPER. A comprehensive study conducted by GIZ provides a preliminary financial plan and tendering documents for a private sector operator to bid (available also upon request at ANPER).

(iii) Existing minigrid feasibility study

Various development partners supported, in close collaboration with ANPER, the development of pre-feasibility studies for a large number of villages. To facilitate the timely uptake of the CEMG market in Niger, AMP will leverage existing pre-feasibility studies for its pilot site selection. The project would cover, where necessary, the update of this study to ensure a better CEMG sizing, including potentially unrecognized "latent" electricity demand from female producers. The update could also encompass, where deemed necessary, an Environmental and Social Impact Assessment (ESIA) and Environmental and Social Management Plan (ESMP) for each site.

This includes the comprehensive CEMG feasibility studies conducted by Power Africa in 36 localities (available also upon request at ANPER).

(iv) Significant economic/productive use potential

The PDAE selected the isolated MG option wherever a village was relatively close to an 'economic development pole'. The project will go one step further and ensure that selected pilot sites are close to a significant 'economic development pole' (e.g., flourishing agricultural activities, mining sites, etc.) and already have (or the potential to become) their own economic activities which would be enhanced through electricity. The existence of a weekly market densifying economic activities on the market day and beyond along with the existence of anchor loads such as at least one telecom tower will be a must. Power Africa's feasibility studies on 36 localities were identified based on the telecom towers as an anchor load. Special considerations can be taken to ensure that electricity provision for productive use doesn't disproportionately advantage individuals, particularly men, who already have higher asset levels, access to finance, business tracks records, and professional networks; this can be accomplished through strategic partnerships with

organizations actively working on women's empowerment and economic success. It may also be necessary to plan ahead to create sex-segregated business premises where women can work away from the public eye (if that is something culturally important to them and their families, i.e. they practice wife-seclusion, according to the gender assessment in Annex 10).

All pilots will have the productive use of energy (business and social) as the entry point (i.e. a necessary condition). This is at the heart of the project's theory of change, where the economics of minigrids can improve in a virtuous cycle of higher loads resulting in lower LCOEs. Under Output 2.2. economic activities that could be boosted through access to an accessible, reliable, affordable and greener energy across the country will highlight such productive uses. This include:

- Agricultural Loads (e.g. irrigation pumps)
- Productive Loads (e.g. milling, rice dehusking, oil pressing, wood/metal workshops,refriger)
- Commercial Loads (e.g. shops, bars, ice-makers, battery charging and renting, lantern renting, conservation)
- Anchor loads (e.g. telecom towers, mines, green-houses, hotels, loggias)

Productive use players could pay higher tariffs and provide load during the day which supports CEMG investment feasibility.

They could also pay based on kWh while households could pay a service-based fee.

Existing solar minigrids under ANPER's supervision (7) that do not have yet an operator, could benefit from AMP Niger by helping in finding a private sector operator and supporting the CEMG capacity increase using productive use as well as probably changing some batteries.

(v) Social infrastructures

Another criteria to pilot site selection is the existence of social infrastructures in the village, namely at least one school and one health center. Given the fact that Niger is facing a rampant extreme poverty phenomenon and a relatively low level of school attendance and healthcare - reinforced because of COVID-19, the need to incorporate social infrastructure is critical. Social infrastructures do pay their electricity bill as part of each sector's budget. However, there are often some delays - which should be considered in the financial model of the CEMG.
Gender mainstreaming efforts and monitoring will also cover social infrastructures whereby the latter will be asked to provide information on the gender breakdown of individuals served per year.

In addition to these selection criteria, the following 5 aspects need to be taken into consideration in the tendering process as well as at construction, operation and maintenance of the pilot projects:



Private sector involvement

In a minigrid nascent market such as Niger and with a rural electrification rate of only 1%, the private sector should play a crucial role to support universal energy access by 2035 as targeted in the SNAE. The pilots' delivery model will incorporate private sector involvement as Engineering, Procurement and Construction (EPC) contractor, ESCO or as an exclusive electricity player. Energy service providers should largely be local players supported in the tendering process, capacity building and facilitating access to finance (see Component 3).

In a focus group with local private sector CEMG developers and operators, the issue of having the opportunity to bid and gain experience from A to Z with relevant CEMG design and operation is tremendous. The requirements in tendering processes don't leave space for local players to bid as they lack the relevant experience, the necessary guarantees and credit lines. Thus, the project, during pilot site selection, could think of giving the opportunity to a consortium of local developers to design and operate from A to Z a CEMG.

Community engagement

The role of the community (households, businesses and social infrastructures) is critical to the pilot site's success. It will be informed, engaged and committed from the very beginning of the pilot site project, i.e. from the definition of the tendering specifications to the CEMG operations. A community board with male and female representatives shall be created in each pilot site to ensure the needs and capacities of the community are considered including on the tendering process preparation and evaluation as well as regarding operations and maintenance aspects of the CEMG.

Minimum concessionality of GEF INV support to pilots

The final design of the minigrid pilots, as set out in the minigrid pilot plan (output 2.1, activity 2.1.1 below), will ensure the efficient and appropriate use of GEF donor funding to the pilots. It will set out a clear methodological basis, for example to ensure LCOE parity with a reference tariff; or based on willingness/ability to pay (determined by a study - or the update of an existing minigrid feasibility study - during implementation). Such methodological assessments will be part of an overall package of financial due diligence/assessments that will be performed during the tender process to select pilot sites/developers.

Local authorities and communities may provide in-kind contributions themselves which will be evaluated during the implementation phase. This may include land and installation labor, as well as, thanks to capacity building efforts provided by the project, local monitoring and basic maintenance (with technical expertise support of the private sector minigrid operator), and local payment collection support and greater system security. Established local women, with buy-in from their families and community leaders, are good candidates for monitoring and maintenance tasks since they are less likely to move. They may also in some cases achieve higher payment compliance or unit prepaid sales (kWh) in cases where mobile-enabled PAYGO solutions are not prevalent. Financial inputs may also come from the Nigerien diaspora, the private sector (productive use players of the community, KeyMaker model players, etc.), and some technical and financial partners.

Potential Electricity Demand

Pilot sites have to be carefully chosen based on the existing demand for productive use of energy (PUE) and not on the demand to be stimulated.

Digital data

As indicated in the previous chapter, in return for benefiting from GEF INV support as a pilot, the asset owner or operator of the minigrid pilot will have to share digital data from the minigrids performance with the AMP national project thanks to relevant digital equipment. More details can be found in the Box below.

Box 5 - Data sharing for minigrid pilots

Pilot beneficiaries (e.g., minigrid operators) receiving support from the project will be required to share minigrid performance data with the national project.

Specific terms and conditions for data-sharing and how best to operationalize the commitment and its adoption by the beneficiaries will be defined and agreed upon with minigrid operators during project implementation, including details of what data can and cannot be used, based on consultations with industry stakeholders and with support from the AMP Regional Project.

The specifications around the data generation by the demonstration pilots supported by the project will consult and follow guidance/standards provided by the AMP Regional Project. A standardized Quality Assurance and Monitoring Framework (QAMF) for application in all minigrid pilots supported under the project will be developed in year 1 of the regional project and disseminated to all national projects.

A digital platform will be procured by the project (under Component 4, Output 4.2) to serve different purposes including: (1) running digital tenders by which minigrid developers will be selected as beneficiaries to receive support under the project and (2) managing all technical and financial data related to minigrid sites.

Through the implementation of this digital management platform, minigrid developers selected to implement minigrid pilots with support from the project will have access to a set of best-in-industry tools for analyzing minigrids (e.g., demand forecasting, system optimization, distribution network design, detailed financial modeling at the site and portfolio level). Similarly, as part of the roll-out of the data platform, minigrid developers (as well as key government and other stakeholders) will receive capacity-building and in-depth training to use analytical tools and data management technologies.

As indicated above, in Niger, the digital tendering platform will be taken care of by World Bank's new HASKÉ project. During PPG, it has been agreed with World Bank that UNDP as well as other relevant partners (in addition to ANPER under which the platform would be located) will be consulted and actively involved in the tendering and roll-out phases. ANPER will take ownership and the lead to ensure coordination and partnership with all relevant stakeholders. This will also contribute to integrating AMP results to the digital platform supported by the HASKÉ project.

This output includes the following activities:

Activity 2.1.1. **Develop a detailed project plan (the project's 'Minigrid Pilot Plan') for advancing the project's minigrid pilot(s)**

The PMU will lead and develop, in close collaboration with other stakeholders and support from the AMP Regional Project, a detailed project plan (the project's 'Minigrid Pilot Plan') for advancing the minigrid pilot(s). Once prepared, the project's Minigrid Pilot Plan will first be reviewed for clearance by UNDP (CO and BPPS NCE), and then shared with the Project Board. This activity should be completed by the end of year 1.

Building on the initial design information in this project document and its annexes, including the key principles for minigrid pilot implementation described in **Box 3**, the project's Minigrid Pilot Plan (MPP) will determine, among other aspects, the following:

- Clear objective for the pilot(s)
- The minigrid delivery model(s) which will be demonstrated in the pilot(s)
- The proposed type of pilot: greenfield minigrid pilots
- The estimated target number of pilot(s), based on *ex-ante* estimates of available GEF INV and other funds needed to cover the CAPEX costs
- Inputs, as necessary, on site selection, including based on geo-spatial mapping, for the pilot(s)
- Site-specific assessments and other requirements (e.g., demand sizing, social and environmental safeguards assessments (ESIAs/ESMPs), gender assessments, e-waste disposal). Some assessments may be needed to be performed by the project *ex-ante*, to inform follow-up competitive tenders. The project's MPP will rely on existing feasibility studies and will update and complete them where necessary.
- The use of the digital platform for competitive tendering at ANPER, as necessary
- Ongoing data collection from minigrid pilot(s), including data-sharing requirements from minigrid pilot(s) (Refer to Box 5), as well as digital hardware requirements (Refer to Box 6)

- The project's approach to ensure minimal concessionality for the level of GEF INV support to the pilot(s) (when there are private sector beneficiaries)
- Review of the Implementing Partner's (IP's) modalities for transfer of GEF INV support to the pilot(s), ensuring they are aligned with UNDP's policies and financial rules
- If a pilot includes GEF INV support for productive use, ensuring the pilot takes a third party ownership model to productive use equipment
- Coordination and rationale on any associated project technical assistance activities which may benefit the minigrid pilot(s)

Box 6 - Indicative Specifications for Minigrid Pilots' Digital Hardware and Software

Offering	Details
Hardware requirements per site	· Inverter monitoring (monitoring & control) · Distribution monitoring · Optional current transformers for energy meter if more than 10 kW (single phase) or 30 kW (three-phase) · 24V power supply (50€) · Various data cables and installation material · Optional: 24V backup battery (50€) · Optional: Cabinet for the complete monitoring system · Industrial internet router · Industrial or high-quality Ethernet Switches
Hardware requirements per connection	· Smart meter

The types of pilot sites that *could be* considered during implementation could encompass the following:

- Around a Multi-Functional Platform (MFP) of a women's group (gender component) planned by a project and which will consider access to electricity via solar minigrids / adapted ancillary equipment
    - e.g. UNDP / PASDEL project - framework project where a proposal of combining a MFP can be submitted to UNDP
- A model called "Keymaker- 4th generation" where a private operator could have two roles: a) solar minigrid (production, transport and distribution of electricity) in the village with residential, commercial and social customers, b) but also an activity of conservation & agro-food transformation. This could be for example around fishing (like Jumeme in Tanzania around Lake Victoria) or an agricultural value chain (local rice/rice steaming; peanuts/peanut oil, poultry farming, etc.)
    - e.g., with a robust cooperative member of the Federation of market gardening cooperatives in Niger to support their food production, conservation and processing activities (e.g. organic onion, rice, etc.)
- Around a refugee camp and its host community (such as in Ouallam, where an agro-ecological farm and the refugee camp are not electrified) - electrification here will also create income generating activities and improve the living conditions of host communities and refugees
    - Coordinate with the World Bank HASKÉ project as its aim is to cover the electrification of refugee camps
- A hybrid RE model with solar and biogas minigrids around several agricultural value chains including a more "sedentary" livestock component as a basis for the biodigester (issue of security and theft of livestock to be considered in the choice of sites)
- A site where local content will be highly favored for local developers and operators to gain the necessary experience to compete for minigrid tenders in Niger

- A focus group of local operators and developers, organized during the PPG phase, identified this a big hurdle for the scale-up of the minigrids market through private sector with local companies

Activity 2.1.2. **Design the tender process for pilot(s) using a digital platform**

The project's pilot(s) may involve private sector engagement in various forms, including models involving private sector minigrid ownership, private sector EPC, and private sector O&M services. Where there is private sector engagement in the pilot(s), a competitive tender process will be executed using the digital tendering feature of the digital platform procured under Component 4.

Under this activity, the PMU, working with the digital platform vendor, specialist engineering, financial, procurement, and legal expertise, and the AMP regional project, will translate the approach set out in the project's Minigrid Pilot Plan into the design of a customized tendering process on the digital platform (to be implemented with ANPER leadership and coordination under World Bank's HASKÉ project), including requirements, specifications and evaluation criteria. Feedback can be solicited from a gender-diverse selection of private sector actors with experience responding to government tenders (not limited just to electricity) about whether the proposed digital modality and in-person supports will result in a more level or less level playing field for male and female prospective developers. The terms of reference will consider, among other factors, which the PMU will determine with support from the AMP Regional project, the following:

(i) establishing a requirement and incentives for pilots to share data with the project

(ii) including incentives for the proposals to be gender-responsive

(iii) including implementing the ESMP (including a requirement for environmentally-sound collection, storage and disposal of all electronic and electrical waste, including rechargeable batteries, associated with off-grid renewable energy technologies)

(iv) considering the different aspects set above before detailing outputs and activities of Component 2.

At the end of this activity, the tendering process on the digital platform will be ready to launch. The tender process itself should be launched before the end of Year 2.

Activity 2.1.3. **Execute tender, contracting and payments to the selected pilot beneficiaries**

In year 2, the tender will be launched and executed according to the design finalized in activity 2.1.2, resulting in minigrid developers/operators being selected as pilot beneficiaries. Submissions to the tender will be competitively assessed against evaluation criteria (engineering, financial, environmental, social, gender, etc.), with the PMU supported by appropriate expertise.

Following selection of beneficiaries, the PMU/IP will enter into legal contracts with the selected minigrids, again supported by appropriate expertise, and make payments on pre-defined milestones, including on commissioning of minigrid plants and following ESMP. The digital platform will validate payment milestones as part of a result-based financing approach. A top-up payment could be envisaged for operators connecting socially vulnerable groups.

Capacity building for government personnel with the digital platform would have been conducted by HASKÉ. Potentially some capacity building would be offered to private sector actors (if not already done) to engage with the competitive tender and adherence to social and environmental standards.

Activity 2.1.4. **Monitor pilot(s), collect and aggregate data shared by pilot(s)**

Data generated by the pilot(s) will be collected using the digital platform, connecting directly to remote monitoring and smart metering equipment. Data that are not amenable to be collected by remote sensing will be collected either by the minigrid operator or some alternative way to be defined by the PMU supported by appropriate expertise.

Data collected from the pilot(s) will be used at the project level to, among other purposes: (i) track the performance of the minigrid systems in real-time; (ii) validate the underlying pilot(s) assumptions and business case; (iii) track performance enhancement in minigrid capacity utilization; and (iv) generate insights and lessons learned to share with the AMP Regional Project.

Also, data collected from pilot(s) will be shared with the AMP Regional Project for aggregating and analyzing data across all AMP national child projects. The regional project will use these data to: (i) generate insights and lessons learned; and (ii) inform the development of knowledge products, both to be disseminated across AMP participating countries and the broad minigrid sector.

Collected data should also be gender disaggregated.

Output 2.2. National report on opportunities to boost economic activities through electricity access and productive use

This output includes the following activities:

Activity 2.2.1. Conduct an analysis on boosting rural economies

As part of the minigrid market development and scale-up efforts, a mapping of relevant high potential value chains (in economic and social terms) with possible linkages between those and energy use will be focused upon (including stakeholder mapping) at national level. The analysis will include:

- A mapping on primary processing opportunities in rural areas to shift the value addition from existing agricultural processing into rural areas. Much of the agricultural outputs of rural areas is, as a result of non-existent rural electricity availability, transported to, aggregated and processed in areas connected to the national grid (or even abroad).
- An analysis on the role minigrid can play in sustainable rural development. The potential opportunities to boost economic activities through electricity access and productive use will be mapped against the difference roles that can be played by:
    - Developers and operators - data supports the fact that investing in productive uses increases customer ability to pay and site Average Revenue Per User (ARPU)
    - National governments: stimulating productive uses in rural areas boosts the local economy, such as releasing female labor from low productivity traps into higher productivity domestic, agricultural and wage work, and as does improving the health and educational status of children and pregnant women
    - Development Finance Institutions (DFIs) - supporting increased electricity demand is a 'soft subsidy' for developers and a variety of socio-economic benefits
    - Investors - increased ARPU and ability to pay are key drivers of a project's bankability. This is also critical information for due diligence and capital raising
- A benchmark of best practices around minigrid value chain support and suitable innovative CEMG business models (e.g. Jumeme's keymaker model in Tanzania, etc.)
- A market sizing exercise to assess the financial and non-financial impacts (including environmental & social aspects) and the viability of minigrids in rural areas in Niger

Activity 2.2.2. Disseminate findings to facilitate the CEMG market scale-up

The comprehensive analysis and its main findings on boosting rural economies via CEMG will be shared via a workshop, publications and knowledge sharing means in Niger as well as abroad including the regional AMP knowledge sharing platform and relevant communities of practice.

Output 2.3. Capacities of private minigrid developers and communities are strengthened

While local private minigrid developers benefitted from capacity building activities as identified during consultation meetings at PPG phase such as with and by Power Africa, there are some remaining needs to complete the gaps to ensure sustainability.

As indicated, the role of the communities and their engagement from the very beginning of the electrification process is crucial for the CEMG project. Hence their capacities need to be built for an adapted operations and maintenance (O&M) service. Obviously, capacities to be reinforced for minigrid developers and communities vary.

This output includes the following activities:

Activity 2.3.1. Assess the capacity gaps of local private minigrid developers and solar panel installers

An evaluation of the capacities of local developers and installers will be conducted. This will include their familiarity with minigrid technical and service delivery design targeting the full spectrum of individual users (i.e., for men and specifically for women and youth), business opportunities, tender writing, innovative business models, successful and durable O&M of the minigrid, quality assurance, raising funds, environmental and social considerations, etc.

Activity 2.3.2. Offer workshops and webinars to fill these gaps

Based on the assessment in Activity 2.3.1, relevant workshops, webinars and other knowledge sharing efforts will be provided to local developers based in the capital city Niamey. existing training and support materials suitable to each of the gaps within AMP, its partners and beyond will be used as much as possible. This includes AfDB's Green Minigrid Helpdesk, ESMAP minigrid Design manual, Power Africa's raising fund technical assistance or AfDB's SEFA Quality Assurance Framework for Minigrids.

Training should also be offered to local solar panel installers based in Niamey as well as in pilot project localities. These hands-on trainings could be offered in collaboration with the Ministry of Technical Education and Vocational Training and its various vocational training centers across the country.

Activity 2.3.3. Raise awareness and building capacities of communities

Pilot sites communities will benefit from continuous awareness raising at launch and during implementation of the project as a whole, and pilot site construction and operation in particular. This will include general benefits of the minigrid (including climate change aspects), demand profile, forecast and stimulation, support in money collection for the provision of power, etc. A specific training will be provided to targeted youth and women in the communities on becoming rural electricians (including power connection), on basic maintenance of the minigrid, as well as other competencies depending on the implemented business model. As stated in Component 1, the modules developed by Plan International and the Schneider Electric Foundation, and validated by the Ministry in charge of Vocation Training, will be leveraged. Youth and women will be trained based on these training materials and trained trainers Efforts made in other countries including Barefoot College or 2iE in Burkina Faso, ProJeunes.Energie in North Côte d'Ivoire, etc. It is worth mentioning that maintenance requiring higher technical skills will be provided by the private sector operator. In addition, private sector operators will play a role in knowledge transfer, capacity building and regular support of these targeted youth and women.

Output 2.4. Support provided to establish and grow a national industry association for private sector developers

A private sector association for players in the solar industry has existed since May 2013 in Niger called 'APE-Solaire'. Some solar companies in Niger are either members of the association or requested to join. The main activity of APE-Solaire is to participate as a stakeholder in the ongoing development of the GoN's solar energy policy and regulation. Followed consultation during the PPG phase, the association could benefit from some strengthening that could be beneficial to the country in the development of CEMGs. APE solaire could become a powerful advocacy platform by including all relevant stakeholders such as the GoN and technical and financial partners.

This output includes the following activities:

Activity 2.4.1. Develop and operationalize a national association of private sector RE providers and aspiring providers

A rapid capacity gap assessment at APE-Solaire will be conducted based on interviews/focus groups of members and other solar energy products and services providers, as well as a benchmark of such successful associations with the support of AMDA - African Minigrid Developers Association[21]. This will enable local private sector operators to have a voice at the National Dialogue Platform and beyond, nationally as well as continentally and internationally (advocacy). Common interests include taxes and import conditions of solar and solar ancillary equipment, collaborating as a consortium of players to answer specific larger bids and competing with international developers, as well as a better outreach. The association could then have thematic sub-committees (if applicable) including one dedicated to minigrids. The association, also considered as a national chapter of AMDA, will seek collaboration with AMDA – African minigrid

[21] https://africamda.org/

Developers Association. This industry association will become a member of the national chapter (an AMDA consultant located in Niger should become the chair of the minigrid committee of the association and could support advocacy, fund mobilization and interaction with its members (mainly private sector led associations and individual developer companies). Such an association can also play an intentional role providing role models and mentors to aspiring market entrants, especially youth and women, who face higher barriers to entry (e.g., lack of market information, weaker professional networks). APE-Solaire could also benefit from expertise and experience sharing with the AMP regional project and other AMP national projects.

Activity 2.4.2. Strengthen the capacities of the RE private sector association

The industry association would highly benefit from capacity building as an association. Indeed, an association operates in different ways compared to a private company. Capacities to be focused on encompass structuring, managing and moderating the association to ensure its well-functioning and its sustainability.

Component 3. Scaled-up Financing

Outcome 3: Financial sector actors are ready to invest in a pipeline of low-carbon minigrids and concessional financial mechanisms are in place to incentivize scaled-up investment

Access to low-cost, commercial capital (equity and debt), for both supply and demand, ideally in local currency, is key to reducing the cost of minigrids, and the scalability and sustainability of a minigrid market. Being an early stage minigrid market, there are only few financing schemes in place around minigrids in Niger. Thus, developing and scaling-up the CEMG market in Niger requires suitable financing mechanisms both on supply and demand sides.

Output 3.1. Support financing mechanisms to scale-up RE minigrids investment

This output includes the following activities:

Activity 3.1.1. Identify existing financing schemes available in the country supporting access and use of energy

A mapping exercise will be undertaken to identify and characterize all existing minigrid funding and support programs underway and planned nationally. From this a gap analysis will be conducted to identify the opportunities and challenges associated with different funding mechanisms.

During the PPG phase a preliminary mapping of available financing mechanisms related to access and use of energy in general has been undertaken. Some available financing products identified are:

On the supply & demand side:

- For SHS: World Bank's NESAP had created a USD 7 million credit line for some financial institutions in Niger. Sonibank and BSIC offer loans to solar systems providers, whereas Capital Finance has been targeting households, solar pumping systems and very small enterprises. However, the interest rate given to banks for the credit line (4.5%) was relatively high compared to the market and therefore impacted the final interest rate for loan takers. Developers would have to take a loan at 11.5% which was finally reduced to 8% and ultimately few loans were given. The challenge was the high interest rate of the mortgage guarantee and the eligibility criteria which limited the list of eligible solar products (including ancillary equipment).

On the demand side: ·

- *Fonds d'Investissement pour la Sécurité Alimentaire et Nutritionnelle* (FISAN), a fiduciary fund for the promotion of agriculture that supports farmers to acquire equipment (mechanization) including e.g. solar pumping for irrigation purposes etc. to ensure food production and processing.
- PAYGO is offered by some enterprises like Olu Solar or High Tech Solution for household and productive use.
- Asusu: Former support for RE equipment through a former micro-finance institution in Niger that ended. This MFI, which has been dissolved, is planned to be revived under another entity soon.

The assessment will include innovative financing solutions both on supply and demand sides. This includes:

- Crowdfunding and crowdlending (already used in Niger for one private sector minigrid operator)
- Blockchain enables Finance
- Renewable Energy Certificates or Peace Renewable Energy Certificates
- etc.

Capacity building related to these innovative solutions are part of Output 3.2.

Activity 3.1.2. Assess the potential of aggregation of minigrid assets

The potential for national and regional financial aggregation of minigrid assets will be explored. The assessment should include an ESIA/ESMP that takes into consideration cumulative impacts of minigrids to be aggregated.

Aggregation can take the form of:

- Operational aggregation whereby operators cooperate to share access to operational or development resources. This can lead to considerable savings and cost reductions;
- Project aggregation whereby minigrids are bundled together to form larger investable assets. This process creates larger portfolios to crowd in investors that might not consider small projects. This level of aggregation requires upfront and standardized due diligence to be carried out before projects can be bundled. This further reduces transaction costs for investors. Aggregating across multiple developers and markets further reduces the risk. An example of this is the CrossBoundary Energy Access fund that bought a portfolio of sites from PowerGen in Tanzania.
- Connection aggregation exploits the large amount of data that is constantly being generated and uploaded to the cloud in near real time from the smart meters installed on minigrids. This information includes data on energy being consumed and revenue being generated from each individual minigrid connection. The granular nature of this data allows different types of customers to be aggregated into asset classes with different characteristics. For example, all of the high revenue producing, consistent connections (most likely based on a productive use of electricity) across multiple sites can be aggregated into a high performance class suitable for commercial investors willing to pay well for revenue producing assets.

Activity 3.1.3. **Benchmark successful financing mechanisms in other industries and countries**

There are few best practice of funds in Niger such as *"Mécanisme Commun de Financement"* (MCF) for the water sector, the FISAN for the agricultural sector and the locally managed Financing Facilities in Nigeria for instance its Rural electrification Fund and ESMAP-NEP Program both managed by the Rural Electrification Agency. The AMP Regional project could advise on the type of mechanism and approach that are in place in other AMP countries.

Activity 3.1.4. **Determine the main financial barriers**

Local and international private sector players will be engaged to determine key financial barriers for the development of the MG market in Niger. The African Minigrids Developer Association AMDA, ANERSOL, APE Solaire and any other organization with locally relevant private sector knowledge will be interviewed towards the same objective. The DREI analysis undertaken in 2021-2022 offers some de-risking measures that should be considered and implemented to overcome such financial barriers. The preliminary conclusions can be found under Chapter III - Strategy.

Activity 3.1.5. **Select relevant financing mechanisms and provide suitable technical assistance**

Based on activities 3.1.1. and 3.1.2., AMP Niger will select 1-2 relevant financing mechanisms to benefit from technical assistance.

This could include:

- a sectoral fund as initiated in December 2016 a decree to create a Financing Facility to promote RE and EE in Niger was adopted. This facility should have been located under the Ministry of Energy and Renewable Energies and focus on public funding and only accessible for the GoN and its related agencies. However, this facility never materialized. To facilitate the reach of universal electricity access by 2035, the design of a Minigrid Funding Facility (MFF) to support rural electrification and minigrids market in Niger has been welcome during stakeholder consultations. This Fund would receive the *Special Electricity Tax (TSE - Taxe Spécifique d'Électricité* that is collected by NIGELEC whereby 60% of the collected tax is used to support rural electrification and transferred to ANPER. The tax is collected from each kWh paid by electricity users of the national grid to

contribute to a national unified tariff grid. In order to contribute to social justice in terms of electricity access as stated in the Electricity Code, offering a tariff matching the average national grid is considered a must by all public energy related institutions. Thus, the TSE could support the funding for the compensation to be paid to private sector developers and operators for the difference between the average national tariff of about 85F/kWh (≈$ 0.13) and the cost-reflective tariffs that private sector players would request to ensure their sustainable profitability and viability. This MFF should also be nurtured by the existing and upcoming minigrid market initiatives financed by development partners such as World Bank, AfDB as well as the AMP Niger project (for the pilot project sites - see Component 2) and private investments. A set of potential financing schemes will be identified as part of the Facility based on the delivery model that will be more clearly defined during project implementation.

- A result-based financing mechanism approach like the model that the pilots for AMP Niger under Component 2 Its design and governance could be informed by a gender-balanced set of stakeholders. Additional provisions may be required to ensure that women and men can access the mechanism in equitable measure (e.g., using scoring criteria, set-asides, special outreach, or offering women agency-based capacity strengthening services in parallel with the Fund's applications).

Relevant financing mechanisms should ensure that environmental and social considerations are taken into account when disbursing loans (see SESP in Annex 5).

Government stakeholders including ANPER will be engaged to ascertain the appetite for the different funding mechanisms proposed. The capacities of ANPER will be assessed as well as relevant institutions to support the selected financing mechanisms..

Activity 3.1.6. Train relevant ANPER staff on the launch and management of the selected financing mechanisms

Relevant ANPER staff will be trained on the selected financing mechanisms

Output 3.2. Domestic financial sector capacity-building on business and financing models for minigrids

Some efforts have been already conducted towards financial institutions to raise their awareness on CEMG (e.g., NESAP), but remain limited and do not include sufficient suitable as well as innovative financing solutions.

This output includes the following activities:

Activity 3.2.1. Build the capacities of the national financial sector

The local financial sector will only offer suitable and affordable financing solutions (demand and supply) once it gains awareness of and appetite for the minigrids market. De-risking means and lucrative opportunities around lending in the minigrids market in Niger will be put forward. Workshops will be conducted with representatives of the finance community - financial institutions and investors - whereby a variety of business models and financing schemes, as well as the best practices (including environmental, social and gender considerations) will be shared and discussed. Capacity building will be carried out with financial institutions to design and implement adapted financing schemes in close collaboration with the other Component 3 outputs.

Activity 3.2.2. Develop links with relevant stakeholders around financing and costs reductions

Links will be developed between local financial institutions, relevant government representatives (DGEME, agriculture and more) and international donors in order to explore hybrid and innovative schemes focused on unlocking finance and reducing risks and capital costs (e.g., first loss pools, currency hedging facilities, etc.). This includes for instance IRENA's marketplace, an online investment catalyst and connecting platform for technical and commercial RE solutions. In addition, awareness will be raised on climate risks and mitigation measures through the introduction and operationalization of low-carbon minigrids as well as the financial impacts of green solutions.

To ensure a good understanding and ownership of these solutions, capacity building for local financing solutions providers, local developers and energy solutions providers, government representatives, ANPER, NIGELEC, etc. will be provided.

Output 3.3. Replication plan (including investment plan) for scaling up rural energy access developed

This output includes the following activities:

Activity 3.3.1. Develop a replication plan for scaling up investment in minigrids

A plan for scaling up minigrid investments will be developed based on data gathered and lessons learned from implementation of project activities across all AMP countries and from GEF-funded minigrid projects worldwide, knowledge shared by the regional project with the national projects, and insights gained from participating in AMP Communities of Practice. The Program's comprehensive approach to reduce financing, hardware and soft costs will create the enabling environment to attract public and private investments. This coupled with sound knowledge management underpinned by a robust theory of change and a strategic environmental and social assessment is expected to catalyze markets.

Activity 3.3.2. Conduct relevant market survey

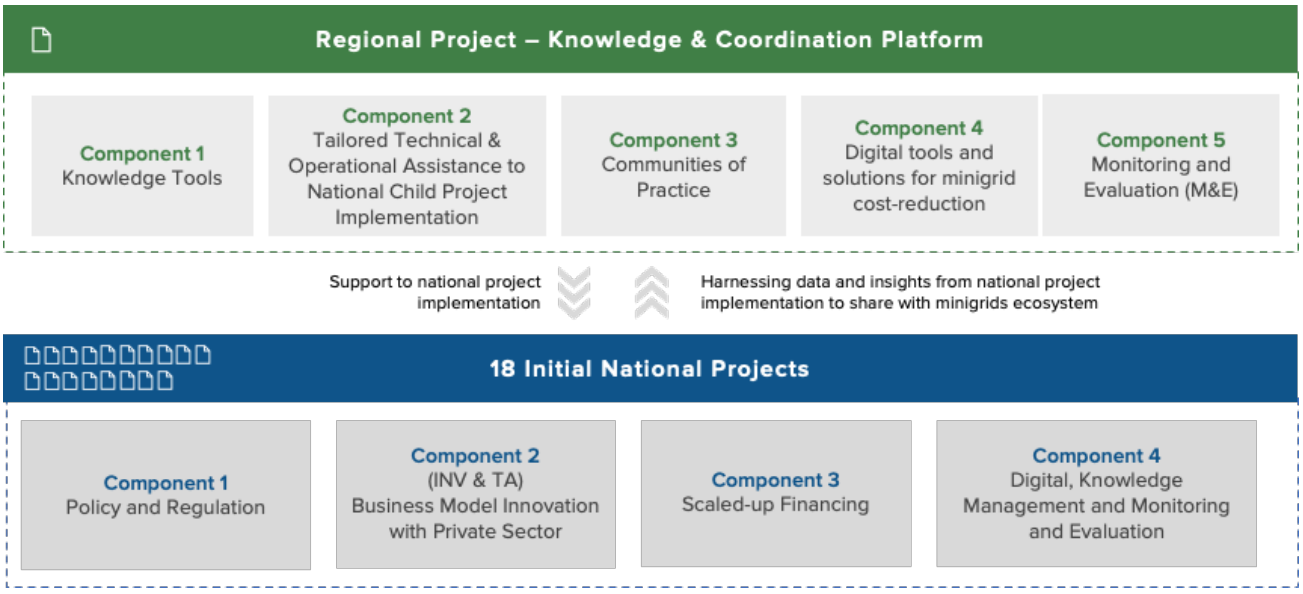
A more detailed market survey will be carried in Niger to assess scaling-up and replication impact potential.

Component 4. Digital & Knowledge Management

Outcome 4: Digitalization and data mainstreamed, across stakeholders, into local minigrid market development. Increased knowledge, awareness and network opportunities in the minigrid market and among stakeholders, including benefitting from linkages to international good practice

Linkages to the AMP Regional Project: Component 4 is a key interface with the AMP Regional Child Project. As such, details on linkages to the regional project as relevant for digital, knowledge management and data performance analytics activities under the project are addressed for each of the outputs described below.

Figure 7- Interactions between AMP regional and national projects



Output 4.1. A project digital strategy is developed and implemented, including linkages to and following guidance from the AMP Regional Project

This output includes the following activities:

Activity 4.1.1. **Develop and implement a project digital strategy (the 'Project Digital Strategy')**

All national child projects will develop a Project Digital Strategy in year 1 which will be implemented thereafter. The Project Digital Strategy will be updated on an annual basis to reflect learnings from project implementation, guidance received from the AMP Regional Project on digital/data tools and solutions, and insights gained from minigrid pilot(s) data. This includes:

- payments data (via the mobile money platform provider)
- energy usage behavior data (kWh consumption patterns - via the smart meter infrastructure)
- generation system data (including metrics like battery voltage levels- via the smart meter infrastructure)

Activity 4.1.2. **Draw recommendations for a national-level digital strategy for minigrid development**

Upon implementation of the Project Digital Strategy and based on lessons learned around opportunities to leverage digital tools and solutions for minigrid sector development, the project will develop a set of evidence-based recommendations for rolling out digital solutions for minigrids at the national level. These recommendations will be shared with key national stakeholders and provide the basis for developing a digital strategy for minigrid development post-project.

Output 4.2. A 'Minigrids Digital and Data Management Platform' is implemented to run tenders and manage data from pilots, and to support minigrids scale-up and cost-reduction

A Minigrids Digital and Data Management Platform is planned to be procured by World Bank's HASKÉ project and implemented by ANPER. At project implementation launch, the PMU should work with the ANPER and World Bank team to assess the platform. Based on these, the activities under this output could be adapted.

This digital platform would serve different purposes including:

Specifically, for minigrid investment pilots under Component 2:

- Running digital tenders by which minigrid developers will be selected as beneficiaries to receive support under the project
- Managing all technical and financial data related to minigrid sites
- Provide minigrid developers - selected to implement minigrid pilots with support from the project - access to a set of best-in-industry digital tools for analyzing minigrids (e.g., demand forecasting, system optimization, distribution network design, detailed financial modeling at the site and portfolio level)
- Capacity building for minigrid developers and government stakeholders around the use of the minigrids data management platform

The implementation of this data management platform by ANPER, the Implementation Partner, to run and manage minigrid tenders and then systematically monitoring minigrid pilots and collected data from pilots, is expected to result in improved project design and system optimization, reduced uncertainties and more transparency in minigrid tenders attracting more bidders and increasing competition, and lower transaction costs associated with bidding.

For the project and minigrid sector more generally:

- Provide a centralized database for all distributed energy projects/programs at national level including sector-wider, large-scale tenders or result-based financing mechanisms.
- Collect, manage and aggregate data from all minigrid sites
- Run digitized tenders and administer grants (other than for Component 2 pilots)
- Performance verification of minigrid systems for improved sector oversight
- Real-time monitoring and evaluation of electrification projects/programs
- Applying advanced analytics of minigrid portfolios to generate critical insights to advance the sector

Similarly, as part of the roll-out of the data platform, minigrid developers (as well as key government and other stakeholders) will receive capacity-building and in-depth training to use analytical tools and data management technologies.

The AMP Regional Project will make its own data management platform available to aggregate data from all national project pilots based on a common M&E framework.

This output includes the following activities:

Activity 4.2.1. **Support the development of Terms of Reference (TORs) for procuring a Minigrids Digital Platform**

The PMU will coordinate with HASKÉ, ANPER and all relevant stakeholders (National Dialogue) to co-develop relevant ToR. An adaptive approach will be used here at project at AMP project launch to assess the status of the procurement of the Minigrids Digital Platform.

The standardized TOR provided by the AMP Regional Project will be shared with ANPER, HASKÉ and any other relevant partner, and tailored to the specific country/project needs. Box 7 below provides indicative specifications for the Digital Platform.

The project will specifically ensure that some specific questions are added to the tendering process of sites, including key value chains on the site and their energy needs (supported by the results of output 2.2.), existence of a telecom tower, a school and a health center, gender-related aspects, etc.

Activity 4.2.2. **Ensure that the selected Minigrids Digital Platform matches the requirements**

The project will ensure that the country-level minigrids digital platform enables (i) convening and capacity building for key stakeholders (public/private), (ii) collecting and managing technical and financial data related to minigrid pilot(s) based on the project's Quality Assurance and Monitoring Framework (QAMF), including linking to the AMP Regional Project, and (iii) acting as the mechanism for running digital tenders for minigrid developers/sites.

Box 7 - Indicative Specifications for the Project's Digital Platform

The project digital platform will provide key functionality for the project in terms of acting as the (i) national digital convening platform for key stakeholders (public/private), (ii) providing ongoing data gathering and M&E on minigrids, including linking to the AMP regional project and (iii) acting as the mechanism for tenders for minigrid developers/sites.

Functionality	Details
National digital convening platform for key stakeholders	· Set up of a country-specific, web-based platform to manage all technical and financial data related to minigrid sites at the site and portfolio level · Single site register of minigrid sites, with geospatial views and technical/financial benchmarks for site assessment · Set of best-in-industry tools for analyzing minigrids, including demand forecasting, minigrid system design and optimization, and financial modeling · Capacity-building and in-depth training of key government and other stakeholders to use analytical tools and data management technologies
Technical data collecting platform: (remote monitoring & analytics)	· Direct integration with smart meters and remote monitoring systems for live data feeds and monitoring (with options to address lack of remote monitoring systems or other restrictions) · Big data analytics and customized reporting to calculate and report on standardized metrics for pilot performance, based on project QAMF · Quality assurance of data quality, accuracy, relevance, consistency · Interactive tools to analyze data, filter, and view at varying levels of granularity · All pilot-specific data can be rolled up into national view, and all country-specific data can be rolled-up into regional view
Financing platform for running tenders to select minigrid pilot beneficiaries	· Complete end-to-end management of e-tenders for minigrids customized to specific project/pilot needs (e.g. customized technology solutions, customized workflow, customized KPIs for pilot monitoring) · Automated proposal analysis for quantitative proposal components · Remote verification of connections through smart meter integrations · Automated M&E analytics for all RBF program indicators (connections deployed, amounts paid, gender/environmental impact metrics, etc.)

Activity 4.2.3. **Develop and operationalize a rural electrification platform**

One of the many findings during PPG was a lack of a centralized repository of documents around rural electrification (including market assessment, regulatory and legal texts, knowledge documents, feasibility studies, training materials and webinars, relevant events and conferences, etc.) and to avoid double work in the future. Relevant documentation would be collected and uploaded by relevant ANPER staff. Such a platform, located under ANPER, and accessible to relevant stakeholders, would be complementary to the specific Minigrids Digital and Data Management Platform. The latter would be integrated into the rural electrification platform.

Output 4.3. A Quality Assurance and Monitoring Framework for measuring, reporting and verification of the sustainable development impacts of all minigrids pilots supported, including GHG emission reductions, is adopted and operationalized based on standardized guidance from the regional project

This output includes the following activities:

Activity 4.3.1. **Provide inputs and feedback to the AMP Regional Project on the development of a standardized Quality Assurance and Monitoring Framework for application across AMP national projects (AMP-QAMF)**

A standardized Quality Assurance and Monitoring Framework for application in all minigrid pilots supported under AMP national projects (AMP-QAMF) will be developed in year 1 of the AMP Regional Project and disseminated to all national project staff. This AMP-QAMF will build upon the minigrid Quality Assurance Framework (QAF), which is a set of technical and financial performance monitoring indicator, developed by NREL, SEFA and others, as well as the considerable data gathering, pooling and analysis work ongoing by partners such as RMI, SE4All and AMDA.

It is expected that national project staff will provide both inputs and feedback on the development of this framework as well as on how best to operationalize the commitment to its adoption by the minigrid operators receiving support from the national project. Concerns around data privacy or sensitive data on the part of minigrid operators will be considered and addressed in each case.

Activity 4.3.2. **Operationalize the AMP-QAMF**

The adoption and utilization of this framework and associated data reporting protocols will be a mandatory requirement for all minigrid pilots supported under AMP (e.g., applicable to all national projects) and each minigrid operator/sponsor who is the beneficiary of investment subsidies and technical support by the project will be required to formally commit to using the QAF as a condition of assistance. The adoption of the QAF by all minigrid operators/sponsors supported under AMP national projects will ensure that the regional project can aggregate common data metrics and track a standardized set of key performance indicators across all minigrid pilots supported by AMP across all partner countries and report this data to the donor on a programmatic level.

To operationalize this, the parties operating the minigrids will be supported with the installation of smart meters and/or remote monitoring equipment as appropriate. Provision will also be made to support the operators to access this data and extract potentially valuable insights on their minigrid operations.
If appropriate, provision will be made to train relevant government agency representatives (and members of the Communities of Practice) in the use of a national minigrid dashboard reporting data on all of Niger's minigrid projects.

Output 4.4. Engage with regional project, including, but not limited to, via (i) participating in Communities of Practice and (ii) capturing and sharing lessons learned

The AMP regional project will support and facilitate knowledge management and information sharing between the regional child project and national child projects, among national child projects, and between the program and the larger minigrid community. Knowledge tools and good practices around minigrid cost-reduction in a variety of regulatory environments, and research and development tools, such as policy packages, template tender documents, and guidelines on productive use program designs will be made available. The toolkits will support both public and private sector (e.g., minigrid developers) and the overall minigrid market.

Activity 4.4.1. **Participate in AMP Communities of Practice (CoP)**

One of the primary ways national project staff will interface with the AMP Regional Project is via the 'Communities of Practice' (CoPs) and associated activities/platforms. While it is expected that many of the activities under the Regional Project Component #3 will be undertaken virtually (via internet-based platforms, webinars or digital platforms) it is also expected that the CoPs will include actual in-person workshops, meetings or training events. Among the topics to be covered, quality standards developed and enforced in other AMP countries around CEMG equipment and ancillary products should be included.

Activity 4.4.2. **Share research and lessons learned with the AMP regional project**

Research and lessons learned will be systematically shared with the regional project based on guidelines that will be defined by the regional project and shared at the project's Inception Workshop. Capacity building will be provided to the national project PMU to compile lessons learned and share knowledge effectively.

Activity 4.4.3. **Collaborate with the regional project on an 'Insight Brief'**

Every AMP national project is expected (during the four-year implementation cycle) to collaborate with regional project staff on the development of at least 1 'insight brief' capturing (in an accessible format) selected key highlights from a successful national project activity. The 'insight brief' can cover any activity of the project and take the form of a written brief or video brief. The regional project has budgeted resources to produce 'insight briefs' (under its Component #1 Knowledge Tools), but the success of regional staff in producing insight briefs highlighting national project activities will be dependent on content and data provided by the national project team and stakeholders.

To facilitate such collaboration, the project will hire a consultant or local firm to gather data and audio-visual content (video footage, photos, etc.) on the subject for the 'insight brief'. The information and data collected at the national level will be provided to the regional project staff who will utilize this content and produce an 'insight brief' according to a standardized communications format for all AMP knowledge products for external audiences. The 'insight brief' will be produced in both the local/national language of the relevant national project as well as English for dissemination by the regional project to regional stakeholders and publishing on the AMP website.

Output 4.5. Awareness raising campaigns, including lessons learned, are developed and disseminated at all levels nationally (including intervention zones) and with the regional project

This output includes the following activities:

Activity 4.5.1. Design a communication strategy

A communication strategy will be elaborated based on awareness raising campaigns and lessons learned. The awareness raising campaign experiences around sustainable energy across Niger and other AMP countries will be identified and leveraged where necessary. Available communication materials will be adapted. The communication strategy should include communication towards members of the GoN as well as the political sphere (National Assembly etc.) to showcase how critical electricity in rural areas is to contribute to sustainable development. It should also cover communication campaigns for the general public and have a yearly communication work plan designed to support the PMU and UNDP Niger to monitor and implement the activities.
This includes radio spots in local languages and banners. In addition, with at least 2 telecom operators (out of 4), SMS campaigns will be rolled out – 1 SMS per operator per month over the entire duration of the project. SMS content will be provided by the PMU. Digital means will also be leveraged to raise awareness for instance with the national digital promotion agency, Agence Nationale pour la Société de l'Information (ANSI), and some youth and women associations.
The project will develop its own website or a dedicated part in the AMP regional website.
Field visits will be organized on pilot sites to see, understand and discuss with local communities on their experience with the minigrids and energy access.
The leveraging role of schools and children will be envisaged to communicate the need to shift to renewable sources of energy and the catalyzing role of access to energy in rural areas to have access to lighting, cooking, productive use, etc.

Activity 4.5.2. Implement and monitor the communication strategy

The communication strategy will be rolled out and specific indicators defined in the communication strategy will be tracked to ensure the expected impacts. The communication strategy will be adapted if necessary, according to potential gaps that may arise during project implementation.

Component 5. Monitoring and Evaluation (M&E)

Output 5.1. Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, (iii) Mid-Term Evaluation and (iv) Terminal Evaluation

The AMP Regional Project will provide support to the project, through its PMU staff or by hiring or recommending subject matter experts, for the project to execute M&E activities such as the inception workshop, ongoing monitoring, and project evaluations. Further details provided in Section VI.

Activity 5.1.1. Hold an inception workshop and prepare a report

A project inception workshop held to officially launch the project and, among other aims, familiarize key stakeholders with the detailed project strategy, roles and responsibilities of the project team, and project planning instruments such as the Total Budget and Work Plan (Section IX), multi-year work plan (Annex 4), Monitoring Plan (Section VI), the Procurement Plan (Annex 11), the communication plan, ESMP (Annex 9) and the gender action plan (Annex 10), among others. The national inception workshop will be carried at the beginning of project implementation (within 60 days of CEO endorsement of this project). The workshop will be organized by the PMU with support from the IP and planned with support from the AMP Regional Project staff. Staff from the AMP Regional Project PMU will participate either remotely or in-person in the Inception Workshop and will provide support to the project PMU to plan the workshop and develop materials and content that will facilitate project planning activities including the template for the Inception Workshop Report. The Inception workshop report will be prepared by the PMU and submitted to UNDP within (within 90 days of CEO endorsement of this project).

Activity 5.1.2. Undertake ongoing project monitoring

As set out in the Monitoring and Evaluation Plan (Section VI), data on Results Framework Indicators will be systematically collected and analyzed to provide decision-makers, managers, and project stakeholders with: (i) information on progress in the achievement of agreed objectives and the use of allocated resources, and (ii) regular feedback on performance of projects and programs taking into account the external environment. Information from systematic monitoring serves as a critical input to ongoing PMU management decisions (adaptive management), evaluation, and learning.

The GEF Core indicators included in the Results Framework (Section V) as per this Project Document (Annex 16) will be used to monitor global environmental benefits and will be updated for reporting to the GEF prior to the project's evaluations. Namely, the mid-term review (MTR) and terminal evaluation (TE) described under Activity 5.1.3 and Activity 5.1.4 below.

The project is accompanied by various plans including Stakeholder Engagement Plan (Annex 8), mitigation plan for project risks (Risk Register in Annex 6), and Gender Action Plan (Annex 10). These plans will be reviewed according to the monitoring and evaluation requirements.

According to the project's social and environmental risk rating, there is a need to carry out continuous monitoring of the social and environmental safeguards as proposed in the Environmental Social Management Framework (ESMF) and other SES frameworks/plans (Annex 9). The environmental and social management plan (ESMP) that will emanate from the application of the ESMF will also be monitored under this activity.

Data collected by monitoring GEF Core indicators, Results Framework indicators, project plans and social and environmental safeguards will be used to prepare the annual Progress Implementation Report (PIR) to report back to UNDP and/or GEF.

Activity 5.1.3. Conduct a Mid-term review (MTR) of the project

An independent mid-term review (MTR) will take place at the half-way mark of project implementation and will be conducted according to guidance, rules and procedures for such evaluations established by UNDP and GEF as reflected in the UNDP Evaluation Guidance for GEF Financed Projects. The MTR will be made widely available to all project stakeholders in the relevant language.

Activity 5.1.4. Conduct a Terminal evaluation (TE) of the project

An independent terminal evaluation (TE) will take place upon completion of all major project outputs and activities. The project's terminal GEF PIR along with the TE report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the Project Board during an end-of-project review meeting to discuss lessons learned and opportunities for scaling up.

Partnerships: The project will work with a variety of partners and initiatives to achieve the project's objective and results (see table below)

Table 4 - Identified partners for the AMP in Niger

Partner name	Description and contributions	Relevant project outputs
Ministry in charge of renewable energy	The Ministry has in charge the setting up of policy, legislative, regulatory, institutional, strategic and planning framework to promote the development of access to electricity in rural areas through renewable energy sources. A steering committee will be set up to monitor the implementation of the project. The Ministry is the beneficiary of component 1.	All outputs relevant to component 1 and even all the overall outputs of the project since the Ministry has in charge the management of the steering committee.
Ministry in charge of environment & Bureau National pour les Etudes et Evaluation (BNEE)	BNEE is responsible for evaluating all development projects and programs in Niger before they are implemented. The developer submits the project notice to the BNEE, which analyzes it. Following this analysis, the project is classified in one of the following categories A, B, C and D. Projects in categories C and D do not need an evaluation study. The evaluation of a project generates additional costs which, in the field of energy, would increase the cost per kwh. Therefore it is important to provide all the necessary information in the project notice so that the project is classified in the appropriate category	Contribute to the achievement of output 1.3 ; 1.5
Ministry in charge of vocational training	In Niger republic each municipality has a vocational training center. There are also in each region technical college and higher vocational training center. Hence different levels of training centers are available. They will contribute to the training of technicians to ensure the maintenance and repair of minigrids.	Contribute to the implementation of output: 1.7 ; 2.3
Ministry in charge of gender promotion	The ministry in charge of gender promotion implements the national gender policy. To this end, it will ensure that gender is taken into account in the implementation of the project.	Outputs: 1.1; 1.5; 2.1; 4.24.4;4.6
National Agency for the Promotion of Rural Electrification (ANPER)	ANPER's mission is to design, implement and monitor rural electrification development programs in Niger. To this end, With the support of the Government, it mobilizes funding and the private sector to promote the development of rural electrification. ANPER is currently in the process of awarding by leasing several minigrids already built to private operators. Currently, it is	ANPER is the implementing institution of the Niger AMP project. Hence it coordinates all the outputs.

Partner name	Description and contributions	Relevant project outputs
	implementing several minigrid projects, financed by development partner including the World Bank, ADB, IDB, EBID, BOAD, Italian Cooperation, etc. With its own funds, ANPER is implementing a project to install two minigrids of 140 kWc each in two localities.	
Africa Green Tec/private enterprise	This enterprise has already built a pilot containerized minigrid solar in the village of Amaloul with a digital monitoring system. Two technicians provide on-site maintenance. The tariff system is not based on cost reflective as stated in the electricity law but based on the ability to pay. It is currently developing a feasibility study for the electrification of 50 rural localities.	It could be a significant player in contributing to the achievement of outputs 1.1, 1.2, all the outputs of component 2, outputs 3.2; 3.3; 4.1; 4.2
National Agency for Solar Energy/ANERSOL	ANERSOL's mission is to promote and develop solar energy in all sectors of the national economy, in particular: Propose and popularize standards and labels for renewable energy equipment. Carry out quality control of renewable energy equipment. Control the compliance of installations and certification of private operating in the field of solar energy. Contribute to capacity building for actors in the field of solar energy. Through the support of partners, it has set up a technical platform used for quality control of renewable energy equipment and has set up communication activities to promote the development of off-grid solar market. This project could strengthen the capacities of the technical platform.	It will contribute to the implementation of the following outputs: 1.1; 1.2; 1.5; 1.6; 1.7; 2.3; 3.3.; 4.2
Professional Solar Association (APE-Solaire) and other non-members operating in solar energy	They have no experience in building solar minigrids but have very good experience in stand-alone systems for domestic, institutional, and productive use. They showed a strong desire to participate in the implementation of the project and specially to acquire additional technical skills required for the development of minigrids. Their participation and contribution would be decisive especially in the regulatory framework, pricing and the mobilization of financial resources. As part of the SEFA project, there are 3 private promoters who have benefited from support in carrying out feasibility studies for solar minigrids. They need to be trained for feasibility study, management and	They will play an important role, among others, in the following outputs: 1.1; 1.2; 1.4; 2.1 2.2; 2.3; 2.4; 3.2; 4.2; 4.5

Partner name	Description and contributions	Relevant project outputs
	operating minigrid and suitable environment and financial mechanisms schemes.	
NGO Plan International Niger	It has built the first solar minigrid in Niger. The minigrid was handed over to ANPER. It supports rural communities in the Tahoua and Maradi regions with access to solar electricity (micro-grid) for both households and productive use. Through a partnership, it also trains beneficiaries women to ensure maintenance and operation of solar equipment/installations.	Contribute to the implementation of outputs 1.1 ; 2.1 ; 2.3 ; 3.2 ; 4.2 ;
Ecole des Mines de l'Industrie et de la Géologie	Train engineers in renewable energy. It could integrate training modules specific to solar mid-grids into its curricula	Could contribute to the achievement of outputs 1.1; 1.5; 2.3; 2.4; 4.2
Fédération des Coopératives Maraîchères du Niger	It carries out activities in the agricultural value chain, in particular IGAs. There is a possibility of taking into account in the selection criteria for the pilot sites the establishment of the production activities of the cooperatives. This association could raise grant funds from its partners such as the FAO or FISAN to contribute to the financing of the projects.	Activities centered on the productive use of energy. Take this into account as a criterion in the choice of pilot sites. Outputs: 1.1; 2.1; 2.3 and 4.2
World Food Organization/FAO	Supports actors in the silvo-agro-pastoral development chain, particularly in irrigation, livestock, farming, production and conservation of market garden products. The source of energy used are diesel or solar power in rural areas. Currently, FAO is implementing regional projects focusing on the value chain. Within the framework of this project implementation, there could be opportunities to replace diesel generators used for water pumping, for example. It would therefore be very interesting to explore potential synergies for the installation of a pilot minigrid on market gardening sites.	Outputs: 1.1 ; 2.1 ; 2.2 ; 2.3 ; 3.2 ; 4.2.
Multifunctional Platform Project/PTFM	Works in the promotion of PTFM for access to energy services, in particular: electrification, mills for processing cereal grains. Training artisans and beneficiaries in the upkeep and maintenance of PTFMs. The training of beneficiaries and the setting up of management committees. Possibility of co-financing if the sites selected correspond to their criteria	Contribute to output 2.1 by integrating the choice of pilot sites including its criteria.
National Financial Institutions	Through partners such as Power Africa and the NESAP project, some banking institutions have benefited from capacity building in energy access. They have also taken an active part as stakeholders in studies in the energy sector. In addition, as partners, for example through the	Contribute to the achievement of outputs 3.1 ; 3.2 ; 3.3

Partner name	Description and contributions	Relevant project outputs
	NESAP project, SONIBANK, BSIC and Capital finance have benefited from credit lines opened for private operators and consumers.	
World Bank	The World Bank is an institution that supports the government of Niger not only to set up an environmental framework favorable to the promotion of the energy sector but also in the financing of electricity access projects. Within the implementation of Haské project, it is planned to revise the electricity code but also to support ANPER in setting up a digital platform to, for example, facilitate the management of tender documents	Contributes to the implementation of outputs 1.2; 3.1 and 4.2
Fonds d'Investissement pour la Sécurité Alimentaire et Nutritionnelle/FISAN	This institution manages trust funds from development partners to promote the development of the silvo-agro-pastoral and tertiary sector. It provides grants to beneficiaries. These interventions indirectly concern the energy sector in the value chain. It could be an important partner in the implementation of this project.	Could help provide subsidies to minigrid beneficiaries in these target sites Output: 3.1;3.2
Power Africa	It has built the capacity of potential promoters of the renewable energy sector, particularly solar, to promote investment and the development of rural electrification. It also supported ANPER in carrying out feasibility studies for the development of solar minigrids built around telecommunication antennas. Possibility of selecting pilot sites among the feasibility studies sites carried out by Power Africa.	The following outputs 2.3; 3.1; 3.2; 3.3 and 4.2

Other ongoing initiatives as listed in the ANPER co-financing letters are supervised and directly implemented by ANPER, the project IP, and thus covered and articulated by ANPER directly.

1. <u>Private sector's engagement in the project</u>:

The project formulation launch workshop held on April 7, 2022, brought together the various stakeholders, including representatives of the private sector. The private sector was represented by the Association of Solar Professional (APE-Solaire) and other non-members of APE-Solaire.

Throughout the process of the Project formulation, representatives of the private sector were consulted. They welcomed this initiative very well. Thus, the stake around the project was shared with them and in return they provided very useful information which contributed to enriching the document.

During the project document validation workshop which took place on June 16, 2022, private sector representatives were also participants. This meeting was an opportunity for the private sector to fully appreciate the objective and expected results of the AMP Niger project and above all to understand the different activities planned by component but also their expectations and their role to be played in the implementation of the project. They noted and remained hopeful that the implementation of the project will make it possible, if not to completely curb, to mitigate the negative impacts of the various obstacles, in particular the regulatory framework, financing, capacity building and dialogue between partners to promote the development of minigrids in Niger. The private sector has also expressed its deep concern about the marginalization to which it is subject in the awarding of contracts for the construction of minigrids under the pretext that it does not have the experience in minigrid. It is therefore recommended that the AMP Niger project considers the adequate involvement of local businesses in the implementation of pilot projects to strengthen their technical skills and ensure proper ownership of minigrid technology.

Co-financed activities

Several of the abovementioned partners have provided letters of co-financing for this project, as attached in Annex 13 to this project document. As further described in Table 5 below, some of these co-financed activities correspond to funds not flowing through UNDP accounts and whose results are not included in the project results framework. In this case, UNDP is accountable to monitor the risk to realization of co-financing amounts and realization amounts annually in the GEF PIR, at mid-term and at terminal evaluation. Specifically, potential risks associated with co-financing that may affect the Project, including safeguards related risks that fall within the project context or area of influence, will be considered in safeguards due diligence and the project risk register, and monitored accordingly. Risk management measures identified will be only those within the control of the UNDP project (e.g., managing reputational risk). See the ESMF (Annex 9) for more details on the management of risks related to the different types of co-financed activities in this project.

Table 5 - List of co-financed activities

Co-financing source	Co-financing type	Co-financing amount USD	Included in project results?	If yes, list the relevant outputs
ANPER	Grant	135,104,356	No	n.a.
ANPER	In-kind	76,490	Yes	PMC
UNDP	Cash	300,000	Yes	2.1., 5.2.,5.3.,5.4., PMC
SIDA	Grant	1,276,171	No	n.a.
TOTAL		**USD 136,757,017**		

N.B.: SIDA's co-financing letter (Annex 13) list it's relevant projects for AMP Niger with the timespan of its projects (e.g., 2020-2024). As AMP Niger will be implemented from 2023-2027, the amounts taken into consideration as co-financing activities will only cover 2023-2027. Thus, the amounts in the co-financing letters and the table do not coincide.

The co-financed amounts have been calculated as follows:

- SIDA: SIDA's "Liptako-Gourma rural electrification project" project lasts from 2020-2024 hence 2 years out of 4 are considered for AMP Niger. The total amount was divided by 4 (years) to have a broad approximation of disbursements per year.

ANPER: The projects breakdown amount listed in the letter has been calculated as from the beginning of AMP Niger, expected to start in 2023. The table below lists the projects and breakdown type (INV/TA) considering only the relevant proportion of the project (in relation to AMP's components and the proportion of these projects implement between 2023 and 2027:

Table 6 – Breakdown of ANPER co-financing letter projects

	Investment in US$	Technical Assistance in US$	Total in US$
Rural electrification project by decentralized solar energy in Niger financing of the Islamic Development Bank (IsDB)	24,870,000	7,130,000	32,000,000
Rural electrification project using hybrid micro power plants in 47 localities in Agadez, Diffa, Dosso, Maradi, Tahoua, Tillabéry and Zinder regions financed by the West African Development Bank (BOAD)	19,316,797	1,502,484	20,819,281
Rural electrification project using solar Photovoltaic systems and individual kits in 100 localities in Niger financed by IRENA/ADFD	13,750,000	1,219,973	14,969,973
Rural Electrification Project by Solar Photovoltaic Systems in the Regions of Dosso, Tahoua and Tillabéry, financed by EBID/Exim Bank	3,115,500	59,350	3,174,850
Photovoltaic rural electrification project in Niger - Keita and Illela departments financed by the Italian Agency for Development Cooperation	978,737	236,733	1,215,470
Electricity Access Acceleration Project (HASKÉ), Component 2. Strengthening the Ecosystem for the Development of Rural Solar Photovoltaic Mini-grids and Sub-component 4.2. Institutional Strengthening and Technical Assistance financed by the World Bank	22,500,000	2,500,000	25,000,000
Niger Solar Electricity Services Access Project (NESAP), Component 2: Rural Electrification through Solar Hybrid Mini-grids following a Delegated Service Operator Model	4,158,698	-	4,158,698
Rural electrification project by solar photovoltaic system of 250 villages in the Republic of Niger under financed by EBID (BIDC in French)	30,995,478	2,770,606	33,766,084
TOTAL in US$	**119,685,210**	**15,419,146**	**135,104,356**

Risks:

The overall risk profile of the Project has been assessed as “substantial”. Based on the risk categorization assigned to the various country projects and the associated environmental and social risks, the following procedures for screening, assessing, and managing those risks must be undertaken during project implementation of each country project. (1) Screening of social and environmental risks and impacts and determining applicable social and environmental standards and requirements (including UNDP SES). The screening process utilizes UNDP’s SESP and develops a specific screening procedure for the forthcoming type of sub-projects/activities. (2) Appropriate types of social and environmental assessment to identify, document and address potential social and environmental risks and impacts. (3) Preparing and approving time-bound action plans for avoiding, and where avoidance is not possible, reducing, mitigating, and managing adverse impacts, including development of specific management plans according to applicable policies and regulations, including UNDP’s SES (i.e., Environmental and Social Management Plans which would be completed post-assessment).

Specifically, the SESP identified 14 risks, 6 of which assessed as “substantial” and 8 as “moderate”. The DREI framework identifies 10 risks, 7 of which assessed as “high”, 2 as “moderate” and 1 as “low”.

Social and environmental risks are primarily linked to human rights given the challenges to secure social inclusiveness and incorporate vulnerable people. Among other root causes, underlying factors include very low-income levels of rural people living in small communities which rely on subsistence farming and collection; undefined or absence of land tenure titles and associative structures enabling people to claim their rights; demographic pressure which may lead to local movements of people as a result of electrification, potentially separating them from their food sources.

While electricity supply is a socio-economic enabler, the nexus between productive uses and electricity supply needs to be further articulated – including according to gender. To address this risk, the Project design incorporates sustained community engagement during the preparation phase of the proposed pilots. This activity will draw upon country knowledge and positive experiences in other sectors (e.g., agricultural development) to make these available to the energy sector in Niger.

For more details, reference is made to the ATLAS Risk Log, Annex 6.

Stakeholder engagement and south-south cooperation:

The participatory and inclusive approach adopted by the project since PPG is part of its DNA. A stakeholder analysis (to be continuously updated) was conducted to identify all relevant stakeholders at the PPG phase. Extensive and numerous stakeholder consultations, one-on-one, focus groups and workshops (inception and validation) were carried out at PPG phase and will be pursued during project implementation to ensure ownership and sustainability during and after project closure. Comments, questions, requests and commitments are stimulated at every project stakeholder.

Particular efforts were put since design phase and will continue during the entire project implementation phase:

- Promote effective engagement, as well as inclusive and meaningful consultation. This encompasses a two-way process between the stakeholders and the PMU, the IP, the RP and UNDP as GEF executing agency to ensure continuous insights, feedback and involvement of project stakeholders. Sharing information and knowledge is critical and supported at PPG as well as during implementation through different specific tools and activities.
- Forge stronger partnerships, particularly with civil society, communities and the private sector. It requires continuous work, based on transparency, engagement and dialogue, outreach, awareness building, listening, and respect for the importance and contributions of all stakeholders.
- Harness the knowledge and expertise of stakeholders which includes acknowledging the fact that stakeholders are a great source of information and knowledge, and that these should be shared through various means including via knowledge management and monitoring & evaluation (Component 4).

Stakeholder engagement since PPG put a high emphasis on conducting it in an inclusive and gender responsive manner. During the PPG phase, relevant stakeholders were asked to provide inputs and comments on the project and their specific role in contributing to overcome the development challenge together. This largely contributed to nourishing the project document and adapting it to local realities and needs as much as possible. At the validation workshop and after sharing the final draft of the project document and annexes, feedback was provided by stakeholders and integrated into the final version of the document. During project implementation, stakeholders will continue to actively contribute and engage in achieving the overall goal of the project. The Stakeholder Engagement Plan in Annex 9 provides further details on the stakeholder groups, their roles and responsibilities.

Different actions are provided to stimulate and sustain stakeholder engagement throughout project duration (and even after):

- Inception workshop at project launch gathering all key stakeholders to present the project and the inception report to facilitate project implementation. Participants will be invited to share their insights and updates to adapt the workplan. The official project launch and main outcomes of the inception workshop will be broadcasted to a larger audience as part of the awareness raising campaign (Component 4).
- The National dialogue platform (Component 1) is a powerful tool to support exchanges between project stakeholders and with the PMU on various topics (especially through sub-committees). This multi-stakeholder media is critical to effectively and efficiently contribute to the development of a nascent minigrids market in Niger. Sub-committees will be created to further facilitate implementation, ownership and engagement.
- The national Community of Practice is another means to engage with all relevant stakeholders and motivate them to be involved in the project, sharing their experience, learning, and growing.
- Continuous stakeholder interactions and consultations through meetings, workshops, training, awareness raising campaigns, etc. will also be used to foster stakeholder engagement. This includes during preparation, construction and implementation of the pilot sites under Component 2.
- The Project Board/Steering Committee is the ultimate platform for stakeholder engagement and decision-making including beneficiary representatives, the project executive and the development partner (see chapter on governance below). Topics such as workplan, activities and results will be debated, decisions taken together, and necessary corrective measures proposed.

ANPER, the project IP, will directly coordinate with all ongoing (and future) initiatives with other partners (as listed in its co-financing letter). As such, it will ensure that AMP Niger brings the necessary additionality to ongoing (and future) initiatives in Niger's (green) minigrid market.

South-South cooperation is particularly fostered in this project. AMP is a regional programme in Sub-Saharan Africa with a minimum of 18 countries participating. Various knowledge sharing tools among AMP countries and AMP's regional

project will be provided (Component 4). Cooperation and partnerships with other initiatives and organizations promoting South-South cooperation will be offered under the AMP umbrella including RMI, AfDB, IRENA, SE4All, AMDA, etc. Discussions with ECREEE should be developed at project launch. Cooperation with AMP Burkina Faso, Mali (and beyond) for francophone training on specific topics as well as with Senegal's successful national dialogue platform on energy will be fostered.

In addition, to bring the voice of Niger to global and regional fora, the project will explore opportunities for meaningful participation in specific events where UNDP could support engagement with the global development discourse on low carbon minigrid and rural electrification. The project will, furthermore, provide opportunities for regional cooperation with countries that are implementing initiatives on low carbon minigrids in geopolitical, social and environmental contexts relevant to the proposed project in Niger.

Gender equality and Women's Empowerment:

According to the 2020 UN Human Development Reports data, Niger ranks 189 of 189 on the Human Development Index and 154 of 162 on the Gender Inequality Index[22]. In the rural settings where the Niger minigrid program will be implemented, women and girls are central to three key electricity service delivery opportunities that can reduce gender gaps, increase human capital, and provide foundations for economic growth: Water, agriculture, and social institutions.

- **Water:** Much of Niger exhibits a harsh climate with water scarcity, and the rural population has a high degree of susceptibility to climate variability. Women and girl children bear the majority of the daily burden of fetching water in Niger[23] with water fetching for household use among the reasons that female children are withdrawn from school. Water (irrigation) is also an extremely important agriculture input that women have less access to than men. And it is needed in school and health care settings for washing, sanitation, drinking, and cooking. Only 22.7% of schools have access to drinking water and 26.7% access to sanitation facilities[24]. Girls and women especially may avoid going to schools or healthcare settings where water, sanitation and hygiene (WASH) services are absent. Girls, in particular, may miss school when they are menstruating. Fee-for-service water pumping and/or purification can be added elements to minigrid design and, if water can be stored, can also aid in load profile management.
- **Agriculture:** 97% of rural households in Niger earn at least some of their income from agriculture[25], but women supply only about 24% of the labor for crop production (excluding tending livestock or processing harvests)[26]. Women own agricultural land at lower rates than men (35% compared to 63%[27]), and their plots are smaller. This is a result of population pressure and privatization taking place against a mixed backdrop of customary, Islamic, and statutory legal regimes. Woman-managed plots are on average 19% less productive than men's, mostly due to women's challenges using male labor and the large amounts of their time devoted to childcare[28], both of which are a result of prevailing gender norms. Electricity investments can boost women's productivity when used for efficient irrigation of high value off-season crops and watering certain livestock, post-harvest processing, and packaging and conservation—but discriminatory gender norms can impede the success of these initiatives. The cooperative-based model of mechanized processing, notably adopted by most multi-functional platforms, addresses or sidesteps multiple barriers faced by women in agriculture.
- **Social Institutions:** Health clinics and schools constitute critical infrastructure in a country where 50% of the population is under the age of 14 and the Human Capital Index, measuring the amount of human capital (health

[22] https://hdr.undp.org/sites/default/files/Country-Profiles/NER.pdf, visited in June 2022

[23] Graham, Jay P., Mitsuaki Hirai, and Seung-Sup Kim. 2016. "An Analysis of Water Collection Labor among Women and Children in 24 Sub-Saharan African Countries."

[24] UNICEF Niger. N.D. "Water, Sanitation & Hygeine."

[25] Backiny-Yetna, Prospere, and Kevin McGee. 2015. "Gender Differentials and Agricultural Productivity in Niger."

[26] Palacios-Lopez, Amparo, Luc Christiaensen, and Talip Kilic. 2017. "How Much of the Labor in African Agriculture Is Provided by Women?"

[27] Slavchevska, Vanya, Cheryl R. Doss, Ana Paula de la O Campos, and Chiara Brunelli. 2021. "Beyond Ownership: Women's and Men's Land Rights in Sub-Saharan Africa."

[28] Backiny-Yetna, Prospere, and Kevin McGee. 2015. "Gender Differentials and Agricultural Productivity in Niger."

and education) a child born today can be expected to attain over 18 years, is 0.32, among the lowest in the world[29]. The mean average schooling for women is 1.4 years, while men's is twice as high[30], though overall education attainment is increasing and the gender gap is closing. Electrification not only enables lighting, cooling, medical equipment, and information technology in these settings for the benefit of students and patients seeking healthcare, but it also plays a role in attracting and retaining professionals who work in these institutions.

These three spheres of opportunity—water, agriculture, social institutions and Income Generating Activities (IGA)—will be included as gender focus areas of Component 2 (Business Model Innovation with Private Sector) and Component 3 (Scaled-up Financing), especially as it relates to demand-side solutions. Cost-reduction levers (and benefit maximization ones) are critical to these areas, and perhaps more so for women because women have less disposable income, access to finance, and decision-making power. The relationship between intra-household dynamics (e.g., fertility rates and dependency ratios, gender norms governing time use and bargaining power) and reproductive work on the one hand, and economic productivity on the other, can be further explored through this project.

A workable business model for improved water or cooking fuel provisioning, for example, is hard to achieve so long as the shadow price of those goods is unaccounted for and household members can be compelled to supply "free" labor to fetch them. Alternatively, it may be impossible for women to devote enough time to succeed with productive uses of electricity until they can first free up their time from domestic chores, either by receiving more help from other family members or by adopting new appliances, or have access to multifunctional platforms for instance. Both these examples demonstrate that gender equality, in particular valuing women's time and labor, can function as a key for unlocking the economic potential of electricity sector investments; thus the project can tackle gender equality from the outset rather than assume it will be a natural byproduct of energy access.

For Component 1 (Policy and Regulation), efforts will be made to meaningfully include women in the National Dialogue (for participation and decision-making) and training activities, especially practical, hands-on learning activities. Work on regulations and tariffs will be informed by and grounded in realities faced by women, such as widowhood, divorce, or living in polygynous arrangements. To the extent possible, gender-relevant data layers will be added to complement existing pre-feasibility studies of minigrid sites to inform design choices and appropriate levels of program support/concessionality.

Component 4 (Digital and knowledge management) can seek innovative ways to capture data beyond the meter. This includes understanding users (not just customers) and their behaviors and needs. Possibilities include low-cost phone-based surveys (e.g., via interactive voice recordings or SMS) or the use of female community liaisons to collect information on users, appliances, payment responsibilities, and more. Gender mainstreaming is among the topics where insights can be shared to/from the regional AMP.

Innovativeness, Sustainability and Potential for Scaling Up:

Innovativeness

Innovation is inherent to AMP's DNA and applied in various aspects:

1. The project's rationale focuses on the cost-reduction niche of minigrids – hardware, soft and financial costs - as an efficient and effective solution to attract private investment and *in fine* to enable rural communities in Niger to have access to sustainable, reliable, affordable and greener energy.
2. A particular focus and *sine qua none* condition of minigrid projects under AMP is energy for productive use (PUE). Given the rampant extreme poverty in Niger's rural areas, productive use is the first entry point for any pilot site. PUE is a pillar of the financial viability of minigrids in rural areas. Productive uses include both

[29] https://databank.worldbank.org, visited in June 2022

[30] https://hdr.undp.org/data-center/documentation-and-downloads visited in June 2022

commercial (e.g., solar pumping, agricultural produce conservation and processing, telecom towers, artisans and shops) and social (e.g., health and education facilities).

3. The cost optimization aspect is combined with innovative minigrid business models. The innovative aspect of business models relies, on the one hand, on cleantech solutions such as the key maker model and all-inclusive containerized solutions (see Component 2), and on a hybridized setup whereby a close collaboration between communities and private sector, as well as with the government and public sector (development, management and operations) is created.
4. Innovative suitable financing solutions targeting minigrids both for supply and demand are put forward including crowdfunding and -lending, mobile money, etc. (see Component 3) under a Minigrid Funding Facility umbrella.
5. A market-based intervention is applied to effectively contribute to sustainability of the minigrids. As such the private sector plays a key role from PIF and PPG phase up to implementation and post-project phase.
6. A participatory approach is applied since the preparation of the project to ensure collaboration with and between all relevant stakeholders, commitment and ownership, as well as the project's sustainability (including post -project. Adaptive project management shall be applied as several months fall between project formulation and implementation, and while some efforts impacting the minigrid environment are undergoing with other partners' support (see chapters above). In other words, some activities could be adapted and modified based on a quick assessment of the new baseline at project inception.
7. Gender mainstreaming and social and environmental safeguards are incorporated to the entire project design and implementation to ensure social inclusivity, gender equity and environmental protection.
8. The regional programmatic approach, with a regional chapeau project, facilitates access and sharing of expertise, best practices, lessons learned, training materials as south south cooperation etc., *inter alia* through a pool of experts, AMP Communities of Practice, a regional digital platform, etc.

Sustainability

The project sustainability is safeguarded through the active participation of the private sector to establish viable and innovative business models in the minigrid industry in Niger. By seeking durable profitability, private sector players will ensure the activities continue well after the end of the project. The market-based approach around pilot projects (Component 2) along with technical and financial assistance provided by the GEF and its partners (including co-financiers) will highly contribute to the sustainability of the project outcomes and objective. The replicability plan (including an investment plan under Output 4.7.) developed during project implementation will further support the project's exit strategy and continuity. Hands-on capacity building (technical and managerial) of private sector players (developers, solar installers and other energy services providers as well as businesses using energy for productive uses) and communities (including youth and women rural electricians) is another lever contributing to sustainability. The Minigrid Funding Facility and its various innovative financial mechanisms from various financial institutions and potential donors is designed to last.

The conducive environment created by a suitable policy and regulatory framework and the relevant institutional capacities around minigrids will also support the market scale-up and durability at national level.

At institutional level, the creation of a structured national RE services providers association (with industry associations, such as AMDA) as well as the National Dialogue platform should be able to survive after the project because of its utility. The platform shall be integrated at the Ministry of Energy and Renewable Energies (Responsible Party of Project Component 1).

The structural anchorage of RE and electrification at lower cost and increased reliability as priorities for the GoN and its various strategies and plans at national level as well as in international conventions signed by Niger should also enable the long-term thinking and implementation of minigrids as a solution for universal, reliable, affordable and green energy access across the country.

Potential for scaling-up

Scaling-up the minigrids market, especially through private sector investment and innovative business models, is the main objective of AMP. As such the potential for scaling-up has been thought through during project formulation phase (PIF and PPG) and a holistic scaling-up plan is elaborated below.

Figure 8 - Scaling-up approach



- *Vertical scaling up* is envisaged in the context of actions targeting institutionalized capacity building at policy, political, legal, regulatory, and budgetary planning levels as well as via the creation and operationalization of a functioning private sector association for RE service providers in Niger. Component 1 envisages creating a National Dialogue around minigrids and rural electrification, which will facilitate synergies and common action through a multi-stakeholder dialogue, a cross-sectoral approach as well as offering targeted needs-based capacity building. In Component 2, vertical scale-up is ensured by structuring the private sector along the RE value chain by establishing an umbrella association of all these players. Dissemination and advocacy will be ensured (Component 4) and will provide the necessary scale-up effect.
- *Horizontal scale-up* – expansion/ replication – will be fostered and promoted especially through the support to innovative and viable business models, and CEMG pilot projects considering cost-reduction and energy for productive use (commercial and social). Thanks to capacity building, education and large-scale public awareness campaigns, the outreach and impact will be bigger and contribute to the expansion and replication of demonstrated business models and pilot projects. Awareness raising, capacity building, technical and financial assistance of businesses along with the catalyzing role of the private sector will facilitate the replication of the innovative business models and pilot projects to other geographical areas and to other communities in Niger. The replication plan (output 4.7.) will crystalize and support market development and business model replication.
- *Diversification* in the scaling-up potential is also sought by the project. Business models that are evaluated to be viable would be promoted in the context of the project strategy for diversification or functional scaling-up. For example, adding energy efficiency measures and appliances was discussed. Continuous improvement and additional innovations will be further added in the context of the pilot projects' implementation (Component 2) as well as best practices and knowledge sharing with AMP Communities of Practice or any other knowledge network, trainings and community outreach activities (Component 4).
- *Spontaneous scaling-up* may also be realized as part of the direct project results and beyond, and these will be documented through the knowledge management actions under Component 4. Therefore, as part of project implementation, the scaling-up strategy and its roll-out will be drawn. The project will also support systematic monitoring of planned activities. It will, based on achieved results, document the profiles of minigrid business models which could be promoted for scaling-up, and to be included in the replication plan (Output 4.7.).

V. PROJECT RESULTS FRAMEWORK

This project will contribute to the following Sustainable Development Goal (s):

- SDG7: Ensure access to affordable, reliable, sustainable and modern energy for all
  - SDG 7.1 By 2030, ensure universal access to affordable, reliable and modern energy services
  - SDG 7.2 By 2030, increase substantially the share of renewable energy in the global power mix
- SDG13: Take urgent action to combat climate change and its impacts;
- SDG5: Achieve gender equality and empower all women and girls

This project will contribute to the following country outcome (UNDAF[31]/CPD[32], RPD[33]):

UNDAF Outcome : By end of 2022, the target rural populations will have access to innovative supply chains that generate decent jobs, participate in crisis and disaster management mechanisms related to food and nutrition insecurity and sustainable management of natural and energy resources adapted to climate change
By 2022, targeted rural populations have access to innovative value chains for decent job creation, participate in preventive mechanisms for the management of food and nutrition insecurity crises and disasters and the sustainable management of natural resources and energy adapted to climate change.

- *Output 1.2. : Vulnerable communities in targeted rural areas, particularly women and youth, are empowered to sustainably manage natural resources by improving access to renewable energy and alternative technologies Vulnerable communities in targeted rural areas, particularly women and youth, have required capacities to sustainably manage natural resources with increased access to renewable energy and alternative technologies.*

	Objective and Outcome Indicators (no more than a total of 20 indicators)	**Baseline**	**Mid-term Target**	**End of Project Target**
Project Objective:	Supporting access to clean energy by increasing the financial viability, and promoting scaled-up commercial investment, in renewable energy minigrids in Niger with a focus on cost-reduction levers and innovative business models.			
	<u>*Mandatory GEF Core Indicators*</u> ***Indicator 1:* Greenhouse gas emissions mitigated** Units of measure: metric tons of carbon dioxide equivalent (tCO2e)	Zero, since the project has not yet started	Zero, since the project pilot(s) have not yet been commissioned	Direct: 17,015 tCO2e Indirect: 2,052,000 tCO2e
	***Indicator 2:* Number of direct beneficiaries benefitting from energy access via minigrids, disaggregated by gender and by customer segment (residential, social, commercial/productive use) as co-benefit of GEF investment** Units of measure: number of people	Zero, since the project has not yet started	Zero, since the project pilot(s) have not yet been commissioned	13,534 people (of which 50% women) ------- 13,250 people (residential) 44 people (social) 240 people (commercial/PUE) ------- 2,650 connections (residential) 11 connections (social)

[31] United Nations Development Assistance Frameworks (UNDAF)
[32] Country Programme Document (CPD)
[33] Regional Programme Document (RPD)

				80 connections (commercial/PUE) 2,741 connections (total)
	Indicator 3: **Increase in installed solar PV capacity and battery storage** *Units of measure: MW(solar PV) and MWh (battery storage)*	Zero, since the project has not yet started	Zero, since the project pilot(s) have not yet been commissioned	Solar PV: 0.334 MW Battery storage: 0.814 MWh
	Indicator 4 Local residents trained in different aspects of minigrid development and operation (e.g. sales, distribution, operations, management) disaggregated by gender. *Units of measure: Absolute number of people employed*	Zero, since the project has not yet started	Female: 80 [people] Male: 60 [people] Total: 140 [people]	Female: 140 [people] Male: 100 [people] Total: 240 [people]
Project Component 1	**Policy and Regulation**			
Outcome 1 ***Stakeholder ownership in a national minigrid delivery model is advanced, and appropriate policies and regulations are adopted to facilitate investment in low-carbon minigrids***	*Indicator 5:* A minigrid delivery model to enable minigrid development is endorsed/adopted by the national government through a consultative process involving key stakeholders (e.g. relevant ministries, local authorities, rural populations, private sector, media, etc.) Units of measure: binary (1/0)	Zero, since the project has not yet started	Multi-stakeholder, national dialogue platform on minigrid delivery models established and active.	At least one minigrid delivery model is identified and endorsed by the government through the work of the multi-stakeholder platform and dialogue.
	Indicator 6: Number of policy derisking instruments for minigrid investments - whose development has been supported by the project - are endorsed/adopted by the national government Units of measure: Absolute number of policy derisking instruments	DREI analysis finalized in summer 2022 hence no derisking instruments implemented yet	3 policy derisking instrument ------- Promotion of productive use of energy National industry association created and operational Technical assistance provided to financial institutions	6 policy derisking instrument(s) ------- Public programmes developed for competitive, skilled labour market in minigrids Standards developed for equipment Access to finance for end-users (ancillary products etc.) Promotion of productive use of energy National industry association created and operational Technical assistance provided to financial institutions
Outputs to achieve Outcome 1	Output 1.1: An inclusive national dialogue to identify minigrid delivery models is facilitated, clarifying priority interventions for an integrated approach to off-grid electrification Output 1.2: A minigrid regulatory framework, including tariff model, tax regime, and grid expansion risk, is developed in close coordination with the			

	authorities concerned and other development partners Output 1.3: Analysis of existing (pre)-feasibility studies conducted for selected minigrid sites to enhance sector planning and decision-making on a delivery model for minigrid development Output 1.4: Targeted policy and financial derisking instruments of the Minigrid DREI techno-economic analyses implemented and contributing to AMP Flagship Report on Cost Reduction Output 1.5: Capacity building provided to public officials (regulator, ministries) to support cost-reduction levers and innovative business models Output 1.6: Domestication of quality standards and norms for solar minigrid components, and institutional capacity of ANERSOL strengthened Output 1.7: Public programmes (apprenticeships, certificates, university programs) to develop competitive, skilled labour market in minigrids facilitated			
Project Component 2	Business Model Innovation with Private Sector			
Outcome 2 Innovative business models based on cost reduction are operationalized, with strengthened private sector participation in renewable energy minigrid development	*Indicator 7:* Minigrid pilots implemented that demonstrate a delivery model, cost-reduction measure(s) and/or productive use of electricity Units of measure: binary (1/0)	Some pilot projects have been implemented under ANPER's supervision/collaboration with other partners. However, not yet by the projects (0)	The project's detailed design plan (the 'Minigrid Pilot Plan') for advancing the minigrid pilots is developed and cleared by UNDP and the Project Board. (1) Any project tendering process, as applicable, for minigrid pilots is launched. (1)	100% of the planned minigrid pilots, as identified in the project's Minigrid Pilot Plan, are commissioned. (1)
	Indicator 8: Capacity of minigrid developers and/or operators and communities is enhanced Units of measure: binary (1/0) Units of measure: binary (1/0)	Some capacity building efforts have been undertaken with some local minigrid developers and operators but some capacities need to be reinforced and additional ones are lacking (e.g. operations, maintenance)	Planned capacity building activities for year 1 and 2 are implemented. (1) The capacity of targeted recipients is assessed by survey towards the end of year 2. On a scale of 1 to 5, an average score of at least 2 is achieved. - 1 represents a low level of capacity - 5 represents a strong capacity to understand relevant issues and apply knowledge and skills to find effective solutions. (1)	Planned capacity building activities for year 3 and 4 are implemented. (1) The capacity of targeted recipients is assessed by survey towards the end of the project. On a scale of 1 to 5, an average score of at least 4 is achieved. - 1 represents a low level of capacity - 5 represents a strong capacity to understand relevant issues and apply knowledge and skills to find effective solutions. (1)
Outputs to achieve Outcome 2	Output 2.1 Pilots developed, including on productive use/innovative appliances and modular hardware/system design, leading to cost-reduction in minigrids Output 2.2 National report on opportunities to boost economic activities through electricity access and productive use Output 2.3 Capacities of private minigrid developers and communities are strengthened Output 2.4 Support provided to establish and grow a national industry association for private sector developers			
Project Component 3	Scaled-up Financing			
Outcome 3 Financial sector actors are ready	*Indicator 9:* Capacity of financial institutions is enhanced through training, knowledge sharing, and/or	Local financial institutions barely benefited so far from capacity building to	Planned capacity building activities for year 1 and 2 are implemented. (1)	Planned capacity building activities for year 3 and 4 are implemented. (1)

to invest in a pipeline of low-carbon minigrids and concessional financial mechanisms are in place to incentivize scaled-up investment.	awareness raising events aimed at increasing the financial sector's capacity to evaluate investments in minigrids. Units of measure: binary (1/0)	better understand and invest in CEMGs	The capacity of targeted recipients is assessed by survey towards the end of year 2. On a scale of 1 to 5, an average score of at least 2 is achieved. - 1 represents a low level of capacity - 5 represents a strong capacity to understand relevant issues and apply knowledge and skills to find effective solutions. (1)	The capacity of targeted recipients is assessed by survey towards the end of the project. On a scale of 1 to 5, an average score of at least 4 is achieved. - 1 represents a low level of capacity - 5 represents a strong capacity to understand relevant issues and apply knowledge and skills to find effective solutions. (1)
	Indicator 10: Number of government- or impact investor-supported financing mechanisms offering concessional finance for low-carbon minigrids Units of measure: binary (1/0)	Zero, since the project has not yet started	At least one complementary funding instrument is designed and operational. (1)	At least one complementary funding instrument is designed and operational. (1)
Outputs to achieve Outcome 3	Output 3.1 Support financing mechanisms to scale-up RE minigrids investment Output 3.2. Domestic financial sector capacity-building on business and financing models for minigrids Output 3.3.: Replication plan (including investment plan) for scaling up rural energy access developed			
Project Component 4	Digital, Knowledge Management and Monitoring and Evaluation			
Outcome 4 **Digital, Knowledge Management and Monitoring & Evaluation**	*Indicator 11:* A project digital strategy is prepared and implemented by the PMU to contribute to project implementation and local minigrid market development. *Units of measure: binary (1/0)*	Not applicable, since the project has not yet started	The project digital strategy is developed and being implemented. (1)	The project digital strategy is implemented. (1) Recommendations for rolling out digital solutions for minigrids at national level have been shared with key national stakeholders. (1)
	Indicator 12: Number of minigrid pilots sharing data on minigrid performance with the regional project and other stakeholders following best practices and received from the AMP Regional Project. *Units of measure: binary (1/0)*	Zero, since the project has not yet started	The project's 'Minigrids Digital and Data Management Platform' is procured and operational, ready for data collection from the project's minigrid pilot(s), and for data sharing with the AMP regional project's digital platform. (1)	100% of the planned minigrid pilots, as identified in the project's Minigrid Pilot Plan, are collecting and sharing data with the project's digital platform (1)
Outputs to achieve Outcome 4	Output 4.1 A project digital strategy is developed and implemented, including linkages to and following guidance from the AMP Regional Project Output 4.2 A 'Minigrids Digital and Data Management Platform' is implemented to run tenders and manage data from pilots, and to support minigrids scale-up and cost-reduction Output 4.3: A Quality Assurance and Monitoring Framework for measuring, reporting and verification of the sustainable development impacts of all minigrids			

	pilots supported, including GHG emission reductions, is adopted and operationalized based on standardized guidance from the regional project Output 4.4: Engage with regional project, including, but not limited to, via (i) participating in Communities of Practice and (ii) capturing and sharing lessons learned Output 4.5. Awareness raising campaigns, including lessons learned, are developed and disseminated at all levels nationally (including intervention zones) and with the regional project
Project Component 5	*Monitoring and Evaluation*
Outputs under Component 5	Output 5.1: M&E and Reporting, including (i) Conducting inception workshop and preparing report, (ii) Ongoing M&E, (iii) Mid Term Evaluation and (iv) Terminal Evaluation

VI. MONITORING AND EVALUATION (M&E) PLAN

Project-level monitoring and evaluation will be undertaken in compliance with UNDP requirements as outlined in the UNDP POPP (including guidance on GEF project revisions) and UNDP Evaluation Policy **The UNDP Country Office is responsible for ensuring full compliance with all UNDP project M&E requirements including project monitoring, UNDP quality assurance requirements, quarterly risk management, and evaluation requirements.**

Additional mandatory GEF-specific M&E requirements will be undertaken in accordance with the GEF Monitoring Policy and the GEF Evaluation Policy and other relevant GEF policies[34]. The M&E plan and budget included below will guide the GEF-specific M&E activities to be undertaken by this project.

In addition to these mandatory UNDP and GEF M&E requirements, other M&E activities deemed necessary to support project-level adaptive management will be agreed – including during the Project Inception Workshop - and will be detailed in the Inception Report.

Minimum project monitoring and reporting requirements as required by the GEF:

1. Inception Workshop and Report: A project inception workshop will be held within 2 months from the First disbursement date, with the aim to:
    a. Familiarize key stakeholders with the detailed project strategy and discuss any changes that may have taken place in the overall context since the project idea was initially conceptualized that may influence its strategy and implementation.
    b. Discuss the roles and responsibilities of the project team, including reporting lines, stakeholder engagement strategies and conflict resolution mechanisms.
    c. Review the results framework and monitoring plan.
    d. Discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E budget; identify national/regional institutes to be involved in project-level M&E; discuss the role of the GEF OFP and other stakeholders in project-level M&E.
    e. Update and review responsibilities for monitoring project strategies, including the risk log; SESP report, Social and Environmental Management Framework (where relevant) and other safeguard requirements; project grievance mechanisms; gender strategy; knowledge management strategy, and other relevant management strategies.
    f. Review financial reporting procedures and budget monitoring and other mandatory requirements and agree on the arrangements for the annual audit.
    g. Plan and schedule Project Board meetings and finalize the first-year annual work plan. Finalize the TOR of the Project Board.
    h. Formally launch the Project.

2. GEF Project Implementation Report (PIR): The annual GEF PIR covering the reporting period July (previous year) to June (current year) will be completed for each year of project implementation. UNDP will undertake quality assurance of the PIR before submission to the GEF. The PIR submitted to the GEF will be shared with the Project Board. UNDP will conduct a quality review of the PIR, and this quality review and feedback will be used to inform the preparation of the subsequent annual PIR.

3. GEF Core Indicators: The GEF Core indicators included as Annex will be used to monitor global environmental benefits and will be updated for reporting to the GEF prior to MTR and TE. Note that the project team is responsible for updating the indicator status. The updated monitoring data should be shared with MTR/TE consultants prior to required evaluation missions, so these can be used for subsequent ground truthing. The methodologies to be used in data collection have been defined by the GEF and are available on the GEF website.

[34] See https://www.thegef.org/gef/policies_guidelines

Box 8 - GEF-7 Core Indicators

As reflected in the Results Framework, the project contributes to the following GEF-7 Core Indicators:

Core indicator 6: Greenhouse Gas Emissions Mitigated captures the amount of GHG emissions expected to be avoided through the GEF project's investment in renewable energy minigrids. It should be measured above a baseline value. Mitigation benefits include:

Direct emissions reductions attributable to the investments made during the project's supervised implementation period, totaled over the respective lifetime of the investments.

Indirect emissions reductions that could result from a broader adoption of the outcomes of a GEF project plus longer-term emission reductions from behavioral change in the post-project period. Broader adoption of a GEF project proceeds through several processes including sustaining, mainstreaming, replication, scaling-up and market change.

Context Sub-indicator 6.4: Increase in installed renewable energy capacity per technology captures the increase in renewable energy generation or storage capacity and should be disaggregate by type of renewable energy technology (biomass, geothermal, ocean, small hydro, solar photovoltaic, solar thermal, wind power, and storage).

Core indicator 11: Number of direct beneficiaries disaggregated by gender as co-benefit of GEF investment captures the total number of direct beneficiaries including the proportion of women beneficiaries. Direct beneficiaries are all individuals receiving targeted support from a given project.

4. *Independent Mid-term Review (MTR):* The terms of reference, the review process and the final MTR report will follow the standard UNDP templates and UNDP guidance for GEF-financed projects available on the UNDP Evaluation Resource Center (ERC). The evaluation will be 'independent, impartial and rigorous'. The evaluators that UNDP will hire to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project under review. The GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the evaluation process. Additional quality assurance support is available from the BPPS/NCE-VF Directorate. The final MTR report and MTR TOR will be publicly available in English and will be posted on the UNDP ERC by December 2023. A management response to MTR recommendations will be posted in the ERC within six weeks of the MTR report's completion.

5. Terminal Evaluation (TE): An independent terminal evaluation (TE) will take place upon completion of all major project outputs and activities. The terms of reference, the evaluation process and the final TE report will follow the standard templates and guidance for GEF-financed projects available on the UNDP Evaluation Resource Center. TE should be completed 3 months before the estimated operational closure date, set from the signature of the ProDoc and according to the duration of the project. Provisions should be taken to complete the TE in due time to avoid delay in project closure. Therefore, TE must start no later than 6 months to the expected date of completion of the TE (or 9 months prior to the estimated operational closure date). The evaluation will be 'independent, impartial and rigorous'. The evaluators that UNDP will hire to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project being evaluated. The GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the terminal evaluation process. Additional quality assurance support is available from the BPPS/NCE-VF Directorate. The final TE report and TE TOR will be publicly available in English and posted on the UNDP ERC by September 2026. A management response to the TE recommendations will be posted to the ERC within six weeks of the TE report's completion.

6. Final Report: The project's terminal GEF PIR along with the terminal evaluation (TE) report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the Project Board during an end-of-project review meeting to discuss lessons learned and opportunities for scaling up.

7. Agreement on intellectual property rights and use of logo on the project's deliverables and disclosure of information: To accord proper acknowledgement to the GEF for providing grant funding, the GEF logo will appear together with the UNDP logo on all promotional materials, other written materials like publications

developed by the project, and project hardware. Any citation on publications regarding projects funded by the GEF will also accord proper acknowledgement to the GEF. Information will be disclosed in accordance with relevant policies notably the UNDP Disclosure Policy[35] and the GEF policy on public involvement[36].

8. M&E linkages to the AMP Regional Project

Box 9 - M&E linkages to the AMP Regional Project

National AMP Projects will provide on an annual basis (and to the extent feasible if requested on an ad-hoc basis) the following M&E information to the AMP regional project staff:

Standard reporting on all indicators in the results framework for aggregation and reporting to GEF (by the regional project) on the impacts of all participating national projects for the program as a whole; and
Reporting on any and all additional Key Performance Indicators (KPIs) adopted by the project under the common M&E framework.

The **AMP Regional Project** will provide support and guidance to the AMP National Projects for conducting M&E activities as follows:

Ongoing project monitoring. The AMP Regional Project PMU will:

Develop a 'common monitoring and evaluation (M&E) framework' against which GHG emission reductions and broader SDG impacts and program objectives can be measured, and work closely with national child projects to ensure operationalization and harmonization.

Provide support to National Project PMUs for updating 'key project planning instruments' at least on an annual basis as required to comply with UNDP project monitoring, quality assurance, and risk management requirements, and ensure adequate project planning and adaptive management. This may entail developing common templates for 'key project planning instruments'.

Review and provide feedback on reports submitted by the national project PMUs seeking to continuously improve the quality and ease of reporting by national projects.

Aggregate M&E data from all national projects, including Results Framework and all additional Key Performance Indicators (KPIs) adopted by the project under the common M&E framework, and report back to GEF at the program level.

Evaluations (MTR and TE). The AMP Regional Project PMU will:

Make available to national projects standardized terms of reference for MTR and TE as well as a roster of vetted evaluation consultants.

Review and provide feedback on terms of reference and draft evaluation reports shared by the project PMU to ensure project-level evaluation will be undertaken in compliance with UNDP requirements.

Make themselves available for interviews and consultation in the context of national project mid-term and terminal evaluations.

Monitoring Plan: The project results, corresponding indicators and mid-term and end-of-project targets in the project results framework will be monitored by the Project Management Unit annually, and will be reported in the GEF PIR every year, and will be evaluated periodically during project implementation. If baseline data for some of the results indicators is not yet available, it will be collected during the first year of project implementation. Project risks, as outlined in the risk register, will be monitored quarterly.

Table 7 – Monitoring Plan

[35] See http://www.undp.org/content/undp/en/home/operations/transparency/information_disclosurepolicy/

[36] See https://www.thegef.org/gef/policies_guidelines

Results Monitoring	Indicators	Mid-term Target	End of Project Target	Description of indicators and targets	Data source/Collection Methods[37]	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
Project objective from the results framework		Supporting access to clean energy by increasing the financial viability, and promoting scaled-up commercial investment, in renewable energy minigrids in Niger with a focus on cost-reduction levers and innovative business models.							
Objective Indicators	*Indicator 1: Greenhouse gas emissions mitigated* *Units of measure: metric tons of carbon dioxide equivalent (tCO2e)*	*Zero, since the project pilot(s) have not yet been commissioned*	*Direct: 17,015 tCO2e* *Indirect: 2,052,000tCO2e*	*Direct: cumulative GHG emission reductions from the minigrid pilots' generation of renewable electricity accruing during the minigrids 20-year technology lifetime, as calculated in Annex 12.* *Indirect: Cumulative GHG emission reductions from the generation of renewable electricity from post-project investments in minigrids as enabled by the broader*	*Baseline: taken as zero relative to the AMP project.* *Direct: Installed capacity and clean energy consumption for pilot projects to be monitored via the digital platform and emissions reductions calculated using the AMP Model (refer Annex 12)*	Annually Reported in the GEF PIR	*PMU, leveraging digital platform.* *Minigrid pilot operators share data via the digital platform.*	*Project's 'Minigrids Digital and Data Management Platform' developed and operationalized under Component 4.*	<u>*Assumptions*</u> *For assumptions and methodology used, determined target values refer to Annex 12.* <u>*Risks*</u> *Development of the pilot projects delayed or not implemented as planned.*

[37] Data collection methods should outline specific tools used to collect data and additional information as necessary to support monitoring. The PIR cannot be used as a source of verification.

Results Monitoring	Indicators	Mid-term Target	End of Project Target	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
				adoption of project outcomes, as calculated in Annex 12. *10% of the indirect GHG impacts calculated at the Niger project level are allocated to the AMP regional project, in line with the apportioning of the overall program budget. To avoid double counting, this 10% is removed from the indirect totals for the Niger project.*					
	Indicator 2: Number of direct beneficiaries benefitting from energy access via minigrids, disaggregated by gender and by	Zero, since the project pilot(s) have not yet been commissioned	2,564 people (of which 50% women) ------- 2,500 people (residential) 10 people (social) 54 people (commercial/PUE)	*The indicator captures the total number of direct beneficiaries from the minigrid pilots disaggregated by gender (male, female)*	*Baseline is taken as zero since the project has not yet started.* *The target is stated as additional*	Annually Reported in the GEF PIR	*PMU, leveraging digital platform.* *Minigrid pilot operators share data*	*Project's 'Minigrids Digital and Data Management Platform' developed and*	<u>*Assumptions*</u> *For assumptions and methodology used, determined target values refer to Annex 12.*

Results Monitoring	Indicators	Mid-term Target	End of Project Target	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
	customer segment (residential, social, commercial/productive use) as co-benefit of GEF investment Units of measure: number of people		2,564 people (total)	*and by customer segment* as calculated in Annex 12.	*beneficiaries who are directly impacted by new and/or improved access to electricity via minigrids across the customer categories defined in Annex 12.*		*via the digital platform.*	*operationalized under Component 4.*	<u>*Risks*</u> *Reach of minigrids not as wide as foreseen, limiting the number of direct beneficiaries.* *Development of the pilot projects delayed or not implemented as planned.*
	Indicator 3: Increase in installed solar PV capacity and battery storage *Units of measure: MW(solar PV) and MWh (battery storage)*	Zero, since the project pilot(s) have not yet been commissioned	Solar PV:0.334MW Battery storage: 0.814MWh	*The project will support investments in solar PV-battery minigrids. This indicator will track the deployment of installed PV capacity, as well as the size of battery storage for meeting peak demand and night-time energy uses.*	*These data will be collected directly from the minigrid developers and operators based on minigrid sizing and solar PV and battery installed capacities.*	Annually Reported in the GEF PIR	*PMU, leveraging digital platform.* *Minigrid pilot operators share data via the digital platform.*	*Project's 'Minigrids Digital and Data Management Platform' developed and operationalized under Component 4.*	<u>*Assumptions*</u> *For assumptions and methodology used, determined target values refer to Annex 12.* <u>*Risks*</u> *Development of the pilot projects delayed or not implemented as planned.*

Results Monitoring	Indicators	Mid-term Target	End of Project Target	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
	Indicator 4 : Local residents trained in different aspects of minigrid development and operation (e.g. sales, distribution, operations, management) disaggregated by gender. *Units of measure: Absolute number of people employed*	Female: 80 [people] Male: 60 [people] Total: 140 [people]	Female: 140 [people] Male: 100 [people] Total: 240 [people]		*Training reports*	Annually Reported in the GEF PIR	*PMU, leveraging digital platform.* *Minigrid pilot operators share data via the digital platform.*	*Project's 'Minigrids Digital and Data Management Platform' developed and operationalized under Component 4.*	*<u>Assumption</u>* *Local resident will to be trained and to work in the energy field (including as an additional income generating activity)* *<u>Risks</u>* *Local residents are not interested* *Training cannot be rolled out due to security reasons or else*
Project Outcome 1	*Indicator 5:* A minigrid delivery model to enable minigrid development is endorsed/adopted by the national government through a consultative process involving key stakeholders (e.g. relevant ministries, local authorities, rural	The project's detailed design plan (the 'Minigrid Pilot Plan') for advancing the minigrid pilots is developed, and cleared by UNDP and the Project Board. (1) Any project tendering	100% of the planned minigrid pilots, as identified in the project's Minigrid Pilot Plan, are commissioned. (1)	This is a binary indicator expressing whether a target has been fulfilled (score 1) or not (score 0).	Progress Report	Annually Reported in the GEF PIR	PMU	*Relevant reform of the policy & regulatory framework*	*<u>Assumption</u>* *National Dialogue's efforts combined with IP and RP's support push toward the definition and adoption of a national delivery model* *<u>Risk:</u> Reform is not passed/approved*

Results Monitoring	Indicators	Mid-term Target	End of Project Target	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
	populations, private sector, media, etc.) Units of measure: binary (1/0)	process, as applicable, for minigrid pilots is launched. (1)							
	Indicator 6: Number of policy derisking instruments for minigrid investments - whose development has been supported by the project - are endorsed/adopted by the national government Units of measure: Absolute number of policy derisking instruments	3 policy derisking instrument ------- Promotion of productive use of energy National industry association created and operational Technical assistance provided to financial institutions	6 policy derisking instrument(s) ------- Public programmes developed for competitive, skilled labour market in minigrids Standards developed for equipment Access to finance for end-users (ancillary products etc.) Promotion of productive use of energy National industry association created and operational Technical assistance	This is a binary indicator expressing whether a target has been fulfilled (score 1) or not (score 0).	*Progress reports*	Annually Reported in the GEF PIR	PMU UNDP	Derisking policy instruments are implemented / in use	*<u>Assumption:</u>* *Relevant stakeholders are engaged and committed in applying these instruments* *<u>Risks:</u>* *Project implementation delays*

Results Monitoring	Indicators	Mid-term Target	End of Project Target	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
			provided to financial institutions						
Project Outcome 2	*Indicator 7*: Minigrid pilots implemented that demonstrate a delivery model, cost-reduction measure(s) and/or productive use of electricity Units of measure: binary (1/0)	The project's detailed design plan (the 'Minigrid Pilot Plan') for advancing the minigrid pilots is developed, and cleared by UNDP and the Project Board. (1) Any project tendering process, as applicable, for minigrid pilots is launched. (1)	100% of the planned minigrid pilots, as identified in the project's Minigrid Pilot Plan, are commissioned. (1)	This is a binary indicator expressing whether a target has been fulfilled (score 1) or not (score 0).	Progress reports Minigrids Digital and Data Management Platform	Every 6 month	PMU Private sector operators	Site visits Data collected by smart meters & platform	<u>*Assumption*</u> *The Minigrid Pilot Plan is relevant, realistic and validated* <u>*Risks:*</u> *Delays or failure of providing the minigrid and its operations*
	Capacity of minigrid developers and/or operators and communities is enhanced Units of measure: binary (1/0)	Planned capacity building activities for year 1 and 2 are implemented. (1) The capacity of	Planned capacity building activities for year 3 and 4 are implemented. (1) The capacity of targeted recipients is assessed by	This is a binary indicator expressing whether a target has been fulfilled (score 1) or not (score 0).	*Training reports*	*Annually*	*Trainers* *PMU*	*Attendance lists* *Satisfaction surveys*	<u>*Assumption*</u> *Beneficiaires are willing to build their capacities* *Their needs are evaluated and provided training is adapted to the latter*

Results Monitoring	Indicators	Mid-term Target	End of Project Target	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
		targeted recipients is assessed by survey towards the end of year 2. On a scale of 1 to 5, an average score of at least 2 is achieved. - 1 represents a low level of capacity - 5 represents a strong capacity to understand relevant issues and apply knowledge and skills to find effective solutions. (1)	survey towards the end of the project. On a scale of 1 to 5, an average score of at least 4 is achieved. - 1 represents a low level of capacity - 5 represents a strong capacity to understand relevant issues and apply knowledge and skills to find effective solutions. (1)						<u>*Risks:*</u> *Training cannot be rolled out due to various reasons including security issues, availability, etc.*
Project Outcome 3	*Indicator 9*: Capacity of financial institutions is enhanced through training, knowledge sharing, and/or awareness raising events aimed at increasing the financial sector's	Planned capacity building activities for year 1 and 2 are implemented. (1) The capacity of targeted recipients is	Planned capacity building activities for year 3 and 4 are implemented. (1) The capacity of targeted recipients is assessed by survey towards the end of the	This is a binary indicator expressing whether a target has been fulfilled (score 1) or not (score 0).	*Training reports*	*Annually*	*Trainers* *PMU*	*Attendance lists* *Satisfaction surveys*	<u>*Assumption*</u> *Beneficiaries are willing to build their capacities* *Their needs are evaluated and provided training is adapted to the latter*

Results Monitoring	Indicators	Mid-term Target	End of Project Target	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
	capacity to evaluate investments in minigrids. Units of measure: binary (1/0)	assessed by survey towards the end of year 2. On a scale of 1 to 5, an average score of at least 2 is achieved. - 1 represents a low level of capacity - 5 represents a strong capacity to understand relevant issues and apply knowledge and skills to find effective solutions. (1)	project. On a scale of 1 to 5, an average score of at least 4 is achieved. - 1 represents a low level of capacity - 5 represents a strong capacity to understand relevant issues and apply knowledge and skills to find effective solutions. (1)						<u>*Risks:*</u> *Training is not really adapted, or participants do not understand the value of it and the industry.*
	Indicator 10: Number of government- or impact investor-supported financing mechanisms offering concessional finance for low-carbon minigrids	At least one complementary funding instrument is designed and operational. (1)	At least one complementary funding instrument is designed and operational. (1)	*The implementation of a minimum of one funding instrument (MFF or equivalent) is designed and operational*					

Results Monitoring	Indicators	Mid-term Target	End of Project Target	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
Project Outcome 4	*Indicator 11:* A project digital strategy is prepared and implemented by the PMU to contribute to project implementation and local minigrid market development. *Units of measure: binary (1/0)*	The project digital strategy is developed and being implemented. (1)	The project digital strategy is implemented. (1) Recommendations for rolling out digital solutions for minigrids at national level have been shared with key national stakeholders. (1)	This is a binary indicator expressing whether a target has been fulfilled (score 1) or not (score 0).	Digital strategy document exists	Yearly (updates)	PMU	Strategy document	
	Indicator 12: Number of minigrid pilots sharing data on minigrid performance with the regional project and other stakeholders following best practices and received from the AMP Regional Project. *Units of measure: binary (1/0)*	*The project's 'Minigrids Digital and Data Management Platform' is procured and operational, ready for data collection from the project's minigrid pilot(s), and for data sharing with the AMP regional project's digital*	*100% of the planned minigrid pilots, as identified in the project's Minigrid Pilot Plan, are collecting and sharing data with the project's digital platform (1)*	*This is a binary indicator expressing whether a target has been fulfilled (score 1) or not (score 0).* *The target for this indicator is for all of the project's minigrid pilots to share data and information with the project and the AMP regional*	*Examination of digital platform and data inputs from minigrid operators*	*Quarterly* *The AMP regional project will aggregate data form all projects* *Reported in DO progress tab of the GEF PIR*	*Project Manager, as well as minigrid operators using the platform.* *AMP Regional Project Manager will confirm ability to collect/aggregate data form the project.*		*It is relatively simple to determine whether this binary indicator has been fulfilled. Assessment of the **quality** of the data collected would require more detailed assessment.*

Results Monitoring	Indicators	Mid-term Target	End of Project Target	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
		platform. (1)		*project, using the 'Minigrids Digital and Data Management Platform'. The data to be shared will be defined in consultations with the minigrid developers and reflected in the Digital Strategy for the project (Component 4).*					
Gender action plan	***Female economic empowerment***	*None*	*60%*	*Percent of female economic actors supported by project with sustained (>1 yr) income increases*	*Survey of program participants*	*Monthly, post intervention, during group/solidarity meetings*	*Project manager, or appointed community member*	*Participant financial diaries, videotaped interviews*	*Risk: Women's income may fluctuate based on seasons, or they may start and stop business activities to give birth or care for family members. Baseline and post-intervention income estimates must account for this.*
	Female decision-making	*None*	*75%*	*Percent of female economic actors*	*Survey of program participants*	*Monthly, post interventi*	*Project Manager, or*	*Videotaped interviews*	*Risk: Without peer-led, agency-based*

Results Monitoring	Indicators	Mid-term Target	End of Project Target	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
				reporting increased frequency of decision-making in home and business		*on, during group/solidarity meetings*	*appointed community member*		*interventions, very few women may report increases in decision-making*
	Female workforce development	*None*	*50%*	*Percent of female project trainees (1.6 & 1.7, 2.1 & 2.3) going on to secure employment or higher education in minigrid-related sector*	*Survey of program participation*	*Annual*	*Project manager*	*Email domain, LinkedIn page, institutional letter, etc.*	*Risk: The program could select and benefit women and youth who were already highly determined to succeed in the energy sector (i.e., limited additionality)*
	Female representation	*30%*	*30%*	*Running percentage of women in project-supported groups, meetings, events, trainings, etc.*	*Membership and attendance records*	*At each occurrence*	*Project manager or delegated individual*	*Signatures, photos, meeting minutes*	*Risk: Smaller or more informal encounters may not be properly recorded*

Table 8 - Monitoring and Evaluation Budget for project execution

GEF M&E requirements to be undertaken by Project Management Unit (PMU)	Indicative costs (US$)	Time frame
Inception Workshop and Report	*3,000 (including PMU work to organize and held the workshop_ Covered by GEF Funds)*	Inception Workshop within 2 months of the First Disbursement
M&E required to report on progress made in reaching GEF core indicators and project results included in the project results framework	*None (part of the PMUs work)*	Annually and at mid-point and closure.
Preparation of the annual GEF Project Implementation Report (PIR)	*4,000 (part of the PMUs work(covered by GEF Funds)*	Annually typically between June-August
Monitoring of environmental and social risks, gender action plan, and corresponding management plans as relevant	*n.a. as under Output 4.3. QAMF*	On-going.
Supervision missions	*20,000 (covered by GEF Funds)*	Annually
Learning missions	*None (part of AMP regional budget)*	As needed
Independent Mid-term Review (MTR):	*30,000 (covered by UNDP)*	1 November 2025
Independent Terminal Evaluation (TE):	50,000 (covered by UNDP)	31 April 2027
Project Closure workshop including TE presentation	5,256 (covered by UNDP)	
TOTAL indicative COST	*112,256 (out of which $27,000 covered by GEF funding)*	*Equivalent to TBWP component (M&E)*

VII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

Section 1: General roles and responsibilities in the projects' governance mechanism

1. Implementing Partner: The Implementing Partner for this project is ANPER.

The Implementing Partner is the entity to which the UNDP Administrator has entrusted the implementation of UNDP assistance specified in this signed project document along with the assumption of full responsibility and accountability for the effective use of UNDP resources and the delivery of outputs, as set forth in this document.

The Implementing Partner is responsible for executing this project. Specific tasks include:

- Project planning, coordination, management, monitoring, evaluation and reporting. This includes providing all required information and data necessary for timely, comprehensive and evidence-based project reporting, including results and financial data, as necessary. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes and is aligned with national systems so that the data used and generated by the project supports national systems.
- Overseeing the management of project risks as included in this project document and new risks that may emerge during project implementation.
- Procurement of goods and services, including human resources.
- Financial management, including overseeing financial expenditures against project budgets.
- Approving and signing the multiyear workplan.
- Approving and signing the combined delivery report at the end of the year; and,
- Signing the financial report or the funding authorization and certificate of expenditures.

2. Responsible Parties:

The Ministry of Energy and Renewable Energies is responsible for implementing component 1. To this end, it is responsible for the implementation of activities related to component 1. All activities will be implemented in close collaboration with the stakeholders concerned by component 1. It will mobilize all the stakeholders concerned by the activities of this component. It reports on the implementation of the component to ANPER through the project management unit. The Ministry will also set up the steering committee, which will be under its chairmanship. Indeed, the Ministry is the administrative tutor of ANPER and has the ability to develop an order to create the project steering committee.

A HACT micro-assessment of the Ministry can be found in Annex 17C attached.

3. Project stakeholders and target groups:

The participatory and inclusive approach adopted by the project since PPG is part of its DNA. Extensive and numerous stakeholder consultations, one-on-one, in groups and at workshops (inception and validation) were carried out at PPG phase and will be pursued during project implementation to ensure ownership and sustainability of the project after closure. Comments, questions, requests and commitments are stimulated at every project stakeholder.

During PPG phase, relevant stakeholders were asked to provide inputs and comments on the project and their specific role in contributing to overcome the development challenge together. This largely contributed to nourishing the project document and adapting it to local realities and needs as much as possible. At the validation workshop and after sharing the final draft of the project document and annexes, feedback was provided by stakeholders and integrated into the final version of the document.

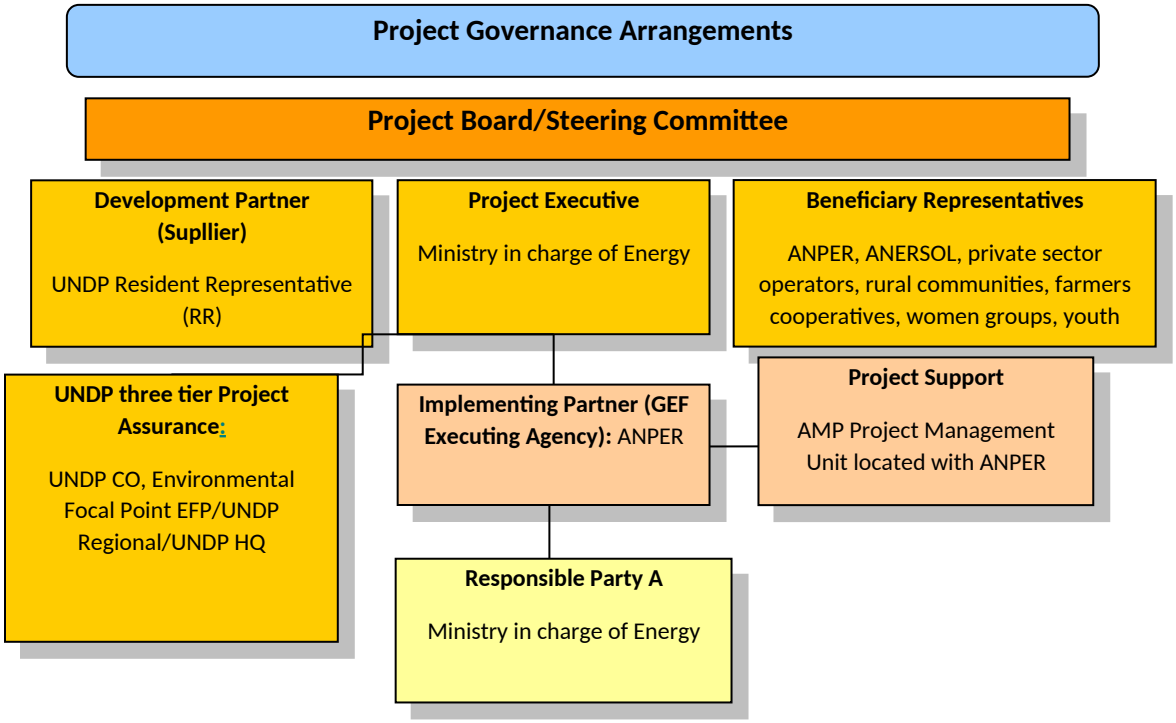
During project implementation, stakeholders and target groups will continue to actively contribute and influence the decision making for the project. Different media will be used to do so:

- Inception workshop at project launch gathering all key stakeholders to present the project and the inception report to facilitate project implementation. Participants will be invited to share their insights and updates to adapt the workplan. The official project launch and main outcomes of the inception workshop will be broadcasted to a larger audience as part of the awareness raising campaign (Component 4).
- The National dialogue platform (Component 1) is a powerful tool to support exchanges between project stakeholders and with the PMU on various topics (especially through sub-committees). This multi-stakeholder media is critical to effectively and efficiently contribute to the development of a nascent minigrids market in Niger. Sub-committees will be created to further facilitate implementation and decision making.
- The Project Board/Steering Committee is the ultimate platform for decision making including beneficiary representatives, the project executive and the development partner (see below). Topics such as workplan, activities and results will be debated, decisions taken together, and necessary corrective measures proposed.
- Continuous stakeholder interactions and consultations through meetings, workshops, trainings, awareness raising campaigns, etc. will also be used to facilitate decision making based on stakeholders' insights.
- Regular progress and monitoring reports that can be disseminated to specific partners and players

4. UNDP: UNDP is accountable to the GEF for the implementation of this project. This includes overseeing project execution undertaken by the Implementing Partner to ensure that the project is being carried out in accordance with UNDP and GEF policies and procedures and the standards and provisions outlined in the Delegation of Authority (DOA) letter for this project. **The UNDP GEF Executive Coordinator, in consultation with UNDP Bureaus and the Implementing Partner, retains the right to revoke the project DOA, suspend or cancel this GEF project.** UNDP is responsible for the Project Assurance function in the project governance structure and presents to the Project Board and attends Project Board meetings as a non-voting member.

Section 2: Project governance structure

Table 9 – Project Governance arrangements



Second line of defense:

- *Regional Bureau oversees RR and Country Office compliance at portfolio level*
- *BPPS NCE RTA oversees technical quality assurance and GEF compliance. BBPS NCE PTA oversees RTA function*
- *UNDP GEF Executive Coordination and Regional Bureau Deputy Director can revoke DOA/cancel/suspens project or provide enhanced oversight*

5. The UNDP Resident Representative assumes full responsibility and accountability for oversight and quality assurance of this Project and ensures its timely implementation in compliance with the GEF-specific requirements and UNDP's Programme and Operations Policies and Procedures (POPP), its Financial Regulations and Rules and Internal Control Framework. A representative of the UNDP Country Office will assume the assurance role and will present assurance findings to the Project Board, and therefore attends Project Board meetings as a non-voting member.

Section 3: Segregation of duties and firewalls vis-à-vis UNDP representation on the project board:

6. As noted in the Minimum Fiduciary Standards for GEF Partner Agencies, in cases where a GEF Partner Agency (i.e. UNDP) carries out both implementation oversight and execution of a project, the GEF Partner Agency (i.e. UNDP) must separate its project implementation oversight and execution duties, and describe in the relevant project document a: 1) Satisfactory institutional arrangement for the separation of implementation oversight and executing functions in different departments of the GEF Partner Agency; and 2) Clear lines of responsibility, reporting and accountability within the GEF Partner Agency between the project implementation oversight and execution functions.

7. In this case, UNDP is only performing an implementation oversight role in the project vis-à-vis our role in the project board and in the project assurance function and therefore a full separation of project implementation oversight and execution duties has been assured.

Section 4: Roles and Responsibilities of the Project Organization Structure:

8. **Project Board:** All UNDP projects must be governed by a multi-stakeholder board or committee established to review performance based on monitoring and evaluation, and implementation issues to ensure quality delivery of results. The Project Board (also called the Project Steering Committee) is the most senior, dedicated oversight body for a project.

9. The two main (mandatory) roles of the project board are as follows:

   1) **High-level oversight of the execution of the project by the Implementing Partner** (as explained in the "Provide Oversight" section of the POPP). This is the primary function of the project board and includes annual (and as-needed) assessments of any major risks to the project, and decisions/agreements on any management actions or remedial measures to address them effectively. The Project Board reviews evidence of project performance based on monitoring, evaluation and reporting, including progress reports, evaluations, risk logs and the combined delivery report. The Project Board is responsible for taking corrective action as needed to ensure the project achieves the desired results.
   2) **Approval of strategic project execution decisions of the Implementing Partner** with a view to assess and manage risks, monitor and ensure the overall achievement of projected results and impacts and ensure long term sustainability of project execution decisions of the Implementing Partner (as explained in the "Manage Change" section of the POPP).

10. **Requirements to serve on the Project Board**:

- ✔ Agree to the Terms of Reference of the Board and the rules on protocols, quorum and minuting.
- ✔ Meet annually; at least once.
- ✔ Disclose any conflict of interest in performing the functions of a Project Board member and take all measures to avoid any real or perceived conflicts of interest. This disclosure must be documented and kept on record by UNDP.
- ✔ Discharge the functions of the Project Board in accordance with UNDP policies and procedures.
- ✔ Ensure highest levels of transparency and ensure Project Board meeting minutes are recorded and shared with project stakeholders.

11. **Responsibilities of the Project Board**:

- ✔ Consensus decision making:
  - The project board provides overall guidance and direction to the project, ensuring it remains within any specified constraints, and providing overall oversight of the project implementation.
  - Review project performance based on monitoring, evaluation and reporting, including progress reports, risk logs and the combined delivery report;
  - The project board is responsible for making management decisions by consensus.
  - In order to ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency and effective international competition.
  - In case consensus cannot be reached within the Board, the UNDP representative on the board will mediate to find consensus and, if this cannot be found, will take the final decision to ensure project implementation is not unduly delayed
- ✔ Oversee project execution:
  - Agree on project manager's tolerances as required, within the parameters outlined in the project document, and provide direction and advice for exceptional situations when the project manager's tolerances are exceeded.
  - Appraise annual work plans prepared by the Implementing Partner for the Project; review combined delivery reports prior to certification by the implementing partner.
  - Address any high-level project issues as raised by the project manager and project assurance;

- Advise on major and minor amendments to the project within the parameters set by UNDP and the donor and refer such proposed major and minor amendments to the UNDP BPPS Nature, Climate and Energy Executive Coordinator (and the GEF, as required by GEF policies);
- Provide high-level direction and recommendations to the project management unit to ensure that the agreed deliverables are produced satisfactorily and according to plans.
- Track and monitor co-financed activities and realization of co-financing amounts of this project.
- Approve the Inception Report, GEF annual project implementation reports, mid-term review and terminal evaluation reports.
- Ensure commitment of human resources to support project implementation, arbitrating any issues within the project.

✔ Risk Management:

- Provide guidance on evolving or materialized project risks and agree on possible mitigation and management actions to address specific risks.
- Review and update the project risk register and associated management plans based on the information prepared by the Implementing Partner. This includes risks related that can be directly managed by this project, as well as contextual risks that may affect project delivery or continued UNDP compliance and reputation but are outside of the control of the project. For example, social and environmental risks associated with co-financed activities or activities taking place in the project's area of influence that have implications for the project.
- Address project-level grievances.

✔ Coordination:

- Ensure coordination between various donor and government-funded projects and programmes.
- Ensure coordination with various government agencies and their participation in project activities.

12. **Composition of the Project Board**: The composition of the Project Board must include individuals assigned to the following three roles:

1. **Project Executive:** This is an individual who represents ownership of the project and chairs (or co-chairs) the Project Board. The Executive usually is the senior national counterpart for nationally implemented projects. The Project Executive is the Director General of ANPER.
2. **Beneficiary Representative(s):** Individuals or groups representing the interests of those groups of stakeholders who will ultimately benefit from the project. Their primary function within the board is to ensure the realization of project results from the perspective of project beneficiaries. Often representatives from civil society, industry associations, or other government entities benefiting from the project can fulfill this role. There can be multiple beneficiary representatives in a Project Board. The Beneficiary representatives include (non-exclusive list that will be adapted at project launch):
    - Representative of the pilot communities (possibly via an association in the community and on a rotational basis)
    - Renewable energy service providers
    - Consumers Association
    - Ministry of Energy and Renewable Energy
    - Ministry of Technical Education and Vocational Training
    - FISAN
    - Ministry of Plan
    - Ministry of Finance
    - Ministry of Promotion of Women and Child Protection
    - ANERSOL
    - Representative of University or High learning school delivering renewable energy courses
    - Representative of NGOs operating in the field of minigrids & CSOs
    - Representative of Financial Institutions
    - Representative of technical and financial partners

3. **Development Partner(s):** Individuals or groups representing the interests of the parties concerned that provide funding, strategic guidance and/or technical expertise to the project. The Development Partner(s) is the UNDP Niger Resident Representative who will ensure the policies of UNDP and the GEF are complied with.

13. **Project Assurance:** Project assurance is the responsibility of each project board member; however, UNDP has a distinct assurance role for all UNDP projects in carrying out objective and independent project oversight and monitoring functions. UNDP performs quality assurance and supports the Project Board (and Project Management Unit) by carrying out objective and independent project oversight and monitoring functions, including compliance with the risk management and social and environmental standards of UNDP. The Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. Project assurance is totally independent of project execution.

14. A designated representative of UNDP playing the project assurance role is expected to attend all board meetings and support board processes as a non-voting representative. It should be noted that while in certain cases UNDP's project assurance role across the project may encompass activities happening at several levels (e.g. global, regional), at least one UNDP representative playing that function must, as part of their duties, specifically attend board meeting and provide board members with the required documentation required to perform their duties. The 2 UNDP representative playing the main project assurance function are the Deputy Resident Representative - representing the management side at UNDP Niger- and the CO programme officer - representing the relevant technical expertise around the project at UNDP Niger.

15. **Project Management – Execution of the Project:** The Project Manager (PM) (also called project coordinator) is the senior most representative of the Project Management Unit (PMU) and is responsible for the overall day-to-day management of the project on behalf of the Implementing Partner, including the mobilization of all project inputs, supervision over project staff, responsible parties, consultants and sub-contractors. The project manager typically presents key deliverables and documents to the board for their review and approval, including progress reports, annual work plans, adjustments to tolerance levels and risk registers.

    A designated representative of the PMU is expected to attend all board meetings and support board processes as a non-voting representative.

    The primary PMU representative attending board meetings is the Project Manager supported where needed by the Project Coordinator or the Administrative Assistant.

VIII. FINANCIAL PLANNING AND MANAGEMENT

1. The total cost of the project is USD 138,358,393. This is financed through a direct contribution to the project from GEF grant of USD 1,601,376 administered by UNDP and additional support of USD 300,000 in cash co-financing from UNDP with USD 136,457,017in cash and in-kind from development partners through ANPER AfDB and USD 1,276,171[38] from SIDA. UNDP, as the GEF Implementing Agency, is responsible for the oversight of the GEF resources and the cash co-financing transferred to UNDP bank account only.

2. **Co-financing**: The actual realization of project co-financing amounts will be monitored by the UNDP Country Office and the PMU on an annual basis in the GEF PIF and will be reported to the GEF during the mid-term review and terminal evaluation process as follows:

Table 10 - Project Co-financing

Co-financing source	Co-financing type	Co-financing amount	Contributing to project results (y/n)
ANPER	grant	USD 135,104,356	n
ANPER	in-kind	USD 76,490	n
SIDA	grant	USD 1,276,171	n
UNDP	Grant	USD 300,000	y
TOTAL		**USD 136,757,017**	

3. **Budget Revision and Tolerance**: As per UNDP POPP, the project board may agree with the project manager on a tolerance level for each detailed plan under the overall multi-year workplan. The agreed tolerance should be written in the project document or approved project board meeting minutes. It should normally not exceed 10 percent of the agreed annual budget at the activity level, but within the overall approved multi-year workplan at the activity level. Within the agreed tolerances, the project manager can operate without intervention from the project board. Restrictions apply as follows:

   Should the following deviations occur, the Project Manager/IP through UNDP Country Office will seek the approval of the BPPS/NCE-VF team to ensure accurate reporting to the GEF. It is **strongly encouraged** to maintain the expenditures within the approved budget at the budgetary account and at the component level:

   a) Budget reallocations must prove that the suggested changes in the budget will not lead to material changes in the results to be achieved by the project. A strong justification is required and will be approved on an exceptional basis. Budget re-allocations among the components (including PMC) of the approved Total Budget and Work Plans (TBWP) that represent a value greater than 10% of the total GEF grant.
   b) Introduction of new outputs/activities (i.e. budget items) that were not part of the agreed project document and TBWP that represent a value greater than 5% of the total GEF grant. The new budget items must be eligible as per the GEF and UNDP policies.
   c) Project management cost (PMC): budget under PMC component is capped and cannot be increased.

   UNDP is not in a position to increase the total budget above the amount approved by the donor, therefore any over-expenditure would have to be absorbed from non-GEF resources by the Implementing Partner (GEF Executing Entity)

[38] This amount is the portion remaining for the duration left till project completion in 2024 for the SIDA Liptako Gourma rural electrification project USD 2,552,342

4. **Project extensions:** The UNDP-BPPS-NCE team Executive Coordinator must approve all requests for extension of the Project Completion Date and for other milestone extensions with hard deadlines. All extensions impose additional time and cost burdens at all levels and the GEF project budget cannot be increased beyond its originally approved amount. A single extension may be granted on an exceptional basis and subject to the conditions and maximum durations set out in the UNDP POPP. The project management costs during the extension period must remain within the originally approved amount, and any increase in PMC costs shall be covered by non-GEF resources; the additional UNDP oversight costs during the extension period must be covered by non-GEF resources, in accordance with UNDP's policy as set out in UNDP POPP.

   For any extension request, UNDP CO and IP will consult and jointly present a clear plan indicating how and from which specific sources the additional oversight costs that will be incurred by UNDP will be covered during the extended period. The BPPS-NCE Executive Coordinator will consult the Regional Bureaux (RBX) and may reject the extension request if no (external co-financing by the IP or internal UNDP CO resources) can be identified.

   All extension requests, along with all supporting documentation, shall be submitted by the IP to the UNDP CO in line with the requirements and within the deadlines set out in the UNDP SOPs and policies in UNDP POPP.

5. **Audit:** The project will be audited as per UNDP Financial Regulations and Rules and applicable audit policies. Audit cycle and process must be discussed during the Inception workshop. If the Implementing Partner is an UN Agency, the project will be audited according to that Agencies applicable audit policies.

6. **Transfer or disposal of assets**: In consultation with the Implementing Partner and other parties of the project, UNDP is responsible for deciding on the transfer or other disposal of assets. Transfer or disposal of assets is recommended to be reviewed and endorsed by the project board following UNDP rules and regulations. Assets may be transferred to the government for project activities managed by a national institution at any time during the life of a project, however **must be done before the operational closure date**. In all cases of transfer, a transfer document must be prepared and kept on file[39]. The transfer should be done before Project Management Unit complete their assignments.

7. **Completion Date:** The project completion date is the date of Project Document Signature plus project duration. This date can only be extended through a formal extension request. Prior to completion date, all UNDP-financed inputs must be provided and related activities for the Project completed. No activities, except for the final clearance of the Terminal Evaluation Report and the corresponding management response and the end-of-project review Project Board Meeting should take place after the Completion Date.

8. **Project Closure**: Project closure will be conducted as per UNDP requirements outlined in the UNDP POPP. All costs incurred to close the project must be included in the project closure budget and reported as final project commitments presented to the Project Board during the final project review. The only costs a project may incur following the final project review are those included in the project closure budget.

   - **Operational Closure**: **Operational closure must happen within 9 months from project completion date.** Prior to operational closure, the Terminal Evaluation must have been submitted and the corresponding TE management response and the end-of-project review Project Board meeting must have been completed. The Implementing Partner through a Project Board decision will notify the UNDP Country Office when operational closure has been completed. Before Operational Closure, the project must have completed the transfer or disposal of any equipment that is still the property of UNDP.
   - **Financial Closure: Financial closure must happen within 6 months of operational closure or after the date of cancellation.** The project will be financially closed when the following conditions have been met: a) the project is operationally completed or has been cancelled; b) the Implementing Partner has reported all financial transactions to UNDP; c) UNDP has closed the accounts for the project; d) UNDP and the

[39] See https://popp.undp.org/ https://popp-prod.acquia.undp.org/policy-page/close-and-transition.

Implementing Partner have certified a final Combined Delivery Report (which serves as final budget revision).
Between operational and financial closure, the implementing partner will identify and settle all financial obligations and prepare a final expenditure report. The UNDP Country Office will send the final signed closure documents including confirmation of final cumulative expenditure and unspent balance to BPPS/NCE for confirmation before the project will be financially closed in Quantum by the UNDP Country Office.

9. **Cancellation and Suspension:** All projects considering going through cancellation or suspension must follow UNDP and GEF requirements. Guidance can be found in the UNDP POPP (SOPs for management actions of Vertical Fund projects escalated to the Executive Coordinator and Guidance for GEF project revisions).

10. **Refund to GEF:** Should a refund of unspent funds to the GEF be necessary, this will be managed directly by the BPPS/NCE team Directorate in New York. No action is required by the UNDP Country Office on the actual refund from UNDP project to the GEF. Unspent project balance is not permitted to be transferred to any other projects.

IX. TOTAL BUDGET AND WORK PLAN

Total Budget and Work Plan			
Quantum Business Unit	UNDP-Niger		
Quantum Project ID:	00143728	Quantum Project Title:	*As in Quantum*
Quantum Award ID:	00131725	Quantum Award Title:	*As in Quantum*
UNDP-GEF PIMS No.	NER10		
Implementing Partner	*ANPER - Agence Nigérienne de Promotion de l'Electrification en milieu Rural*		

Table 11 – AMP Niger Budget

Atlas Activity (GEF Component)	**Atlas Implementing Agent (Responsible Party[2] , IP or UNDP)**	**Atlas Fund ID**	**Donor Name**	**Atlas Budgetary Account Code [3]**	**ATLAS Budget Account Description[3]**	**Amount Y1 2023**	**Amount Y2 2024**	**Amount Y3 2025**	**Amount Y4 2026**	**Total (USD)**	***See Budg Not***
						(USD)	**(USD)**	**(USD)**	**(USD)**		
Component 1: Policy and Regulation	ANPER	**62000**	**GEF**	71200	International Consultants	39,050	32,150	-	-	71,200	*1*
				71300	Local Consultants	30,375	20,625	1,500	1,500	54,000	*2*
				72100	Contractual Services-Companies	52,250	1,250	1,250	1,250	56,000	*3*
				71600	Travel	12,049	8,179	-	-	20,228	*4*
				75700	Training, Workshops and Confer	16,750	13,750	4,750	4,750	40,000	*5*
				72300	Materials & Goods	-	6,000	-	-	6,000	*6*

Atlas Activity (GEF Component)	Atlas Implementing Agent (Responsible Party[2	Atlas Fund ID	Donor Name	Atlas Budgetary Account Code [3]	ATLAS Budget Account Description[3]	Amount Y1 2023	Amount Y2 2024	Amount Y3 2025	Amount Y4 2026	Total (USD)	See Budg Not
				74200	Audio Visual&Print Prod Costs	900	2,100	-	-	3,000	7
				sub-total GEF		151,374	84,054	7,500	7,500	**250,428**	
		Total Outcome 1				**151,374**	**84,054**	**7,500**	**7,500**	**250,428**	
Component 2: Business Model Innovation with Private Sector	ANPER	62000	GEF	71200	International Consultants	18,000	6,000	-	-	24,000	8
				71300	Local Consultants	12,000	9,360	9,360	9,360	40,080	9
				72100	Contractual Services-Companies	34,800	4,800	16,801	13,800	70,201	10
				72800	Information Technology Equipmt	7,000	-	800	-	7,800	11
				71600	Travel	10,808	7,625	7,625	7,625	33,683	12
				75700	Training, Workshops and Confer	8,500	4,500	3,000	3,000	19,000	13
				72300	Materials & Goods	6,000	400,000	162,275	-	568,275	14
				72200	Equipment and Furniture	5,144	750	750	750	7,394	15
				74200	Audio Visual&Print Prod Costs	2,000	-	1,000	-	3,000	16
				sub-total GEF		104,252	433,035	201,611	34,535	**773,433**	
		Total Outcome 2				**104,252**	**433,035**	**201,611**	**34,535**	**773,433**	
Component 3: Scaled-up Financing	ANPER	62000	GEF	71200	International Consultants	10,500	5,250	5,250	9,000	30,000	17
				71300	Local Consultants	-	-	-	12,000	12,000	18
				72100	Contractual Services-Companies	42,800	33,900	-	-	76,700	19
				71600	Travel	2,248	2,248	2,248	5,400	12,144	20

Atlas Activity (GEF Component)	Atlas Implementing Agent (Responsible Party[2	Atlas Fund ID	Donor Name	Atlas Budgetary Account Code [3]	ATLAS Budget Account Description[3]	Amount Y1 2023	Amount Y2 2024	Amount Y3 2025	Amount Y4 2026	Total (USD)	See Budg Not
				75700	Training, Workshops and Confer	8,167	4,833	1,500	2,500	17,000	21
				sub-total GEF		63,715	46,231	8,998	28,900	**147,844**	
		Total Outcome 3				**63,715**	**46,231**	**8,998**	**28,900**	**147,844**	
Component 4: Digital, Knowledge Management	ANPER	62000		71300	Local Consultants	17,400	17,400	17,400	17,400	69,600	22
				72100	Contractual Services-Companies	57,000	10,960	9,960	21,960	99,880	23
				72800	Information Technology Equipmt	2,500	1,000	1,000	1,000	5,500	24
				71600	Travel	5,120	20,120	20,120	20,120	65,480	25
				75700	Training, Workshops and Confer	2,500	-	-	-	2,500	26
				74200	Audio Visual&Print Prod Costs	12,000	3,000	3,000	3,000	21,000	27
				sub-total GEF		96,520	52,480	51,480	63,480	**263,960**	
		Total Outcome 4				**96,520**	**52,480**	**51,480**	**63,480**	**263,960**	
Component 5: Monitoring & Evaluation	ANPER	62000	GEF	74200	Audio Visual&Print Prod Costs	1,000	1,000	1,000	1,000	4,000	28
				71600	Travel	5,000	5,000	5,000	5,000	20,000	29
				75700	Training, Workshops and Confer	3,000	0	0	0	3,000	30
				sub-total GEF		9,000	6,000	6,000	6,000	27,000	
		4000	UNDP	71200	International Consultants	-	20,000	-	30,000	50,000	31
				71300	Local Consultants	-	10,000	-	20,000	30,000	32
				75700	Training, Workshops and Confer	-	-	-	5,256	5,256	33
				sub-total UNDP			30,	-	55,256	**85,**	

Atlas Activity (GEF Component)	Atlas Implementing Agent (Responsible Party[2] , IP or	Atlas Fund ID	Donor Name	Atlas Budgetary Account Code [3]	ATLAS Budget Account Description[3]	Amount Y1 2023	Amount Y2 2024	Amount Y3 2025	Amount Y4 2026	Total (USD)	See Budg Not
				[3]		-	000			256	
					Total Outcome 5	**9,000**	**36,000**	**6,000**	**61,256**	**112,256**	
Project Management Cost (PMC)	ANPER	62000	GEF	71300	Local Consultants	28,960	30,616	32,416	34,000	125,992	*34*
				74100	Professional Services	3,000	3,000	3,000	3,000	12,000	*35*
				72500	Supplies	180	180	180	180	720	*36*
					sub-total GEF	32,140	33,796	35,596	37,180	**138,712**	
		4000	UNDP	71300	Local Consultants	23,520	23,880	24,384	24,960	96,744	*37*
				72200	Equipment and Furniture	40,000	0	0	0	40,000	*38*
				72800	Information Technology Equipmt	7,000	0	2,000	0	9,000	*39*
				71500	UN Volunteers	15,000	15,000	15,000	15,000	60,000	*40*
				73400	Rental & Maint of Other Equip	0	3,000	3,000	3,000	9,000	*41*
					sub-total UNDP	85,520	41,880	44,384	42,960	**214,744**	
					Total Project Management	117,660	75,676	79,980	80,140	**353,456**	
					SUB-TOTAL GEF	**457,001**	**655,596**	**311,185**	**177,595**	**1,601,377**	
					UNDP	**85,520**	**71,880**	**44,384**	**98,216**	**300,000**	
					PROJECT TOTAL	**542,521**	**727,476**	**355,569**	**275,811**	**1,901,377**	

Table 12 – Budget Notes

Budget Notes:	
No.	**Component 1: Policy and Regulation**
1	1.1. Senegalese delegation of 2 consultants paid at USD 500/day for 5+2travel days to provide training and advisory for the national dialogue platform 1.2. 25 days for necessary studies based on short assessment under activity 1.2.1. (provision) 1.5. International consultants for knowledge gap analysis, developing comprehensive modular training materials, conducting training and ToT - 40 days 1.6. 30 days to conduct a review (and benchmark) and to develop the relevant standards, 5 days for the preparation & holding of the workshop 1.7. 12 days of Training facilitator/Capacity building specialist to train the trainers of relevant institutions in Niger
2	1.1. $300 per meeting x 20 meetings over 4 years - acting as local coordinator for the National Dialogue Platform 1.2. 35 days for necessary studies based on short assessment under activity 1.2.1. (provision) + 20 days for adapting the PP law to facilitate CEMG investments 1.4. 15 days Selection & implementation of DREI instruments to be covered by AMP 1.5. 60 days national consultants knowledge gap analysis, developing comprehensive modular training materials, conducting training 1.6. 31 days to conduct a review (and benchmark) and to develop the relevant standards, 5 days for the preparation & holding of the workshop
3	1.1. 4 trainings 1.3. Analysis to cluster existing (pre-)-feasibility studies and recommendations 1.7. Local company to conduct study on existing training offer and relevant gaps, as well as recommendations to overcome then
4	1.1. One Exchange visit with Senegal's successful national dialogue platform. 2 people for 5 days (excluding travel time). DSA at USD 187 in Niger in June 2021; USD 1,000 for the travel Dakar-Niamey. No local travel costs for participants located in Niamey but for representatives in relation to pilot sites 1.2. 1 10 day-mission for international consultant 1.5. 2 missions to Niger - 1 for initial phase and ToT, 1 for ToT on Year 2 1.6. 5 day mission for international consultant 1.7. 2 training missions of 4 days each
5	1.1. Various meetings, trainings and workshops for National Dialogue Platform members including workshop with Senegalese delegation for 2 days and various expenses to organize workshops, dissemination/communication materials, etc. 1.2. Stakeholder validation workshop for the regulatory framework and a workshop on customs procedures etc. 1.4. DREI Workshop in Year 1 1.5. 1 training session incl. ToT each year and related training supplies 1.6. 2 workshops around standards 1.7. 2 ToT

6	1.6. Various training equipment incl. solar PV panels etc.
7	1.5. Printing training materials 1.6. Printing training materials & infographics/leaflets on quality standards for users 1.7. Printing training materials
No.	**Component 2: Business Model Innovation with Private Sector**
8	2.3. 40 days - Pedagogical engineering, Training and ToT on business models, cost-reduction levers, new technologies depending on the topics, tariff negotiation, understanding of anchor loads, remote monitoring, etc. all services - several modules ; and knowledge transfer
9	2.3. 100 days - Capacity building for the communities incl. becoming rural electricians (60 days) and for local private sector developers (40 days) with pedagogical engineering, Training and ToT on business models, cost-reduction levers, new technologies depending on the topics, tariff negotiation, understanding of anchor loads, remote monitoring, etc. all services - several modules ; and knowledge transfer 2.4. An assistant (or similar) in charge of the bureau and association, updates the website, organizes meetings etc. / part time for 3 years
10	2.2. Analysis on key national rural economic outputs and their value chains, and social activities incl. field visits on key value chains and stakeholder consultations 1 mission to Niger for the study, field visits inside the country 2.3. Provision for additional capacity building needed for communities/local private sector developers & operators 2.4. Development of the Private Sector RE association and relevant communications strategy and roll-out, and 2Accounting, audit, etc. incl. for funding purposes - 2000 per year
11	2.3. IT hardware and software useful for trainings 2.4. laptop, printers, software for a functioning association office
12	2.3. 1 mission to Niger; local travels on pilot sites 2.4. Travel costs for study tours, and relevant fairs and conferences for 5 people of the association
13	2.2. Half day workshop on the analysis and recommendations to boost economic and social activities through electricity access and productive use 2.3. Training and ToT sessions throught project duration for communities & local private sector developers & operators 2.4. Venue, breaks for meetings & study tour, fees to attend conferences, fairs etc.
14	2.1. CAPEX subsidies for the 4 CEMG pilot sites 2.3. Various training equipment incl. solar PV panels etc.
15	2.4. Furnitures and other goods for the association's office
16	2.2. Printing of the report executive summary 2.3. Printing training materials
No.	**Component 3: Scaled-up Financing**
17	3.2. 35 days to develop and conduct relevant training on business models and innovative finance solutions for national financial institutions 3.3. 15 days of work - remotely on replication plan
18	3.3. 40 days with field visits incl. pilot sites and some eligible communities for replication plan

19	3.1. Conduct a benchmarking of existing financial products supporting access and use of energy in Niger and aborad, and other industries,Identify innovative financing schemes that are relevant and applicable to support the development of the minigrid market in Niger. Selection of 1-2 financing mechanisms and technical assistance around these
20	3.2. 3 visits for 3 trainings in Niger 3.3. Field visits over 15 days incl. vehicle with driver and gas
21	3.1. Conduct workshops related to the study on financing mechanisms and technical assistance on 1-2 selected financing mechanisms 3.2. Venue and related costs for the 3 training sessions 3.3. Replication plan validation workshop
No.	**Component 4: Digital and Knowledge management**
22	4.3. 2 consultants - 1 SES and 1 Gender - 29 days per year per consultant for 4 years in charge of monitoring and recommending corrective measures where needed
23	4.1. Digital strategy 4.2. National monitoring and evaluation platform (remote monitoring & analytics) 300/site/year data collection, storage, and management fee (can vary by number of connections at site); additional training and support of IP & co (3k) and Rural electrification platform development (year 1) and maintenance fee for Years 2-4 4.3. Setting up the Quality Assurance and Monitoring Framework for the project complying and in coordination with the AMP regional requirement. Training the PMU and other targeted consultants 4.5. Gather data and audio-visual content for the "insight brief" (local company), incl. translation into English 4.6. Awareness raising campaign targeting different segments including general public. Communication strategy and roll out (including short video production and digital marketing)
24	4.2. Server, domain name and other costs for rural electrification platform
25	4.3. Field visits and stakeholder consultations, incl. DSA - 2 missions on the ground on pilot sites per year - and car with driver for the 2 experts doing the missions together 4.5. Regional AMP workshops, meetings or training events
26	4.1. Data Strategy validation workshop
27	4.6. For the GoN & political sphere for the 1st year and for the national promotion campaign for the entire project duration (4 years)
No.	**Component 5: Monitoring and Evaluation**
28	Monitoring & Evaluation related printing costs
29	Travel for PMU for M&E related activities
30	Inception workshop
31	Evaluations: Costs including lump sum for travels (international & national)
32	Evaluations: Costs including lump sum for travels (national)
33	Project closure workshop

No.	Project Management Cost
34	PMU - full time for 4 years (Grille harmonisée de rémunération brute + taxes of 20%): Project Manager, Project Assistant/Coordinator (incl. M&E),- PC is split between GEF & UNDP (UNDP covers $11,000 per year) - Year 1: $1560 (PM), $1215/2 (PC); Year 2: $1625 (PM); $1265/2 (PC);; Year 3 $1700 (PM), $1315 (PC); Year 4 : $1755 (PM); $ 1370 (PC)
35	Audit costs
36	PMU Office supplies
37	PMU - full time for 3 years Administrative Assistant (incl. financial aspects) - (Grille harmonisée de rémunération brute + taxes of 20%)- Year 1: $800 (AA); Year 2: $825 (AA); Year 3 $860; Year 4 : $900 (AA); + $11,000 a year for Project coordinator Additional local consultant support provision of $1,000 per year to support PMU activities
38	2 pickups and 1 motorbike for the VNU
39	Laptops, software, etc.
40	1 full time UNV
41	Maintenance costs for the 3 vehicles for the project (Year 2-4)

Summary of funds

Table 13 – Summary of funds

	Amount	Amount	Amount	Amount	Total
	Year 1 (USD)	Year 2 (USD)	Year 3 (USD)	Year 4 (USD)	(USD)
GEF grant administered by UNDP	460,001	658,596	302,184	180,595	1,601,376
Grant (cash) co-finance by GEF Agency: UNDP	85,520	71,880	44,384	98,216	300,000
Rural Electrification Agency ANPER (grant)	45,723,918	41,565,219	41,565,219	6,250,000	135,104,356
Rural Electrification Agency ANPER (in-kind)	19,123	19,123	19,122	19,122	76,490
SIDA (grant)	638,085	638,086	0	0	1,276,171
TOTAL	**47,017,556**	**42,952,904**	**41,930,909**	**6,547,933**	**138,358,393**

X. Legal Context

This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of (country) and UNDP, signed on (date). All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

This project will be implemented by ANPER ("Implementing Partner") in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

XI. Risk Management

1. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:
   a) put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
   b) assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

2. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document.

3. The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the United Nations Security Council Consolidated Sanctions List, and that no UNDP funds received pursuant to the Project Document are used for money laundering activities. The United Nations Security Council Consolidated Sanctions List can be accessed via https://www.un.org/securitycouncil/content/un-sc-consolidated-list.

4. The Implementing Partner acknowledges and agrees that UNDP will not tolerate sexual harassment and sexual exploitation and abuse of anyone by the Implementing Partner, and each of its responsible parties, their respective sub-recipients and other entities involved in Project implementation, either as contractors or subcontractors and their personnel, and any individuals performing services for them under the Project Document.

   (a) In the implementation of the activities under this Project Document, the Implementing Partner, and each of its sub-parties referred to above, shall comply with the standards of conduct set forth in the Secretary General's Bulletin ST/SGB/2003/13 of 9 October 2003, concerning "Special measures for protection from sexual exploitation and sexual abuse" ("SEA").

   (b) Moreover, and without limitation to the application of other regulations, rules, policies and procedures bearing upon the performance of the activities under this Project Document, in the implementation of activities, the Implementing Partner, and each of its sub-parties referred to above, shall not engage in any form of sexual harassment ("SH"). SH is defined as any unwelcome conduct of a sexual nature that might reasonably be expected or be perceived to cause offense or humiliation, when such conduct interferes with work, is made a condition of employment or creates an intimidating, hostile or offensive work environment. SH may occur in the workplace or in connection with work. While typically involving a pattern of conduct, SH may take the form of a single incident. In assessing the reasonableness of expectations or perceptions, the perspective of the person who is the target of the conduct shall be considered.

5. a) In the performance of the activities under this Project Document, the Implementing Partner shall (with respect to its own activities), and shall require from its sub-parties referred to in paragraph 4 (with respect to their activities) that they, have minimum standards and procedures in place, or a plan to develop and/or improve such standards and procedures in order to be able to take effective preventive and investigative action. These should include: policies on sexual harassment and sexual exploitation and abuse; policies on whistleblowing/protection against retaliation; and complaints, disciplinary and investigative mechanisms. In line with this, the Implementing Partner will and will require that such sub-parties will take all appropriate measures to:

    i. Prevent its employees, agents or any other persons engaged to perform any services under this Project Document, from engaging in SH or SEA;

    ii. Offer employees and associated personnel training on prevention and response to SH and SEA, where the Implementing Partner and its sub-parties referred to in paragraph 4 have not put in place its own training regarding the prevention of SH and SEA, the Implementing Partner and its sub-parties may use the training material available at UNDP;

    iii. Report and monitor allegations of SH and SEA of which the Implementing Partner and its sub-parties referred to in paragraph 4 have been informed or have otherwise become aware, and status thereof;

    iv. Refer victims/survivors of SH and SEA to safe and confidential victim assistance; and

    v. Promptly and confidentially, record and investigate any allegations credible enough to warrant an investigation of SH or SEA. The Implementing Partner shall advise UNDP of any such allegations received and investigations being conducted by itself or any of its sub-parties referred to in paragraph 4 with respect to their activities under the Project Document, and shall keep UNDP informed during the investigation by it or any of such sub-parties, to the extent that such notification (i) does not jeopardize the conduct of the investigation, including but not limited to the safety or security of persons, and/or (ii) is not in contravention of any laws applicable to it. Following the investigation, the Implementing Partner shall advise UNDP of any actions taken by it or any of the other entities further to the investigation.

    b) The Implementing Partner shall establish that it has complied with the foregoing, to the satisfaction of UNDP, when requested by UNDP or any party acting on its behalf to provide such confirmation. Failure of the Implementing Partner, and each of its sub-parties referred to in paragraph 4, to comply of the foregoing, as determined by UNDP, shall be considered grounds for suspension or termination of the Project.

6. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

7. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

8. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

9. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, responsible parties, subcontractors and sub-recipients in implementing the project or using UNDP funds.

10. In the implementation of the activities under this Project Document, UNDP places reasonable reliance upon the Implementing Partner for it to apply its laws, regulations and processes, and applicable international laws regarding anti money laundering and

countering the financing of terrorism, to ensure consistency with the principles of then in force the UNDP Anti-Money Laundering and Countering the Financing of Terrorism Policy.

11. The Implementing Partner will ensure that its financial management, anti-corruption, anti-fraud and anti-money laundering and countering the financing of terrorism policies are in place and enforced for all funding received from or through UNDP.

12. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

13. In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP projects and programmes in accordance with UNDP's regulations, rules, policies and procedures. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.

14. The signatories to this Project Document will promptly inform one another in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

    Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). The Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

15. UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud, corruption or other financial irregularity, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail the Implementing Partner's obligations under this Project Document.

    Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud, corruption or other financial irregularity, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

    Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

16. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

17. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

18. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled "Risk

Management Standard Clauses" are included, mutatis mutandis, in all sub-contracts or sub-agreements entered into further to this Project Document.

XII. MANDATORY ANNEXES

1. GEF Budget Template (available from BPPS NCE-VF)
2. GEF Execution Support Letter
3. Project Map and geospatial coordinates of the project area
4. Multiyear Workplan
5. Social and Environmental Screening Procedure (SESP)
6. UNDP Atlas Risk Register
7. Overview of technical consultancies/subcontracts
8. Stakeholder Engagement Plan
9. Environmental Social Management Framework (ESMF) or other SES frameworks/plans, if required
10. Gender Analysis and Gender Action Plan
11. Procurement Plan – for first year of implementation especially
12. GEF focal area specific annexes (e.g. METT, GHG calculations, target landscape profile, feasibility study, other technical reports)
13. Co-financing letters
14. Additional agreements: such as cost sharing agreements, project cooperation agreements signed with NGOs (where the NGO is designated as the "executing entity").
15. GEF and/or LDCF/SCCF Core indicators (see template below)
16. GEF Taxonomy (see template below)
17. Partners Capacity Assessment Tool and HACT assessment
18. UNDP Project Quality Assurance Report (to be completed in UNDP online corporate planning system)

ANNEX 1: GEF BUDGET TEMPLATE

Expenditure Category	Detailed Description	Component (USDeq.)								Responsible Entity
		Component 1	*Component 2*	*Component 3*	*Component 4*	*M&E*	*Sub-Total*	*PMC*	Total (USDeq.)	(Executing Entity receiving funds from the GEF Agency)[1]
Equipment	2.4. Furnitures and other goods for the association's office		7,394				7,394		7,394	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Equipment	1.6. Various training equipment incl. solar PV panels etc.	6,000					6,000		6,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Equipment	2.1. CAPEX subsidies for the 4 CEMG pilot sites2.3. Various training equipment incl. solar PV panels etc.		568,275				568,275		568,275	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Equipment	2.3. IT hardware and software useful for trainings2.4. laptop, printers, software for a functioning association office		7,800				7,800		7,800	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Equipment	4.2. Server, domain name and other costs for rural electrification platform				5,500		5,500		5,500	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Sub-contract to executing partner	2.1. 50% of Project Assistant/Coordinator of the PMU covering pilot project related topics from A to Z		37,188				37,188		37,188	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Contractual services-Company	1.1. 4 trainings1.3. Analysis to cluster existing (pre-)-feasibility studies and recommendations1.7. Local company to conduct study on existing training offer and relevant gaps, as well as recommendations to overcome then	56,000					56,000		56,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)

Contractual services-Company	2.2. Analysis on key national rural economic outputs and their value chains, and social activities incl. field visits on key value chains and stakeholder consultations 1 mission to Niger for the study, field visits inside the country2.3. Provision for additional capacity building needed for communities/local private sector developers & operators2.4. Development of the Private Sector RE association and relevant communications strategy and roll-out		50,200				50,200		50,200	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Contractual services-Company	3.1. Conduct a benchmarking of existing financial products supporting access and use of energy in Niger and aborad, and other industries,Identify innovative financing schemes that are relevant and applicable to support the development of the minigrid market in Niger. Selection of 1-2 financing mechanisms and technical assistance around these			76,700			76,700		76,700	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Contractual services-Company	4.1. Digital strategy4.2. National monitoring and evaluation platform (remote monitoring & analytics) 300/site/year data collection, storage, and management fee (can vary by number of connections at site); additional training and support of IP & co (3k) and Rural electrification platform development (year 1) and maintenance fee for Years 2-44.3. Setting up the Quality Assurance and Monitoring Framework for the project complying and in coordination with the AMP regional requirement. Training the PMU and other targeted consultants4.5. Gather data and audio-visual content for the "insight brief" (local company), incl. translation into English4.6.				99,880		99,880		99,880	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)

	Awareness raising campaign targeting different segments including general public. Communication strategy and roll out (including short video production and digital marketing)									
International Consultants	1.1. Senegalese delegation of 2 consultants paid at USD 500/day for 5+2travel days to provide training and advisory for the national dialogue platform1.2. 25 days for necessary studies based on short assessment under activity 1.2.1. (provision)1.5. International consultants for knowledge gap analysis, developing comprehensive modular training materials, conducting training and ToT - 40 days1.6. 30 days to conduct a review (and benchmark) and to develop the relevant standards, 5 days for the preparation & holding of the workshop1.7. 12 days of Training facilitator/Capacity building specialist to train the trainers of relevant institutions in Niger	71,200					71,200		71,200	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
International Consultants	2.3. 40 days - Pedagogical engineering, Training and ToT on business models, cost-reduction levers, new technologies depending on the topics, tariff negotiation, understanding of anchor loads, remote monitoring, etc. all services - several modules ; and knowledge transfer		24,000				24,000		24,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
International Consultants	3.2. 35 days to develop and conduct relevant training on business models and innovative finance solutions for national financial institutions3.3. 15 days of work - remotely on replication plan			30,000			30,000		30,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Local Consultants	1.1. $300 per meeting x 20 meetings over 4 years - acting as local coordinator for the National Dialogue Platform1.2. 35 days for necessary studies based on short assessment under activity 1.2.1.	54,000					54,000		54,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)

	(provision) + 20 days for adapting the PP law to facilitate CEMG investments1.4. 15 days Selection & implementation of DREI instruments to be covered by AMP1.5. 60 days national consultants knowledge gap analysis, developing comprehensive modular training materials, conducting training1.6. 31 days to conduct a review (and benchmark) and to develop the relevant standards, 5 days for the preparation & holding of the workshop									
Local Consultants	2.3. 100 days - Capacity building for the communities incl. becoming rural electricians (60 days) and for local private sector developers (40 days) with pedagogical engineering, Training and ToT on business models, cost-reduction levers, new technologies depending on the topics, tariff negotiation, understanding of anchor loads, remote monitoring, etc. all services - several modules ; and knowledge transfer2.4. An assistant (or similar) in charge of the bureau and association, updates the website, organizes meetings etc. / part time for 3 years		40,080				40,080		40,080	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Local Consultants	3.3. 40 days with field visits incl. pilot sites and some eligible communities for replication plan			12,000			12,000		12,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Local Consultants	4.3. 2 consultants - 1 SES and 1 Gender - 29 days per year per consultant for 4 years in charge of monitoring and recommending corrective measures where needed				69,600		69,600		69,600	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Local Consultants	PMU - full time for 4 years (Grille harmonisée de rémunération brute + taxes of 20%): Project Manager, Project Assistant/Coordinator (incl. M&E),-						-	132,804	132,804	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)

	PC is split between UNDP & GEF (UNDP covering USD 11,000 a year)- Year 1: $1560 +20% (PM), $1215/2 +20% (PC); Year 2: $1625 +20% (PM); $1265/2 +20% (PC);; Year 3 $1700 +20% (PM), $1315/2 +20% (PC); Year 4 : $1755 +20% (PM); $ 1370/2 +20% (PC)									
Training, Workshops, Meetings	1.1. Various meetings, trainings and workshops for National Dialogue Platform members including workshop with Senegalese delegation for 2 days and various expenses to organize workshops, dissemination/communication materials, etc.1.2. Stakeholder validation workshop for the regulatory framework and a workshop on customs procedures etc.1.4. DREI Workshop in Year 11.5. 1 training session incl. ToT each year and related training supplies1.6. 2 workshops around standards1.7. 2 ToT	40,000					40,000		40,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Training, Workshops, Meetings	2.2. Half day workshop on the analysis and recommendations to boost economic and social activities through electricity access and productive use2.3. Training and ToT sessions throught project duration for communities & local private sector developers & operators2.4. Venue, breaks for meetings & study tour, fees to attend conferences, fairs etc.		19,000				19,000		19,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Training, Workshops, Meetings	3.1. Conduct workshops related to the study on financing mechanisms and technical assistance on 1-2 selected financing mechanisms3.2. Venue and related costs for the 3 training sessions3.3. Replication plan validation workshop			17,000			17,000		17,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Training, Workshops, Meetings	4.1. Data Strategy validation workshop				2,500		2,500		2,500	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)

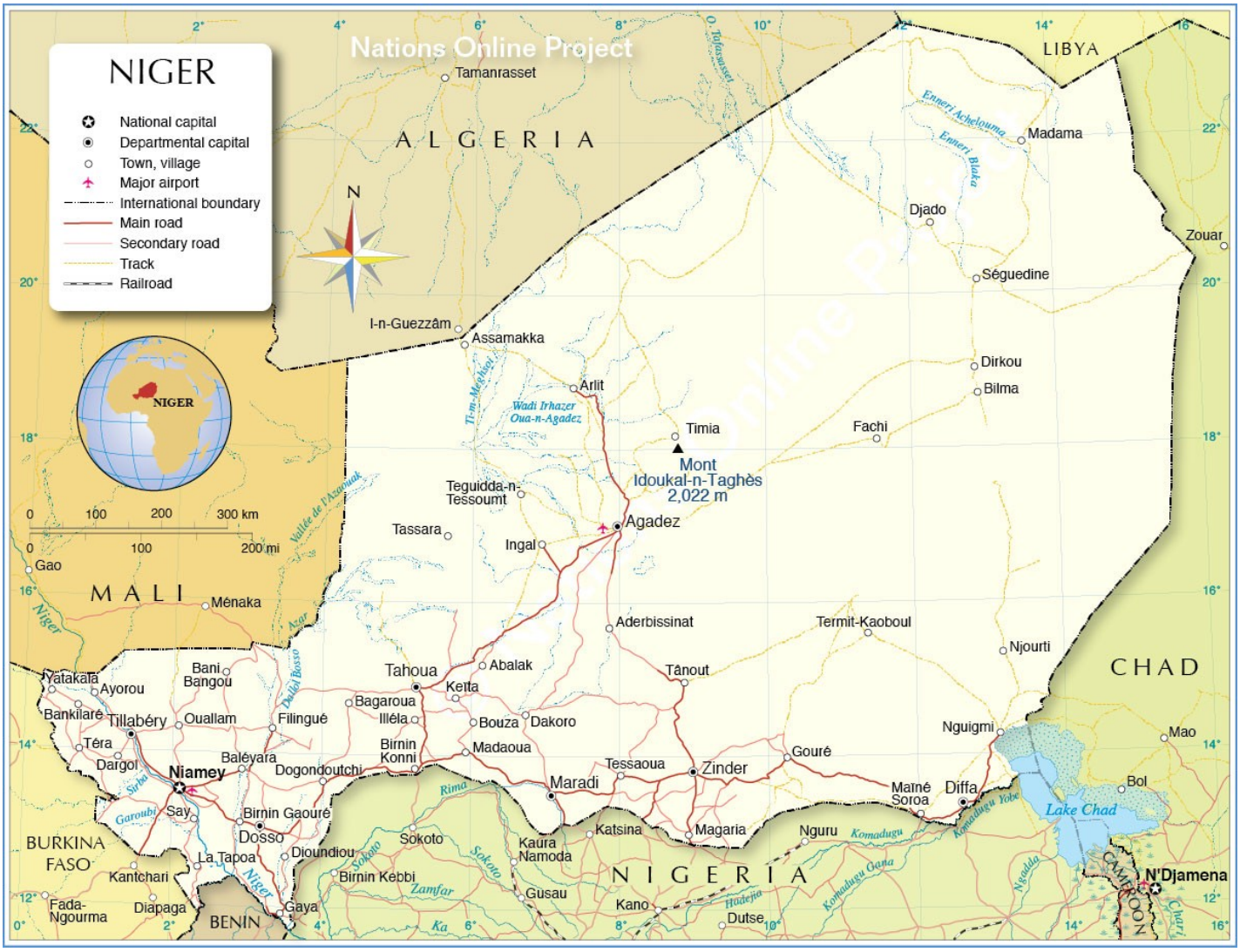
Training, Workshops, Meetings	Inception workshop					3,000	3,000		3,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Travel	1.1. One Exchange visit with Senegal's successful national dialogue platform. 2 people for 5 days (excluding travel time). DSA at USD 187 in Niger in June 2021; USD 1,000 for the travel Dakar-Niamey. No local travel costs for participants located in Niamey but for representatives in relation to pilot sites1.2. 1 10 day-mission for international consultant1.5. 2 missions to Niger - 1 for initial phase and ToT, 1 for ToT on Year 21.6. 5 day mission for international consultant1.7. 2 training missions of 4 days each	20,228					20,228		20,228	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Travel	2.3. 1 mission to Niger; local travels on pilot sites2.4. Travel costs for study tours, and relevant fairs and conferences for 5 people of the association		21,683				21,683		21,683	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Travel	3.2. 3 visits for 3 trainings in Niger3.3. Field visits over 15 days incl. vehicle with driver and gas			12,144			12,144		12,144	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Travel	4.3. Field visits and stakeholder consultations, incl. DSA - 2 missions on the ground on pilot sites per year - and car with driver for the 2 experts doing the missions together4.5. Regional AMP workshops, meetings or training events				65,480		65,480		65,480	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Office Supplies	PMU Office supplies						-	720	720	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Other Operating Costs	2.4. Accounting, audit, etc. incl. for funding purposes - 2000 per year		8,000				8,000		8,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale

										(ANPER)
Other Operating Costs	Audit costs						-	12,000	12,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Other Operating Costs	1.5. Printing training materials1.6. Printing training materials & infographics/leaflets on quality standards for users1.7. Printing training materials	3,000					3,000		3,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Other Operating Costs	2.2. Printing of the report executive summary2.3. Printing training materials		3,000				3,000		3,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Other Operating Costs	4.6. For the GoN & political sphere for the 1st year and for the national promotion campaign for the entire project duration (4 years)				21,000		21,000		21,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
Other Operating Costs	Monitoring & Evaluation related printing costs					4,000	4,000		4,000	Nigérienne Agency for the Promotion of Rural Electrification / Agence Nationale Pour l'Électrification Rurale (ANPER)
		250,428	**786,620**	**147,844**	**263,960**	**7,000**	**1,455,852**	**145,524**	**1,601,376**	

ANNEX 2: GEF EXECUTION SUPPORT LETTER SERVICES

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ANNEX 3: PROJECT MAP AND GEOSPATIAL COORDINATES OF PROJECT SITES



ANNEX 4: MULTI YEAR WORK PLAN

Table 14 – Multi-year workplan

Components	Outcomes	Outputs	Year 1 (2023/2024)				Year 2 (2024/2025)				Year 3 (2025/2026)				Year 4 (2026/2027)			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Component 1: Policy and regulation	**Outcome 1: Stakeholder ownership in a national minigrid delivery model is advanced, and appropriate policies and regulations are adopted to facilitate investment in low-carbon minigrids**	Output 1.1 *An inclusive national dialogue to identify minigrid delivery models is facilitated, clarifying priority interventions for an integrated approach to off-grid electrification*																
		Output 1.2. *A minigrid regulatory framework, including tariff model, tax regime, and grid expansion risk, is developed in close coordination with the authorities concerned and other development partners*																
		Output 1.3. *Analysis of existing (pre)-feasibility studies conducted for selected minigrid sites to enhance sector planning and decision-making on a delivery model for minigrid development*																
		Output 1.4 *Targeted policy and financial derisking instruments of the Minigrid DREI techno-economic analyses implemented and contributing to AMP Flagship Report on Cost Reduction*																
		Output 1.5. *Capacity building provided to public officials (regulator, ministries) to support cost-reduction levers and innovative business models*																
		Output 1.6 *Domestication of quality standards for solar minigrid components, and institutional capacity of ANERSOL strengthened*																
		Output 1.7. *Public programmes (apprenticeships, certificates, university programs) to develop competitive, skilled labour market in minigrids*																
Component 2: Business Model innovation with private sector	**Outcome 2: Innovative business models based on cost reduction are operationalized, with strengthened private**	Output 2.1 *Pilots developed, including on productive use/innovative appliances and modular hardware/system design, leading to cost-reduction in minigrids* hardware/system design, leading to cost-reduction in minigrids																

Components	Outcomes	Outputs	Year 1 (2023/2024)				Year 2 (2024/2025)				Year 3 (2025/2026)				Year 4 (2026/2027)			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
	sector participation in low-carbon/renewable energy minigrid development	Output 2.2 *National report on opportunities to boost economic activities through electricity access and productive use*																
		Output 2.3. *Capacities of private minigrid developers and communities are strengthened*																
		Output 2.4 *Support provided to establish and grow a national industry association for private sector developers*																
Component 3: Scaled-up financing	**Outcome 3: Financial sector actors are ready to invest in a pipeline of low-carbon minigrids and concessional financial mechanisms are in place to incentivize scaled-up investment**	Output 3.1. *Support financing mechanisms to scale-up RE minigrids investment*																
		Output 3.2. *Domestic financial sector capacity-building on business and financing models for minigrids*																
		Output 3.3.: *Replication plan (including investment plan) for scaling up rural energy access developed*																
Component 4: Digital, and Knowledge Management (KM)	**Outcome 4: Digital and data are mainstreamed, across stakeholders, into local minigrid market development. Increased knowledge, awareness and network opportunities in the minigrid market and among stakeholders, including benefitting from linkages to international good practice**	Output 4.1. *A project digital strategy is developed and implemented, including linkages to and following guidance from the AMP Regional Project*																
		Output 4.2. *A 'Minigrids Digital and Data Management Platform' is implemented to run tenders and manage data from pilots, and to support minigrids scale-up and cost-reduction*																
		Output 4.3. *A Quality Assurance and Monitoring Framework for measuring, reporting and verification of the sustainable development impacts of all minigrids pilots supported, including GHG emission reductions, is adopted and operationalized based on standardized guidance from the regional project*																
		Output 4.4: *Engage with regional project, including, but not limited to, via (i) participating in Communities of Practice and (ii) capturing and sharing lessons learned*																

Components	Outcomes	Outputs	Year 1 (2023/2024)				Year 2 (2024/2025)				Year 3 (2025/2026)				Year 4 (2026/2027)			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
		Output 4.5. *Awareness raising campaigns, including lessons learned, are developed and disseminated at all levels nationally (including intervention zones) and with the regional project*																
Component 5: Monitoring and Evaluation (M&E)		M&E and Reporting, including (i) Conducting inception workshop and preparing report, (ii) Ongoing M&E, (iii) Mid-term Review (MTR), and (iv) Terminal Evaluation (TE).																

ANNEX 5: UNDP SOCIAL AND ENVIRONMENTAL SCREENING PROCEDURE (SESP)

Project Information

Project Information	
1. Project Title	Niger national child project under the Africa Minigrids Program
2. Project Number r (i.e. Atlas project ID, PIMS+)	6659
3. Location (Global/Region/Country)	Niger
4. Project stage (Design or Implementation)	Design stage
5. Date	22-07-2022

2. Part A. Integrating Programming Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Programming Principles in order to Strengthen Social and Environmental Sustainability?

Briefly describe in the space below how the project mainstreams the human-rights based approach

Rights holders are women and men, a great number belongs to the poor and marginalized sector such as customary community groups, rural population and resource dependent groups. This project will ensure that their rights are exercised by facilitating their own capacity to think, act, organize, and advocate these rights. Primary duty-bearers comprise the State, with all its provincial agencies and institutions, and the staff dedicated to the project. This project will ensure their mandate will respect, protect, promote and fulfill the rights of the poor and marginalized sectors/groups in all spheres of life.

The project addresses the human rights to sustainable development through the provision of measures to prevent the potential pollution from batteries and e-waste used at the project, as well as the monitored reduction of greenhouse gases emissions. Likewise, the project addresses the human rights to poverty alleviation and sustaining peace by taking into account the local communities as a workforce, including the fuel/energy sellers from the informal sector. Similarly, the project will ensure fair distribution of development opportunities and benefits through the empowerment of disadvantaged groups for example by capacity building.

Altogether, the project fully incorporates the human Leave No One Behind approach, in particular through ensuring the participation, inclusion, equality and non-discrimination of disadvantaged groups (marginalized, discriminated and excluded), including the informal sector. This is achieved by design in the project, to empower them as active agents of the development process, facilitating their participation on the project design and implementation through the requirements established in this report. Similarly, the requirements here include actions to be taken related to advocacy, creating enabling environments, capacity development and support for civil society, community empowerment, and enhancing the quality and accessibility of services.

Across all project components, activities include the participation of varied stakeholders through capacity building strategies at the policy, program, monitoring and evaluation, knowledge management on environmental conservation, human rights, gender equality, and social protection perspectives so that the intended project results are achieved also beyond the project

cycle.
Briefly describe in the space below how the project is likely to improve gender equality and women's empowerment
As the implications of gender in the sector are not fully understood or appreciated, a gender analysis has been conducted during project preparation to fully gauge the gender implications, identify possible interventions that can meaningfully improve and enhance women's participation, and develop specific indicators and targets related to gender equality. Based on that, a gender action plan has been established at the same phase for the preparation of specific investment interventions that will include along the whole project cycle special attention for vulnerable groups, especially women and girls, who face multiple and intersecting forms of discrimination in the energy sector and in general in the society. Women are often marginalized and excluded from other forms of formal participation in the sector and the economy; often, they are reduced to the lower positions in the job market and as beneficiaries.
Briefly describe in the space below how the project mainstreams sustainability and resilience
The project is primarily focused on environmental sustainability. It should be noted that no activities that could cause harm may proceed until assessments are undertaken and management plans are in place for specific sites. The monitoring, reporting and verification (MRV) system that will be set up by the project will include social, environmental and financial indicators to safeguard the improvement of individuals and local communities, with an emphasis on the most vulnerable groups and individuals identified. Additionally, a comprehensive Quality Assurance Framework (QAF) is expected to be operationalized through technical support from the regional AMP. Finally, the mechanisms established in this report will help to strengthen the enforcement of existing laws interacting with the energy sector in order to fulfil public services while promoting the vulnerable groups and their human rights involved to achieve such task.
Briefly describe in the space below how the project strengthens accountability to stakeholders
The Stakeholder Engagement Plan, the information disclosure process, the Grievance Redress Mechanism (GRM) and the Accountability Mechanism will strengthen remarkably the accountability of the most vulnerable groups and individuals affected by the Project both directly and indirectly at a fair level to the conventional groups. These processes and mechanisms have been established at the design phase and will continue along the project cycle. For example, to achieve this a multi-stakeholder platform will be set up to enhance horizontal participation and will include representatives from a varied range of groups in society.

3. Part B. Identifying and Managing Social and Environmental Risks

QUESTION 2: What are the Potential Social and Environmental Risks? Note: Complete SESP Attachment 1 before responding to Question 2.	QUESTION 3: What is the level of significance of the potential social and environmental risks? Note: Respond to Questions 4 and 5below before proceeding to Question 5			QUESTION 6: Describe the assessment and management measures for each risk rated Moderate, Substantial or High
Risk Description (broken down by event, cause, impact)[40]	***Impact and Likelihood (1-5)***	***Significance (Low, Moderate, Substantial, High)***	***Comments (optional)***	***Description of assessment and management measures for risks rated as Moderate, Substantial or High***
Risk 1: Discrimination or marginalization of vulnerable communities through the proposed tariff model in the proposed minigrid regulation Related to: • Human Rights; P.4, P.5, P.6	I = 4 L = 2	Moderate	The project will propose a tariff model for isolated minigrids based on the equity principle that is enshrined in the legal and regulatory framework as well as the recent switch to an affermage-lease MG delivery model and ANPER's strategy. Therefore, there is a low likelihood that the project may lead to setting tariffs that would discriminate against vulnerable communities (including the poor) and prevent them from having access to essential energy services.	As the project risk category has been rated as Substantial, an Environmental and Social Management Framework (ESMF) has been prepared and annexed to the ProDoc. The ESMF requires that formulation of mini grid regulation (Activity 1.2.3) undergo a Strategic Environmental and Social Assessment (SESA) that would take this risk into consideration in the decision-making process. In addition, a Stakeholder Engagement Plan (SEP) has been prepared to ensure that stakeholders have an opportunity to provide feedback on decisions that may affect them. Depending on the identified impacts, a Livelihood Action Framework/Resettlement Policy Framework may also be developed if needed. The project will also put in place a project-level GRM to provide meaningful means for local communities and affected populations to raise concerns and/or grievances when activities may adversely impact them.
Risk 2: Risk on lack of ability for people	I = 3	Moderate	Lack of transparency and tedious or costly	Through the Stakeholder Engagement Plan, the Project

[40] See "SESP Summary" for detailed breakdown by event, cause, impact.

to claim their rights within the areas served by the pilot minigrids Related to: • Human Rights; P.2, P.3 • Accountability; P.14	L = 4		procedures of people/customers to claim their rights may exist within the pilot areas and often the legal or contractual basis for claiming these rights is not well defined or even absent.	shall give priority to community engagement to ensure that No-one is Left Behind. This will imply a proactive attitude to reach out to vulnerable people and groups and treat people equally. Output 2.3. focuses especially on building the capacities of communities in pilot sites. Output 4.5. envisages to raise awareness of the general public and electricity users at large during the entire project duration.
Risk 3: Marginalization of vulnerable groups when developing standards and selecting the pilot minigrids Related to: • Human Rights; P.4, P.5, P.6 • Accountability; P.13, P.14 • Standard 6: Indigenous Peoples; 6.1, 6.2, 6.3, 6.6	I = 3 L = 3	Moderate	Development of standards for Clean Energy Minigrids may marginalize stakeholders from participating in this sector, or from having access to energy by setting stringent technical criteria. Selection of the pilot minigrids, if not done in collaboration with all stakeholders also risks marginalizing certain groups, including Indigenous Peoples.	A Stakeholder Engagement Plan has been prepared to manage this risk, which is associated with Activity 1.6.2 and Activity 2.1.1, through engaging stakeholders to ensure that standards do not marginalize any specific group and exclude them from the decision-making process on issues that affect them. Objective selection criteria have been developed and validated by ANPER and stakeholders at the validation workshop in June 2022, complying with the SNAE and PDAE. The Project Board as well as the National Dialogue Platform should support the validation of the selected pilot projects during project implementation. If a minigrid pilot site is located in close proximity to or on land used by Indigenous Peoples, Free and Prior Informed Consent (FPIC) will be obtained prior to validation of the site. Towards that end, an Indigenous Peoples Policy Framework (IPPF) has been prepared to tackle this issue. During preparation of the Environmental and Social Impact Assessment (ESIA) for the pilot minigrid and based on the developed IPPF, an Indigenous Peoples Plan (IPP) (or its equivalent) will be developed (at the site-level or other level, as deemed appropriate) and implemented, along with measures for FPIC, as needed for compliance with the SES. The project GRM will also help in managing this risk by providing a means for affected stakeholders to raise concerns and/or grievances.
Risk 4: Reproducing existing discriminations against women through excluding them from	I = 4 L = 4	Substantial	The male dominated nature of the energy sector and the limited social status and opportunities identified for women. This may pose a challenge to	Measures have been established through the Gender Analysis and Action Plan established at the PPG phase, to manage and reduce the risks identified on women.

decision-making on project activities, benefiting from project outputs and capacity building initiatives Related to: • Gender Equality and Women Empowerment; P.10			ensure that women will have the chance to participate at the decisions-making level.	Gender-balanced representation in the national dialog, deliberate inclusion of electricity end-uses prioritized by community women at the feasibility stage, and outreach and training support to women, youth, and the financial institutions that serve them are examples of actions meant to prevent unbalanced appropriation by men of minigrid technology, services, and benefits. In addition, this risk will be further assessed in the SESAs and Environmental and Social Impact Assessments (ESIAs) that will be undertaken during project implementation as described in the ESMF.
Risk 5: Damage to biodiversity, natural resources and cultural heritage sites due to installation and operation of pilot minigrids Related to: • Standard 1: Biodiversity Conservation and Natural Resource Management; 1.1, 1.2, 1.3, 1.4, 1.7 • Standard 4: Cultural Heritage; 4.1, 4.2, 4.3, 4.4	I = 4 L = 3	Substantial	Pilot minigrids may be located within or near critical habitats, environmentally sensitive areas or cultural heritage sites (including those of value to Indigenous Peoples). As the pilots will mostly entail greenfield activities, this will require changes to the use of lands and resources, affecting natural ecosystems. All minigrids, as well as their aggregation, involve the construction of new infrastructure and operational activities, which will lead to changes in nearby ecosystems or land uses and lead to various impacts including air emissions. Excavation activities may lead to the removal, destruction or displacement of the existing cultural heritage to allow the new structures to be built. Due to the fact that the sites have not yet been selected, the likelihood of this risk is not known and a conservative approach is adopted, rating it a Substantial risk.	Pilot minigrids (Output 2.1) will incorporate SES criteria during the site selection process and adopt the list of exclusion criteria that is found in the ESMF. After selection and before commencement of the pilot activity each pilot minigrid will undergo a scoped ESIA or targeted assessment that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific Environmental and Social Management Plan (ESMP). Details of this process can be found in the ESMF. Once identified, aggregation of minigrids (Activity 3.2.2) will undergo a SESP and based on the additional infrastructure needed, an ESIA/ESMP may be undertaken.
Risk 6: Exposure to electrocution risks for humans and any fauna (ex. animals or birds) using the minigrid area	I = 3 L = 2	Moderate	All mini-grids involve electrical equipment. At the operational stage, the electrical structure alien to pre-existing conditions in the area, may cause the damage/death/fire/et due to the interaction with people living nearby (including Indigenous Peoples), fauna and flora.	Pilot minigrids (Output 2.1) will incorporate SES criteria during the site selection process and adopt the list of exclusion criteria that is found in the ESMF. After selection and before commencement of the pilot activity each pilot minigrid will undergo a scoped ESIA or targeted assessment that will analyze this risk.

Related to: • Standard 1: Biodiversity Conservation and Natural Resource Management; 1.1, 1.2, 1.4 • Standard 3: Community Health, Safety and Security; 3.2				Mitigation measures will then be adopted as described in the pursuant site-specific ESMP. Details of this process can be found in the ESMF.
Risk 7: Climate events and disasters (including floods) on new and existing infrastructure Related to: • Standard 2: Climate Change and Disaster Risks; 2.1, 2.2 • Standard 3: Community Health, Safety and Security; 3.3	I = 4 L = 3	Substantial	Climate change is expected to significantly affect infrastructure in Niger through extreme weather events. High precipitation amounts, expected to increase over time, can lead to the **flooding of** infrastructure. All mini-grids are open air structures exposed to climate events and involve built structures that may be vulnerable to the impacts of climate change or disasters. Due to the fact that the sites have not yet been selected, the likelihood of this risk is not known and a conservative approach is adopted, rating it a Substantial risk.	Pilot minigrids (Output 2.1) will each undergo a scoped ESIA or targeted assessment that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific ESMP. Details of this process can be found in the ESMF. Once identified, aggregation of minigrids (Activity 3.2.2) will undergo a SESP and based on the additional infrastructure needed, an ESIA/ESMP may be undertaken.
Risk 8: Risk on the community due to domestic connections and electricity usage, and presence of hazardous materials (mainly batteries, e-waste). Related to: • Standard 3: Community Health, Safety and Security; 3.2	I = 4 L = 3	Substantial	The novelty of some structures and practices brought about by the project could become a source of harm if not accompanied with concomitant awareness of risks and safe practices. More specifically, the use of potentially hazardous materials by the project, domestic electrical wiring and connection activities and subsequent domestic usage of electricity.	Pilot minigrids (Output 2.1) will each undergo a scoped ESIA or targeted assessment that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific ESMP. Details of this process can be found in the ESMF. In particular, operators, contractors and owners of sites shall be required to abide by the ESMP's requirements on safety measures and minimum qualifications for the handling of hazardous materials. Similarly, those responsible for connecting households should ensure the provision of qualified electrician services to do so. Consumer awareness campaigns should also be performed, including through local workshops, clear signage (pictograms and local language indications) and awareness-raising activities in schools and public spaces to inform communities of risks associated with installations (e.g. prevention of trespassing and/or makeshifts connections attempts, etc.) and of the safe usage of electricity domestically.

Risk 9: Community health and safety risks due to construction of the pilot minigrids and relevant infrastructure Related to: • Standard 3: Community Health, Safety and Security; 3.1, 3.2, 3.3, 3.4, 3.5 • Standard 6: Indigenous Peoples; 6.1, 6.3	I = 3 L = 2	Moderate	Some new activities and/or structures may interact with the surrounding area and/or involve the alteration of the normal functioning of the community health, safety and/or security in the project's area of influence, mainly as noise and physical hazards. This risk may also affect Indigenous Peoples if present nearby.	Pilot minigrids (Output 2.1) will incorporate SES criteria during the site selection process and adopt the list of exclusion criteria that is found in the ESMF. After selection and before commencement of the pilot activity each pilot minigrid will undergo a scoped ESIA or targeted assessment that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific ESMP, which shall include a Pollution Prevention and Management Plan and a Traffic Management Plan. Details of this process can be found in the ESMF. Once identified, aggregation of minigrids (Activity 3.2.2) will undergo a SESP and based on the additional infrastructure needed, an ESIA/ESMP may be undertaken.
Risk 10: Risk on community health, safety and/or security due to the influx of people, mainly project workers Related to: • Standard 3: Community Health, Safety and Security; 3.4, 3.7, 3.8 • Standard 6: Indigenous Peoples; 6.1, 6.3	I = 3 L = 3	Moderate	● New activities in the project's area of influence may attract newcomers affecting community health, safety and/or security as this new influx of people, expected to be mainly men, may interact with the local residents (including Indigenous Peoples) and/or involve the alteration of the normal functioning of the community leading to new diseases and/or gender safety concerns.	Pilot minigrids (Output 2.1) will each undergo a scoped ESIA or targeted assessment that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific ESMP. Details of this process can be found in the ESMF. Once identified, aggregation of minigrids (Activity 3.2.2) will undergo a SESP and based on the additional infrastructure needed, an ESIA/ESMP may be undertaken. The project GRM will provide a means for affected community to report on any incidents that may occur as a result of this risk.
Risk 11: Physical or economic displacement and loss of livelihood due to eviction from land on which pilot minigrid may be installed	I = 4 L = 2	Moderate	All minigrids involve the construction of new infrastructure. New built structures occupy land, and access to the area may be restricted. Expected impacts include the displacement of existing legal or illegal inhabitants (including Indigenous	Pilot minigrids (Output 2.1) will incorporate SES criteria during the site selection process and adopt the list of exclusion criteria that is found in the ESMF. After selection and before commencement of the pilot activity each pilot minigrid will undergo a scoped ESIA or

Related to: • Standard 5: Resettlement and Displacement; 5.1, 5.2, 5.4 • Standard 6: Indigenous Peoples; 6.1, 6.2, 6.3, 6.6			Peoples) to allow the new structures to be built. However, this is very limited in Niger given the setup of villages and the large low-land areas with limited forestation. Due to the fact that the sites have not yet been selected, a conservative approach is adopted and this risk is rated as Moderate.	targeted assessment that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific Environmental and Social Management Plan (ESMP), which may include a Livelihoods Restoration Plan. Details of this process can be found in the ESMF. Once identified, aggregation of minigrids (Activity 3.2.2) will undergo a SESP and based on the additional infrastructure needed, an ESIA/ESMP may be undertaken.
Risk 12: Loss of income for fuel sellers once pilot minigrids are operational. Related to: • Human Rights; P.5	I = 4 L = 4	Substantial	Traditional fuels supplied by local providers, including those from the informal/traditional sectors see their market diminished. Some mini-grid systems and project appliances to be implemented may replace an activity that was fueled with other energy sources like wood charcoal, paraffin, kerosene and diesel. The decrease in fuel demand will lead to the loss of income for fuel suppliers, some of whom may be vulnerable people working in the informal market. Due to the fact that the sites have not yet been selected, the likelihood of this risk is not known and a conservative approach is adopted, rating it a Substantial risk.	Pilot minigrids (Output 2.1) will each undergo a scoped ESIA or targeted assessment that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific ESMP. Details of this process can be found in the ESMF. The efforts under Component 2 to facilitate PUE and build the capacities of inhabitants as rural electricians for instance will be a means for fuel sellers to shift to other income generating revenues. Some traditional fuel sources sold by such sellers will still be needed for transportation and cooking.
Risk 13: Working conditions not in line with national and international standards (by contractor or other entities involved in the project) Related to: • Standard 7: Labour and Working Conditions; 7.1, 7.2, 7.3, 7.4, and 7.5, 7.6	I = 4 L = 4	Substantial	All stages of the pilot minigrids will require labour, some of which may be sourced to unskilled/manual labourers who could be less familiar with the type of installations considered for this project and the concomitant occupational health and safety requirements and risks. Maintenance of the right-of-way and bush-clearing under transmission lines by manual labourers is especially relevant in this context. This may lead to the use of child, forces, discriminatory, under-minimum practices and/or occupational health and safety accidents/incidents. In addition, manufacturers, suppliers, subcontractors and subcontractors within the solar supply chain may not be in line with SES and thus	For each pilot minigrid (Output 2.1), Labour Management Procedures (including requirements and terms/conditions related to the selection, procurement and management of primary suppliers of solar panels) and an Occupational Health and Safety Plan will be prepared and applied for the project to ensure labour standards and rights are upheld for project workers. In addition, the ESIA or targeted assessment will assess the likelihood of this risk and prevalence of child labour within the energy sector in the target area and propose measures to reduce it and find working persons under

			procurement of solar panels for the demonstration pilots may contribute to working conditions that undermine worker human rights, health and safety.	the age of 18 perform tasks appropriate to their age.
Risk 14: Generation of hazardous waste (specifically e-waste) from the pilot minigrids that have been installed Related to: • Standard 1: Biodiversity Conservation and Natural Resource Management; 1.14 • Standard 8: Pollution Prevention and Resource Efficiency; 8.1, 8.2, 8.3, 8.4	I = 3 L = 3	Moderate	While minigrids are small-scale technology, construction and maintenance involves the use of minor amounts of chemicals (paints, solvents, cleaning liquids, solder). Montreal Protocol chemicals can be present in appliances power by minigrids (i.e., cooling equipment). Persistent organic pollutants will not be used under this project. However, proper work procedures and equipment handling are sufficient measures to prevent releases into the environment. In addition, modest amounts of waste will be generated during construction (ground movement and concrete residues); electric wiring and insulator ends; broken or rejected parts and components. Operation of minigrids will lead to the generation of different types of waste, in particular electronic waste ("e-waste") in the form of solar panels and/or batteries at the end of their useful lives will be generated. Without proper handling directives, disposal and/or recycling mandate for obsolete equipment, this could result in additional waste generation, including of hazardous/phase-outs materials, chemicals or other pollutants (e.g. from batteries). Failure to recycle non-hazardous waste could also contribute to additional waste generation.	This risk will be assessed in the ESIA or targeted assessment that will be undertaken for each pilot minigrid (Output 2.1), such that the ESMP will include a Waste Management Plan detailing the procedures for disposal of all types of waste associated with construction and operation of the pilot minigrids.

	QUESTION 4: What is the overall project risk categorization?		
	=		
	Low Risk	☐	
	Moderate Risk	☐	

	Substantial Risk	**X**	Fourteen potential risks have been identified for this project, eight of which are assessed as MODERATE and six as SUBSTANTIAL. As a result, this project is rated overall as a SUBSTANTIAL Risk project. During the PPG, an ESMF, IPPF, Stakeholder Engagement Plan and Gender Action Plan have been prepared to meet SES requirements. During project implementation, SESAs addressing potential environmental and socioeconomic impacts of upstream activities will be performed, and a scoped ESIA or targeted assessment along with a site-specific ESMP will be prepared for each pilot minigrid. The ESMP will include an Occupational Health and Safety Plans, Pollution Prevention and Management Plan, Waste Management Plan, and Traffic Management Plan, and any other plans required for SES
	High Risk	☐	

	QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are triggered? (check all that apply)				
	Question only required for Moderate, Substantial and High Risk projects				
	Is assessment required? (check if "yes")	**X**			***Status? (completed, planned)***
	if yes, indicate overall type and status		**X**	Targeted assessment(s)	Completed during PPG: gender analysis, stakeholder analysis
			X	ESIA (Environmental and Social Impact Assessment)	Planned (during implementation)
			X	SESA (Strategic Environmental and Social Assessment)	Planned (during implementation)
	Are management plans required? (check if "yes)	**X**			
	If yes, indicate overall type		**X**	Targeted management plans (e.g. Gender Action Plan, Emergency Response Plan, Waste Management Plan, others)	Completed during PPG: IPPF, Gender Action Plan, Stakeholder Engagement Plan
			X	ESMP (Environmental and Social Management Plan which may include range of targeted plans)	Planned (for during implementation)
			X	ESMF (Environmental and Social Management Framework)	Completed during PPG

	Based on identified <u>risks</u>, which Principles/Project-level Standards triggered?		**Comments (not required)**
	Overarching Principle: Leave No One Behind		
	Human Rights	**X**	
	Gender Equality and Women's Empowerment	**X**	
	Accountability	**X**	
	1. Biodiversity Conservation and Sustainable Natural Resource Management	**X**	
	2. Climate Change and Disaster Risks	**X**	
	3. Community Health, Safety and Security	**X**	
	4. Cultural Heritage	**X**	
	5. Displacement and Resettlement	**X**	
	6. Indigenous Peoples	**X**	
	7. Labour and Working Conditions	**X**	
	8. Pollution Prevention and Resource Efficiency	**X**	

Final Sign Off

Signature	***Date***	***Description***
QA Assessor		UNDP staff member responsible for the Project, typically a UNDP Programme Officer. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver		UNDP senior manager, typically the UNDP Deputy Country Director (DCD), Country Director (CD), Deputy Resident Representative (DRR), or Resident Representative (RR). The QA Approver cannot also be the QA Assessor. Final signature confirms they have "cleared" the SESP prior to submittal to the PAC.
PAC Chair		UNDP chair of the PAC. In some cases, PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the Project appraisal and considered in recommendations of the PAC.

4. SESP Attachment 1. Social and Environmental Risk Screening Checklist

Checklist Potential Social and Environmental Risks	
INSTRUCTIONS: The risk screening checklist will assist in answering Questions 2-6 of the Screening Template. Answers to the checklist questions help to (1) identify potential risks, (2) determine the overall risk categorization of the project, and (3) determine required level of assessment and management measures. Refer to the SES toolkit for further guidance on addressing screening questions.	
Overarching Principle: Leave No One Behind **Human Rights**	**Answer (Yes/No)**
P.1 Have local communities or individuals raised human rights concerns regarding the project (e.g. during the stakeholder engagement process, grievance processes, public statements)?	*No*
P.2 Is there a risk that duty-bearers (e.g. government agencies) do not have the capacity to meet their obligations in the project?	*Yes*
P.3 Is there a risk that rights-holders (e.g. project-affected persons) do not have the capacity to claim their rights?	*Yes*
Would the project potentially involve or lead to:	
P.4 adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	Yes
P.5 inequitable or discriminatory impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups, including persons with disabilities? [41]	Yes
P.6 restrictions in availability, quality of and/or access to resources or basic services, in particular to marginalized individuals or groups, including persons with disabilities?	Yes

[41] Prohibited grounds of discrimination include race, ethnicity, sex, age, language, disability, sexual orientation, gender identity, religion, political or other opinion, national or social or geographical origin, property, birth or other status including as an indigenous person or as a member of a minority. References to "women and men" or similar is understood to include women and men, boys and girls, and other groups discriminated against based on their gender identities, such as transgender and transsexual people.

P.7	exacerbation of conflicts among and/or the risk of violence to project-affected communities and individuals?	No
Gender Equality and Women's Empowerment		
P.8	Have women's groups/leaders raised gender equality concerns regarding the project, (e.g. during the stakeholder engagement process, grievance processes, public statements)?	No
Would the project potentially involve or lead to:		
P.9	adverse impacts on gender equality and/or the situation of women and girls?	No
P.10	reproducing discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	Yes
P.11	limitations on women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? *For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well being*	No
P.12	exacerbation of risks of gender-based violence? *For example, through the influx of workers to a community, changes in community and household power dynamics, increased exposure to unsafe public places and/or transport, etc.*	No
Sustainability and Resilience: Screening questions regarding risks associated with sustainability and resilience are encompassed by the Standard-specific questions below		
Accountability		
Would the project potentially involve or lead to:		
P.13	exclusion of any potentially affected stakeholders, in particular marginalized groups and excluded individuals (including persons with disabilities), from fully participating in decisions that may affect them?	Yes
P.14	grievances or objections from potentially affected stakeholders?	Yes
P.15	risks of retaliation or reprisals against stakeholders who express concerns or grievances, or who seek to participate in or to obtain information on the project?	No

Project-Level Standards		
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management		
Would the project potentially involve or lead to:		
1.1	adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? *For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes*	Yes
1.2	activities within or adjacent to critical habitats and/or environmentally sensitive areas, including (but not limited to) legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	Yes
1.3	changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	Yes
1.4	risks to endangered species (e.g. reduction, encroachment on habitat)?	Yes
1.5	exacerbation of illegal wildlife trade?	No
1.6	introduction of invasive alien species?	No
1.7	adverse impacts on soils?	Yes
1.8	harvesting of natural forests, plantation development, or reforestation?	No
1.9	significant agricultural production?	No
1.10	animal husbandry or harvesting of fish populations or other aquatic species?	No
1.11	significant extraction, diversion or containment of surface or ground water? *For example, construction of dams, reservoirs, river basin developments, groundwater extraction*	No
1.12	handling or utilization of genetically modified organisms/living modified organisms?[42]	No

[42] See the Convention on Biological Diversity and its Cartagena Protocol on Biosafety.

1.13	utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)[43]	No
1.14	adverse transboundary or global environmental concerns?	Yes
Standard 2: Climate Change and Disaster Risks		
Would the project potentially involve or lead to:		
2.1	areas subject to hazards such as earthquakes, floods, landslides, severe winds, storm surges, tsunami or volcanic eruptions?	Yes
2.2	outputs and outcomes sensitive or vulnerable to potential impacts of climate change or disasters? *For example, through increased precipitation, drought, temperature, salinity, extreme events, earthquakes*	Yes
2.3	increases in vulnerability to climate change impacts or disaster risks now or in the future (also known as maladaptive or negative coping practices)? *For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding*	No
2.4	increases of greenhouse gas emissions, black carbon emissions or other drivers of climate change?	No
Standard 3: Community Health, Safety and Security		
Would the project potentially involve or lead to:		
3.1	construction and/or infrastructure development (e.g. roads, buildings, dams)? (Note: the GEF does not finance projects that would involve the construction or rehabilitation of large or complex dams)	Yes
3.2	air pollution, noise, vibration, traffic, injuries, physical hazards, poor surface water quality due to runoff, erosion, sanitation?	Yes
3.3	harm or losses due to failure of structural elements of the project (e.g. collapse of buildings or infrastructure)?	Yes
3.4	risks of water-borne or other vector-borne diseases (e.g. temporary breeding habitats), communicable and noncommunicable diseases, nutritional disorders, mental health?	Yes
3.5	transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel	Yes

[43] See the Convention on Biological Diversity and its Nagoya Protocol on access and benefit sharing from use of genetic resources.

	and other chemicals during construction and operation)?	
3.6	adverse impacts on ecosystems and ecosystem services relevant to communities' health (e.g. food, surface water purification, natural buffers from flooding)?	No
3.7	influx of project workers to project areas?	Yes
3.8	engagement of security personnel to protect facilities and property or to support project activities?	Yes
Standard 4: Cultural Heritage		
Would the project potentially involve or lead to:		
4.1	activities adjacent to or within a Cultural Heritage site?	Yes
4.2	significant excavations, demolitions, movement of earth, flooding or other environmental changes?	Yes
4.3	adverse impacts to sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: projects intended to protect and conserve Cultural Heritage may also have inadvertent adverse impacts)	Yes
4.4	alterations to landscapes and natural features with cultural significance?	Yes
4.5	utilization of tangible and/or intangible forms (e.g. practices, traditional knowledge) of Cultural Heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement		
Would the project potentially involve or lead to:		
5.1	temporary or permanent and full or partial physical displacement (including people without legally recognizable claims to land)?	Yes
5.2	economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	Yes
5.3	risk of forced evictions?[44]	No
5.4	impacts on or changes to land tenure arrangements and/or community based property	Yes

[44] Forced eviction is defined here as the permanent or temporary removal against their will of individuals, families or communities from the homes and/or land which they occupy, without the provision of, and access to, appropriate forms of legal or other protection. Forced evictions constitute gross violations of a range of internationally recognized human rights.

	rights/customary rights to land, territories and/or resources?	
Standard 6: Indigenous Peoples		
Would the project potentially involve or lead to:		
6.1	areas where indigenous peoples are present (including project area of influence)?	Yes
6.2	activities located on lands and territories claimed by indigenous peoples?	Yes
6.3	impacts (positive or negative) to the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in question)? *If the answer to screening question 6.3 is "yes", then the potential risk impacts are considered significant and the project would be categorized as either Substantial Risk or High Risk*	Yes
6.4	the absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	No
6.5	the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	No
6.6	forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources? *Consider, and where appropriate ensure, consistency with the answers under Standard 5 above*	Yes
6.7	adverse impacts on the development priorities of indigenous peoples as defined by them?	No
6.8	risks to the physical and cultural survival of indigenous peoples?	No
6.9	impacts on the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices? *Consider, and where appropriate ensure, consistency with the answers under Standard 4 above.*	No
Standard 7: Labour and Working Conditions		
Would the project potentially involve or lead to: (note: applies to project and contractor workers)		

7.1	working conditions that do not meet national labour laws and international commitments?	Yes
7.2	working conditions that may deny freedom of association and collective bargaining?	Yes
7.3	use of child labour?	Yes
7.4	use of forced labour?	Yes
7.5	discriminatory working conditions and/or lack of equal opportunity?	Yes
7.6	occupational health and safety risks due to physical, chemical, biological and psychosocial hazards (including violence and harassment) throughout the project life-cycle?	Yes
Standard 8: Pollution Prevention and Resource Efficiency		
Would the project potentially involve or lead to:		
8.1	the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or transboundary impacts?	Yes
8.2	the generation of waste (both hazardous and non-hazardous)?	Yes
8.3	the manufacture, trade, release, and/or use of hazardous materials and/or chemicals?	Yes
8.4	the use of chemicals or materials subject to international bans or phase-outs? *For example, DDT, PCBs and other chemicals listed in international conventions such as the Montreal Protocol, Minamata Convention, Basel Convention, Rotterdam Convention, Stockholm Convention*	Yes
8.5	the application of pesticides that may have a negative effect on the environment or human health?	No
8.6	significant consumption of raw materials, energy, and/or water?	No

ANNEX 6: UNDP (ATLAS) RISK REGISTER

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
Social and Environmental Risks and Mitigation Measures					
1	**Discrimination or marginalization of vulnerable communities through the investment selection in the replication plan**	Social and Environmental	I = 4, L = 3 (substantial)	An Environmental and Social Management Framework (ESMF) has been prepared and annexed (see Annex 9). In addition, a Stakeholder Engagement Plan (SEP) has been prepared to ensure that stakeholders have an opportunity to provide feedback on decisions that may affect them. The project will also put in place a project-level and/or site-level GRM to provide (Annex.). meaningful means for local communities and affected populations	Project Management Unit
2	**Risk of lack of ability for people to claim their rights within the areas served by the pilot minigrids**	Social and Environmental	I = 3; L = 4 (moderate)	Through the Stakeholder Engagement Plan, the Project shall give priority to community engagement to ensure that No-one is Left Behind (Annex 8)	Project Management Unit
3	**Marginalization of vulnerable groups when selecting the pilot minigrids**	Social and Environmental	I = 3; L = 3 (moderate)	Through the Stakeholder Engagement Plan, the Project shall give priority to community engagement to ensure that No-one is	Project Management Unit

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
				Left Behind (Annex 8)	

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
4	**Reproducing existing discriminations against women through excluding them from decision-making on project activities, benefiting from project outputs and capacity-building initiatives**	Social and Environmental	I = 4; L = 4 (substantial)	Measures have been established through the Gender Analysis and Action Plan established at the PPG phase, to manage and reduce the risks identified on women. In addition, this risk will be further assessed in the SESAs and Environmental and Social Impact Assessments (ESIAs) that will be undertaken during project implementation as described in the ESMF.	Project Management Unit
5	**Damage to biodiversity, natural resources and cultural heritage sites due to installation and operation of pilot minigrids or planned minigrids in the investment plan**	Social and Environmental	I = 4; L = 3 (substantial)	Pilot minigrids (Output 2.1) will incorporate SES criteria during the site selection process and adopt the list of exclusion criteria that are found in the ESMF. After selection and before the commencement of the pilot activity each pilot minigrid will undergo a scoped ESIA that will analyze this risk. Pilots in any case have to comply with national environmental regulations. Regarding the minigrids planned under the investment plan (Output 2.2), this will be subjected to a SESA that will address this risk and incorporate the site-selection criteria included in the ESMF.	Project Management Unit

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
6	**Exposure to electrocution risks for humans and any fauna (ex. animals or birds) using the minigrid area**	Social and Environmental	I = 3; L = 2 (moderate)	Pilot minigrids (Output 2.1) will incorporate SES criteria during the site selection process and adopt the list of exclusion criteria that are found in the ESMF. After selection and before the commencement of the pilot activity each pilot minigrid will undergo a scoped ESIA that will analyze this risk. Details of this process can be found in the ESMF	Project Management Unit
7	**Climate events and disasters (including floods) on new and existing infrastructure**	Social and Environmental	I = 4; L = 3 (substantial)	Pilot minigrids (Output 2.1) will each undergo a scoped ESIA that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific ESMP. Details of this process can be found in the ESMF. Regarding the minigrids planned under the investment plan (Output 2.2), this will be subjected to a SESA that will address this risk and incorporate the site-selection criteria included in the ESMF.	Project Management Unit
8	**Risk on the community due to domestic connections and electricity usage and presence of hazardous materials**	Social and Environmental	I = 4; L = 3 (substantial)	Pilot minigrids (Output 2.1) will each undergo a scoped ESIA that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-	Project Management Unit

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
	(mainly batteries, e-waste).			specific ESMP. Details of this process can be found in the ESMF. In particular, operators, contractors and owners of sites shall be required to abide by the ESMP's requirements on safety measures and minimum qualifications for the handling of hazardous materials. Similarly, those responsible for connecting households should ensure the provision of qualified electrician services to do so and they take into account the type of construction of roofs (thatched or not) and walls. Consumer awareness campaigns should also be performed, including through local workshops, clear signage (pictograms and local language indications) and awareness-raising activities in 9schools and public spaces to inform communities of risks associated with installations	

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
9	**Community health and safety risks due to the construction of the pilot minigrids and relevant infrastructure after the new economic activities resulting from productive use of the energy**	Social and Environmental	I = 3; L = 2 (moderate)	Pilot minigrids (Output 2.1) will incorporate SES criteria during the site selection process and adopt the list of exclusion criteria that are found in the ESMF. After selection and before the commencement of the pilot activity each pilot minigrid will undergo a scoped ESIA that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific Environmental and Social Management Plan (ESMP). Electricity access will improve the functioning of the existing health center or clinics	Project Management Unit
10	**Risk on community health, safety and/or security due to the influx of people, mainly project workers and other newcomers subsequent to the new economic activities resulting from the productive use of the energy**	Social and Environmental	I = 3; L = 3 (moderate)	Pilot minigrids (Output 2.1) will each undergo a scoped ESIA that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific ESMP. Details of this process can be found in the ESMF. Contractors including any security personnel shall abide to UNDP's Standards of Conduct and apply best practices at all times. The project GRM will provide a means for affected community to report on any incidents that may occur as a result	Project Management Unit

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
				of this risk	

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
11	**Physical or economic displacement and loss of livelihood due to eviction from land on which pilot minigrids may be installed**	Social and Environmental	I = 4; L = 4 (substantial)	Land needs of solar PV (the bulk of the minigrids) are usually allocated in close communication with the local Chief and community After selection and before the commencement of the pilot activity, each pilot minigrid will undergo a scoped ESIA that will analyze these risks. Mitigation measures will then be adopted as described in the pursuant site-specific Environmental and Social Management Plan (ESMP), which may include a Livelihoods Restoration Plan. Details of this process can be found in the ESMF.	Project Management Unit
12	**Loss of income for fuel sellers once pilot minigrids are operational.**	Social and Environmental	I = 4; L = 4 (substantial)	The effect is small as traditional fuels are seldom bought, while kerosene or diesel use is minimal to have a big impact on sales in the region. Pilot minigrids (Output 2.1) will each undergo a scoped ESIA that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific ESMP	Project Management Unit
13	**Working conditions not in line with national and international standards**	Social and Environmental	I = 4; L = 4 (substantial)	As part of the ESIA/ESMP for each pilot minigrid (Output 2.1), Labour Management Procedures and an	Project Management Unit

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
	(by the contractor or other entities involved in the project)			Occupational Health and Safety Plan will be prepared and applied for the project to ensure labour standards and rights are upheld for project workers. In addition, the ESIA will assess the likelihood of this risk and the prevalence of child labour	

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
14	**Generation of hazardous waste (specifically e-waste) from the pilot minigrids that have been installed**	Social and Environmental	I = 3; L = 5 (moderate)	This risk will be assessed in the ESIA that will be undertaken for each pilot minigrid (Output 2.1), such that the ESMP will include a Waste Management Plan detailing the procedures for disposal of all types of waste associated with the construction and operation of the pilot minigrids.	Project Management Unit
DREI Risk categories					
1	**Energy Market Risk :** **Market access, competition and grid expansion: Limitations and inability, including due to government regulations, of minigrid developers to access the electrification market; uncertainty regarding potential future competition in electrification; unclear, or lack of, grid planning and expansion policies.** **Tariffs: Uncertainty or inflexibility in electricity tariff regulations for minigrids.**	Strategic, Political, Regulatory	High L = 2 I = 3	Build political will and develop realistic and transparent targets, using multi-tier electrification indicators. Establish co-existing (i) light-touch (no license) and (ii) comprehensive (licensed) approaches. "Light-touch regime (no license): Voluntary compliance with comprehensive regime standards. Comprehensive regime (licensed): Develop balanced technical standards/requirements for quality of electricity and grid integration, with active enforcement [45]	Project Management Unit

[45] All the risks mitigation measures should be within the project scope and stated/referred to.

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
	Technical standards: **Lack of clarity, uncertainty and/or inconsistent government technical requirements for minigrids regarding (i) quality of service and (ii) grid integration, should it occur**				

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
2	**Social acceptance Risk** **Resistance by general public and local communities due to unfamiliarity with electricity and renewable energy sources; misinformation/perceptions and lack of awareness for minigrid offerings; resistance from incumbent businesses (e.g., diesel-based generation) and users (e.g., SHS), disrupted by minigrids**	Social	Medium L = 2 I = 4	· Develop and coordinate ongoing community impact and public awareness campaigns. · Pilot models for community involvement	Project Management Unit/ with MERE/Communities/NGO/ANERSOL/ANPER
3	**Hardware Risk** **Quality of hardware: Lack of access to information on quality, reliability (performance) and cost of hardware; lack of clarity or uncertainty regarding government technical standards to ensure safety of minigrid hardware; lack of availability of warranties for components** **Availability of hardware: Lack of a competitive market for buying hardware (from both international and domestic suppliers); where appropriate, lack of locally tailored hardware** **Customs: Cumbersome customs/clearing process for**	Technical	High L = 2 I = 4	Ensure an open, competitive marketplace for buying hardware. · Streamlined and consistent customs procedures; reform of punitive custom tariff system · Development of standards and certifications for equipment ; · Streamlined customs procedures - Clarification of equipment exempt from import taxes and VAT	Project Management Unit with MERE/MF/ANERSOL

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
	importing hardware, leading to delays in delivery; punitively high customs tariffs on minigrid hardware, particularly in comparison to other sectors				

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
4	**Digital risk: Cellular networks and mobile money: lack of cellular coverage in rural areas, where electrification needed; over-dependence on a single operator for reliable cell service and payment processing; lack of mobile money, or limitations relating to fees on mobile money transactions** **Software: Limited standardization of software and interfaces on minigrid developers' back-end data and operations, and mobile money payment platforms** **Abuse of consumer data: possible abuse of consumer data privacy on payments and usage; lack of understanding/clarity on uses of consumer information**	Strategic	High L = 3 I = 4	Policy measures: Well-designed telecom regulations enabling universal, competitive coverage and mobile money. Government support to form industry associations for standard-setting and sharing of best practices. Institute balanced consumer data protection regulations · Telecommunications regulations for universal coverage and access to mobile money ; · Support capacity building for industrial associations for the sharing of best practices and standards.	Government MERE with Telecom Operators
5	**Labour Risk** **Lack of a competitive labor market of educated, skilled and qualified potential employees, leading to higher costs, hiring non-local staff and suboptimal performance**	Organizational	Medium L = 2 I = 4	Programmes to develop competitive, skilled labour market in renewable energy (all roles) · Set up programs to develop a competitive and qualified labor market in renewable energies	Project Management Unit
6	**Developer Risk**	Strategic	High		Government

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
	Management capability: lack of C-suite talent and experience to ensure effective execution (business planning, financial structuring, plant design (resource and demand assessment), installation, operations and maintenance), and to manage challenges (limited information, unforeseen events) **Developer credit worthiness and cash flow strength: Inability of developer to secure low cost financing from investors due to lack of credit worthiness, or insufficient cash flows to meet investors' return requirements**		L = 3 I = 4	Government support to improve information flows and network effects Policy measures : · Improvement of network effects and information flow / technical assistance to promoters Financial measures : · Public loans, guarantees, public equity	

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
7	**End-user credit risk** **Lack of information on end-user credit worthiness: Lack of end-user credit data with which to assess the ability of end-users to pay for the initial connection fees, ongoing electricity bills and ancillary equipment (e.g., lights and appliances)** **Poor credit worthiness and non-payment: Risk of delayed, reduced or non-payment by customers due to poor credit worthiness, lack of funds available, electricity theft and social dynamics**	Financial	High L =2 I = 4	Facilitate access to consumer finance (e.g., government-sponsored digital ID scheme; general consumer finance reform; mobile money); Promote productive use of electricity (e.g, establish network of business development incubators and advisors providing training and guidance covering mini-grid areas)". Government targets and mandates require creditworthy actors, both private (e.g., cell phone towers) and public (e.g., health centers), to obtain their electricity from renewable energy mini-grids Balanced approach to liquidity requirements, assessing trade-offs between financial stability and renewable Policy measures : · Promotion of the productive use of electricity · Facilitation of access to consumer credit / micro-credit · Government mandates to insure solvent key customers (ex: telecommunication tower) Financial measures : · Public loans, guarantees, public equity	Project Management Unit
8	**Financing Risk**	Financial	High	Reform reserve requirements for	Government MERE/Financial institutions

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
	Capital scarcity - liquidity constraints in domestic banking: Limited availability of long-term domestic loans due to high banking reserve requirements **Capital scarcity - under-developed domestic financial sector:** **Low number of well-capitalized actors (debt, equity, insurance, pensions); lack of regulatory clarity on new types of financial products** **Capital scarcity - competing incentives/ mandates: existing policies incentivize or mandate domestic financial sector (banks, pension funds) to invest in alternative, competing sectors to minigrids** **Limited domestic investor experience with minigrids: Lack of information, assessment skills and track-record for minigrid projects amongst domestic investor community; lack of network effects (investors, investment opportunities) found in established markets; lack of familiarity and skills with appropriate finance structures.**			domestic lending to businesses. Liberalize domestic financial sector. Reform financial sector incentives for investing in specific sectors. Strengthen domestic investors' (debt and equity) familiarity with and capacity regarding renewable energy minigrids Policy measures: · Liberalize the national financial sector (reforms for new types of financing, incentives) · Technical assistance to financial institutions and investors Financial measures: · Public loans, guarantees, public equity	

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
9	**Currency Risk** **Uncertainty due to volatile local currency; unfavorable currency exchange rate movements resulting in domestic currency revenues not being sufficient to cover hard currency debt/equity servicing; inability to economically hedge FX exposure due to illiquid FX derivative markets.**	Financial	Low L =1 I = 2	Government support for long term development of liquid domestic FX derivative markets Policy measures : · Support for the development of a foreign exchange market derivative products.	Government with central bank
10	**Sovereign risk** **Limitations and uncertainty related to conflict, political instability, economic performance, weather events/natural disaster, legal governance, ease of doing business, crime and law enforcement, land tenure and infrastructure in the particular country.**	Political	L = 4 I = 4	Where applicable, provision of political risk insurance (PRI) covering (i) expropriation, (ii) political violence, (iii) currency restrictions, (iv) breach of contract Financial measure : · Political risk insurance	Government

ANNEX 7: OVERVIEW OF PROJECT STAFF AND TECHNICAL CONSULTANCIES

Consultant	Time Input	Tasks, Inputs and Outputs
		For Project Management
Local / National contracting		
Project Manager/Coordinator *Rate: $1,560-1,755/month* *(according to a national salary grid voted in Niger for PMU staff of technical & financial partners)*[46]	*Full-time for 4 years* *Total value of about $95,616 (including 20% tax contribution etc.)*	*The Project Manager (PM), together with the Lead Technical Advisor will be responsible for the overall management of the project, including the mobilization of all project inputs, supervision over project staff, consultants and sub-contractors.* *The Terms of Reference (ToR) for this position should include a clear statement indicating that a minimum of 10% of the person's time will be allocated to AMP Regional Project activities. If the PM is also delegated as the 'beneficiary(ies) representative' on the AMP Regional Project board, this should also be included in their ToR.* *Duties and Responsibilities* ● *Manage the overall conduct of the project.* ● *Plan the activities of the project and monitor progress against the approved workplan.* ● *Execute activities by managing personnel, goods and services, training and low-value grants, including drafting terms of reference and work specifications, and overseeing all contractors' work.* ● *Monitor events as determined in the project monitoring plan, and update the plan as required.* ● *Provide support for completion of assessments required by UNDP, spot checks and audits.* ● *Manage requests for the provision of UNDP financial resources through funding advances, direct payments or reimbursement using the FACE form.* ● *Monitor financial resources and accounting to ensure the accuracy and reliability of financial reports.* ● *Monitor progress, watch for plan deviations and make course corrections when needed within project board-agreed tolerances to achieve results.* ● *Ensure that changes are controlled and problems addressed.* ● *Perform regular progress reporting to the project board as agreed with the board, including measures to address challenges and opportunities.* ● *Prepare and submit financial reports to UNDP on a quarterly basis.* ● *Manage and monitor the project risks – including social and environmental risks - initially identified and submit new risks to the Project Board for consideration and decision on possible actions if required; update the status of these risks by maintaining the project risks log;* ● *Capture lessons learned during project implementation.* ● *Prepare revisions to the multi-year workplan, as needed, as well as annual and quarterly plans if required.* ● *Prepare the inception report no later than one month after the inception workshop.* ● *Ensure that the indicators included in the project results framework are monitored annually in advance of the GEF PIR submission deadline so that progress can be reported in the GEF PIR.* ● *Prepare the GEF PIR;* ● *Assess major and minor amendments to the project within the parameters set by UNDP-GEF;* ● *Monitor implementation plans including the gender action plan, stakeholder engagement plan, and any environmental and social management plans;* ● *Monitor and track progress against the GEF Core indicators.*

[46] Grille Harmonisée de rémunération brute du personnel des projets financés par les PTF May 2022

Consultant	Time Input	Tasks, Inputs and Outputs
		● *Support the Mid-term review and Terminal Evaluation process.* ● *Liaise with the AMP Regional Project PMU Staff to request and receive operational and technical support as needed, to participate in activities led by the AMP Regional Project, and share data and information with the AMP Regional Project as required.* ● *Additional tasks as deemed necessary*
Project Assitant Coordinator *Rate: $1,215-1,370/month* *(according to a national salary grid voted in Niger for PMU staff of technical & financial partners)*[47] ***$11,000 a year covered by UNDP funds***	*Full-time for 4 years* *Total value of about $74,376 (including 20% tax contribution etc.)*	*The Terms of Reference (ToR) for this position should include a clear statement indicating that a minimum of 10% of the person's time will be allocated to AMP Regional Project activities.* *<u>Duties and Responsibilities</u>* *Under the guidance and supervision of the Project Manager, the Project Assistant Coordinator will carry out the following tasks:* ● *Assist the Project Manager in day-to-day management and oversight of project activities;* ● *In charge of specific project activities* ● *Assist the M&E efforts in matters related to M&E and knowledge resources management;* ● *Assist in the preparation of progress reports;* ● *Ensure all project documentation (progress reports, consulting and other technical reports, minutes of meetings, etc.) are properly maintained in hard and electronic copies in an efficient and readily accessible filing system, for when required by PB, TAC, UNDP, project consultants and other PMU staff;* ● *Contribute actively to Output 2.1. on pilot projects* ● *Additional tasks as deemed necessary*
Administrative Assistant (including financial aspects) *$800-900/month* *(according to a national salary grid voted in Niger for PMU staff of technical & financial partners)*[48]	*Full-time for 4 years* *Total value of about $48,744 (including 20% tax contribution etc.)*	*The Terms of Reference (ToR) for this position should include a clear statement indicating that a minimum of 10% of the person's time will be allocated to AMP Regional Project activities. The Terms of Reference (ToR) for this position should clearly indicate commitment not only to the national project but also to the AMP Regional Project's M&E protocols as regards provision of timely reporting data to the regional project staff.* *<u>Duties and Responsibilities</u>* *Under the guidance and supervision of the PM and Project Assistant Coordinator, the Administrative Assistant will carry out the following tasks:* ● *Manage all aspects related to the good functioning of the PMU in relation to all secretarial activities* ● *Provide administrative backstopping to the project team, and also manage project finances* ● *Provide logistical assistance* ● *Keep records of project funds and expenditures, and ensure all project-related financial documentation are well maintained and readily available when required by the Project Manager;* ● *Review project expenditures and ensure that project funds are used in compliance with the Project Document and GoI financial rules and procedures;* ● *Validate and certify FACE forms before submission to UNDP;* ● *Provide necessary financial information as and when required for project management decisions;*

[47]as above

[48]as above

Consultant	Time Input	Tasks, Inputs and Outputs
		● *Provide necessary financial information during project audit(s);* ● *Review annual budgets and project expenditure reports, and notify the Project Manager if there are any discrepancies or issues;* ● *Consolidate financial progress reports submitted by the responsible parties for implementation of project activities;* ● *Liaise and follow up with the responsible parties for implementation of project activities in matters related to project funds and financial progress reports.* ● *Liaise with the AMP Regional Project PMU Staff to request and receive operational and technical support as needed, to participate in activities led by the AMP Regional Project, and share data and information with the AMP Regional Project as required* ● *Additional tasks as deemed necessary*
International / Regional and global contracting		
UN Volunteer *$1,250/month*	*Full-time for 4 years* *Total value of about $60,000 (including 20% tax contribution etc.)*	*The Terms of Reference (ToR) for this position should include a clear statement indicating that a minimum of 10% of the person's time will be allocated to AMP Regional Project activities.* *Duties and Responsibilities* *Under the guidance and supervision of the PM and Project Assistant Coordinator, the VNU will carry out the following tasks:* ● *Assist the Project Manager in day-to-day management and oversight of project activities;* ● *In charge of specific project activities* ● *Assist the M&E efforts in matters related to M&E and knowledge resources management;* ● *Assist in the preparation of progress reports;* ● *Ensure all project documentation (progress reports, consulting and other technical reports, minutes of meetings, etc.) are properly maintained in hard and electronic copies in an efficient and readily accessible filing system, for when required by PB, TAC, UNDP, project consultants and other PMU staff;* ● *Additional tasks as deemed necessary*
For Technical Assistance		
Component 1: Policy and regulation		
Local / National contracting		
National Dialogue Platform Organizer & Facilitator *Rate: $300/meeting*	*20 meetings over 4 years* *Total value of $6,000*	*The purpose of this consultancy is to support the project team with planning, organizing, facilitating the operationalization and meetings of the National Dialogue Platform. The consultant(s) will also act as the point of contact for the National Dialogue Platform, facilitating the work between the members of the Platform as well as with other stakeholders and the PMU.*
Minigrid Policy and Regulations Expert *Rate: $300/day*	*55 days over year 1 & 2* *Total value of about $16,500*	*The purpose of this consultancy is to cover costs for necessary studies based on short assessment under activity 1.2.1. (provision). A close collaboration with the PMU, ANPER and the National Dialogue Platform is critical.*

Consultant	Time Input	Tasks, Inputs and Outputs
DREI Expert *Rate: $300/day*	*15 days in year 1* *Total value of about $4,500*	*The purpose of this consultancy is to support the selection & implementation of DREI instruments to be covered by AMP. The DREI Analysis has been conducted under the ALG project financed by SIDA and executed by UNDP from end of 2021 to mid 2022.*
Training Facilitator and Capacity Building Specialist with background in minigrids and energy policy & tendering *Rate: $300/day*	*60 days over year 1 & 2* *Total value of about $18,000*	*The purpose of this consultancy is to support the project team and the international training consultant with conducting a knowledge gap analysis and the necessary training workshops, and other individual and institutional capacity building activities involved in successful achievement of Component 1. In the preparation of training materials and operations manuals, the consultant(s) will collaborate with the AMP Regional Project and follow the guidance issued by the Communities of Practice (CoPs) to ensure harmonization and knowledge sharing on the activities under Component 1.*
Standards Expert *Rate: $300/day*	*30 days in year 1* *Total value of $9,000*	*The purpose of this consultancy is to conduct a review (and benchmark) and to develop the relevant standards, and the preparation & holding of the workshop - and support the international Standards Expert*
Contractual Services	*4 trainings over the entire project duration* *Total value of $5,000*	*Provision of short trainings for the National Dialogue Platform*
Contractual Services	*Year 1* *Total value of $16,000*	*The purpose of this consultancy is to conduct study on existing training offer and relevant gaps, as well as recommendations to overcome then for Public Programmes*
International / Regional and global contracting		
2 National Dialogue Platform Experts *Rate: $500/day*	*7 days in year 1* *Total value of $7,000*	*The purpose of this consultancy is to provide training and advisory on setting up, operationalizing and moderating a national dialogue platform around energy/minigrids. 2 experts from the successful implementation and operations of the Senegalese national dialogue platform will undertake a South-South Cooperation visit and exchange.*
Minigrid Policy and Regulations Expert *Rate: $600/day*	*25 days over year 1 & 2* *Total value of about $15,000*	*The purpose of this consultancy is to cover costs for necessary studies based on short assessment under activity 1.2.1. (provision). A close collaboration with the PMU, ANPER and the National Dialogue Platform is critical.*
Training Facilitator and Capacity Building Specialist with background in minigrids and energy policy & tendering *Rate: $600/day*	*40 days over year 1 & 2* *Total value of about $24,000*	*The purpose of this consultancy is to conduct a knowledge gap analysis of institutional capacities at technical, managerial and regulatory levels, including incorporating cost-cutting levers and innovative business models, developing comprehensive modular training materials, conducting training and ToT*

Consultant	Time Input	Tasks, Inputs and Outputs
Standards Expert *Rate: $600/day*	*30 days in year 1* *Total value of $18,000*	*The purpose of this consultancy is to conduct a review (and benchmark) and to develop the relevant standards, and the preparation & holding of the workshop*
minigrid Education and Vocational Training Expert *Rate: $600/day*	*12 days over year 1 & 2* *Total value of about $7,200*	*The purpose of this consultancy is to act as a Training facilitator/Capacity building specialist to train the trainers of relevant institutions in Niger*
Contractual Services	*Year 1* *Total value of $35,000*	*The purpose of this consultancy is to conduct an analysis to cluster existing (pre-)-feasibility studies and draw recommendations on suitable delivery models, business models and technologies based on village types etc.*
Component 2: Business Model Innovation with Private Sector		
Local / National contracting		
Training Facilitators and Capacity Building Specialist with Engineering background *Rate: $300/day*	*100 days over total project duration* *Total value of 30,000*	*Capacity building for the communities including becoming rural electricians (60 days) and for local private sector developers (40 days) with pedagogical engineering, Training and ToT on business models, cost-reduction levers, new technologies depending on the topics, tariff negotiation, understanding of anchor loads, remote monitoring, etc. all services - several modules ; and knowledge transfer*
Private RE Services Provider Association Assistant *Rate: $180/day*	*Part time from Years 2-4* *Total value of $10,080*	*The purpose of this consultancy is to take care of the bureau and the association, update the website and social media, organize meetings, etc.*
Contractual Services	*Entire project duration* *Total value of $11,200*	*Provision for additional capacity building needed for communities/local private sector developers & operators*
Contractual Services	*Year 4* *Total value of $9,000*	*Development of the Private Sector RE association and relevant communications strategy and roll-out*
International / Regional and global contracting		
Training Facilitators and Capacity Building Specialist with Engineering background *Rate: $600/day*	*40 days over year 1 & 2* *Total value of about $24,000*	*Pedagogical engineering, Training and ToT on business models, cost-reduction levers, new technologies depending on the topics, tariff negotiation, understanding of anchor loads, remote monitoring, etc. all services - several modules ; and knowledge transfer*
Contractual Services	*Year 1* *Total value of $30,000*	*Analysis on key national rural economic outputs and their value chains, and social activities including field visits on key value chains and stakeholder consultations, as well as 1 mission to Niger for the study and field visits inside the country*
Component 3: Scaled-up Financing		
Local/National Contracting		

Consultant	Time Input	Tasks, Inputs and Outputs
Scale-up & Investment Planning Expert *Rate: $300/day*	*40 days in Year 4* *Total value of $12,000*	*Development of the replication plan (includingincl. investment planning).with field visits includingincl. pilot sites and some eligible communities*
International / Regional and global contracting		
Training Facilitators and Capacity Building Specialists with background in Finance *Rate: $600/day*	*Years 1 - 3* *Total value of $21,000*	*Develop and conduct relevant training on business models and innovative finance solutions for national financial institutions*
Contractual Services	*Years 1 & 2* *Total value of $76,700*	*Conduct a benchmarking of existing financial products supporting access and use of energy in Niger and aborad, and other industries,Identify innovative financing schemes that are relevant and applicable to support the development of the minigrid market in Niger. Selection of 1-2 financing mechanisms and technical assistance around these*
Scale-up & Investment Planning Expert *Rate: $300/day*	*15 days in Year 4* *Total value of $9,000*	*Development of the replication plan (includingincl. investment planning) - remote*
Component 4: Digital and Knowledge Management		
Local / National contracting		
SES Expert *Rate: $300/day*	*29 days per year, over 4 years* *Total value of about $34,800*	*The expert will be nationally recruited by UNDP and she/he will be responsible for undertaking social and environmental studies related to the activities of the project.* *<u>Duties and Responsibilities:</u>* *· Monitor progress in development/implementation of the project ESMF ensuring that UNDPs SES policy is fully met and the reporting requirements are fulfilled;* *· Oversee/develop/coordinate implementation of all safeguard related plans;* *· Ensure social and environmental grievances are managed effectively and transparently;* *· Review the SESP annually, and update and revise corresponding risk log; mitigation/management plans as necessary;* *· Ensure full disclosure with concerned stakeholders;* *· Ensure environmental and social risks are identified, avoided, mitigated and managed throughout project implementation;* *· Work with the M&E officer to ensure reporting, monitoring and evaluation fully address the safeguard issues of the project;* *· Assist the finance and administration staff by providing technical inputs during the preparation and revision of the Management Plan, Annual Work Plans, periodic reports such as the Combined Project Implementation Review/Annual Project Report (PIR/APR), inception report, technical reports, quarterly reports for submission to UNDP, the GEF, other donors and Government Departments, as required;* *· Ensure quality control of interventions/outcomes/deliverables;* *· Document lessons learned from project implementation and make recommendations to the Steering Committee for more effective implementation and coordination of project activities.*

Consultant	Time Input	Tasks, Inputs and Outputs
Gender Expert ***Rate: $300/day***	*29 days each per year, over 4 years* *Total value of about $34,800*	*The expert will be nationally recruited by UNDP and she/he will be responsible for undertaking social and environmental studies related to the activities of the project.* *Duties and Responsibilities for the Gender role:* · *Monitor progress in implementation of the project Gender Action Plan ensuring that targets are fully met and the reporting requirements are fulfilled;* · *Oversee/develop/coordinate implementation of all gender-related work;* · *Review the Gender Action Plan annually, and update and revise corresponding management plans as necessary;* · *Work with the M&E officer and Safeguards Officer to ensure reporting, monitoring and evaluation fully address the gender issues of the project.*
Contractual Services	*Year 4* *Total value of $12,000*	*This consultancy shall include the design of the Project's Digital Strategy, National monitoring and evaluation platform (remote monitoring & analytics) 300/site/year data collection, storage, and management fee (can vary by number of connections at site); additional training and support of IP & co (3k) and Rural electrification platform development (year 1) and maintenance fee for Years 2-4* *Setting up the Quality Assurance and Monitoring Framework for the project complying and in coordination with the AMP regional requirement. Training the PMU and other targeted consultants* *(covering Outputs 4.1, 4.2., 4.3.)*
Contractual Services	*Entire project duration* *Total value of about $34,000*	*Awareness raising campaign targeting different segments including general public. Communication strategy and roll out (including short video production and digital marketing)*
International / Regional and global contracting		
Contractual Services	*Entire project duration* *Total value of about $48,600*	*This local contractor will be responsible for data collection and development of communications content (including photos and/or video footage) for the preparation of an 'insight brief' capturing (in an accessible format) selected key highlights from a successful national project activity. This insight brief will be developed in a standard format provided by the AMP Regional Project – noting that the fees include translation of the final output from French to English. The AMP Regional Project will also support the dissemination of the Insight Briefs developed by the national AMP projects.*
Component 5: Monitoring and Evaluation (M&E)		
Local / National contracting		
Local M&E Consultant to support the Mid-term Review (MTR)	*Year 2 (Q4)* *Total value of about $10,000*	*Contracted to support the international M&E Consultant on performing the MTR for the project.*
Local M&E Consultant to support the Terminal Evaluation (TE)	*Year 4 (Q3 & Q4)* *Total value of about $20,000*	*Contracted to support the international M&E Consultant on performing the TE for the project.*
International / Regional and global contracting		
International M&E Consultant	*Year 2 (Q4)*	*Contracted to conduct the MTR for the project, including review of available document, field mission, interviews with*

Consultant	Time Input	Tasks, Inputs and Outputs
	Total value of about $20,000	*stakeholders, providing recommendations, and issuance of the MTR report.* *Liaise with the AMP Regional Project PMU Staff to request and receive operational and technical support as needed, to participate in activities led by the AMP Regional Project, and share data and information with the AMP regional Project as required. The Terms of Reference (ToR) for this position should clearly indicate commitment not only to the national project but also to the Regional Project's M&E protocols as regards provision of timely reporting data to the regional project staff. The ToR should also include a clear statement indicating that a minimum of 10% of the person's time will be allocated to regional project activities.*
International M&E Consultant	*Year 4 (Q3-Q4)* *Total value of about $30,000*	*Contracted to conduct the TE for the project, including review of available document, field mission, interviews with stakeholders, assessment of lessons learned, and issuance of the TE report.* *Liaise with the AMP Regional Project PMU Staff to request and receive operational and technical support as needed, to participate in activities led by the AMP Regional Project, and share data and information with the AMP regional Project as required. The Terms of Reference (ToR) for this position should clearly indicate commitment not only to the national project but also to the Regional Project's M&E protocols as regards provision of timely reporting data to the regional project staff. The ToR should also include a clear statement indicating that a minimum of 10% of the person's time will be allocated to regional project activities.*

ANNEX 8: STAKEHOLDER ENGAGEMENT PLAN

(See Annex Attached)

ANNEX 9: ENVIRONMENTAL AND SOCIAL MANAGEMENT FRAMEWORK (ESMF)

(See Annex Attached)

ANNEX 10: GENDER ANALYSIS AND GENDER ACTION PLAN

(See Annex Attached)

ANNEX 11: PROCUREMENT PLAN

(See Annex Attached)

ANNEX 12: GEF FOCAL AREA SPECIFIC ANNEXES

(e.g. METT, GHG calculations, target landscape profile, feasibility study, other technical reports)

(See Annex Attached)

ANNEX 13: CO-FINANCING LETTERS

(include letters of financial commitments such as letters of cofinancing and UNDP TRAC commitment letters, etc..)

(See Annex Attached)

ANNEX 14: ADDITIONAL AGREEMENTS

(none)

ANNEX 15: GEF CORE INDICATORS

Core Indicator 6	**Greenhouse gas emission mitigated**				***(Metric tons of CO_2e)***
		Expected metric tons of CO_2e (6.1+6.2)			
		PIF stage	Endorsement	MTR	TE
	Expected CO2e (direct)	*4.987*	*17.015*		
	Expected CO2e (indirect)	*2.184.053*	*2.052.000*		

Indicator 6.4	Increase in installed renewable energy capacity per technology					
		Technology	Capacity (MW/MWh)			
			Expected		Achieved	
			PIF stage	Endorsement	MTR	TE
		Solar PV	*0.101 MW*	*0.334 MW*		
		Storage	*n.a. MWh*	*0.814 MWh*		

Core Indicator 11	**Number of direct beneficiaries disaggregated by gender as co-benefit of GEF investment**				***(Number)***
			Number			
			Expected		Achieved	
			PIF stage	Endorsement	MTR	TE
		Female	*2.540*	*6.902*		
		Male	*2.489*	*6.632*		
		Total	*5,029*	*13.534*		

ANNEX 16: GEF 7 TAXONOMY

Level 1	Level 2	Level 3	Level 4
☒**Influencing models**			
	☒**Transform policy and regulatory environments**		
	☒**Strengthen institutional capacity and decision-making**		
	☒**Convene multi-stakeholder alliances**		
	☒**Demonstrate innovative approaches**		
	☒**Deploy innovative financial instruments**		
☒**Stakeholders**			
	☐**Indigenous Peoples**		
	☒**Private Sector**		
		☒Capital providers	
		☒Financial intermediaries and market facilitators	
		☒Large corporations	
		☒SMEs	
		☒Individuals/Entrepreneurs	
		☐Non-Grant Pilot	
		☐Project Reflow	
	☒**Beneficiaries**		
	☒**Local Communities**		
	☒**Civil Society**		
		☒Community Based Organization	
		☐Non-Governmental Organization	
		☐Academia	
		☐Trade Unions and Workers Unions	
	☒**Type of Engagement**		
		☒Information Dissemination	
		☒Partnership	
		☒Consultation	
		☒Participation	
	☒**Communications**		
		☒Awareness Raising	
		☒Education	
		☒Public Campaigns	
		☒Behavior Change	
☒**Capacity, Knowledge and Research**			
	☒**Enabling Activities**		
	☒**Capacity Development**		
	☒**Knowledge Generation and Exchange**		
	☐**Targeted Research**		
	☒**Learning**		
		☒Theory of Change	
		☒Adaptive Management	
		☒Indicators to Measure Change	
	☒**Innovation**		
	☒**Knowledge and Learning**		
		☒Knowledge Management	
		☒Innovation	
		☒Capacity Development	

Level 1	Level 2	Level 3	Level 4
		☒Learning	
	☒**Stakeholder Engagement Plan**		
☒**Gender Equality**			
	☒**Gender Mainstreaming**		
		☒Beneficiaries	
		☒Women groups	
		☒Sex-disaggregated indicators	
		☒Gender-sensitive indicators	
	☒**Gender results areas**		
		☐Access and control over natural resources	
		☒Participation and leadership	
		☒Access to benefits and services	
		☒Capacity development	
		☒Awareness raising	
		☒Knowledge generation	
☒**Focal Areas/Theme**			
	☒**Climate Change**		
		☒**Climate Change Mitigation**	
			☐Agriculture, Forestry, and other Land Use
			☒Energy Efficiency
			☐Sustainable Urban Systems and Transport
			☒Technology Transfer
			☒Renewable Energy
			☒Financing
			☒Enabling Activities
		☒**United Nations Framework on Climate Change**	☒Nationally Determined Contribution
	☒ **Rio Markers**		
		☒Paris Agreement	
		☒Sustainable Development Goals	
		☐Climate Change Mitigation 0	
		☐Climate Change Mitigation 1	
		☒Climate Change Mitigation 2	
		☐Climate Change Adaptation 0	
		☐Climate Change Adaptation 1	
		☐Climate Change Adaptation 2	

ANNEX 17: PARTNERS CAPACITY ASSESSMENT TOOL AND HACT ASSESSMENT

(See Annexes Attached: 17A, 17B and 17C)

ANNEX 18: UNDP PROJECT QUALITY ASSURANCE REPORT

(See Annex Attached)